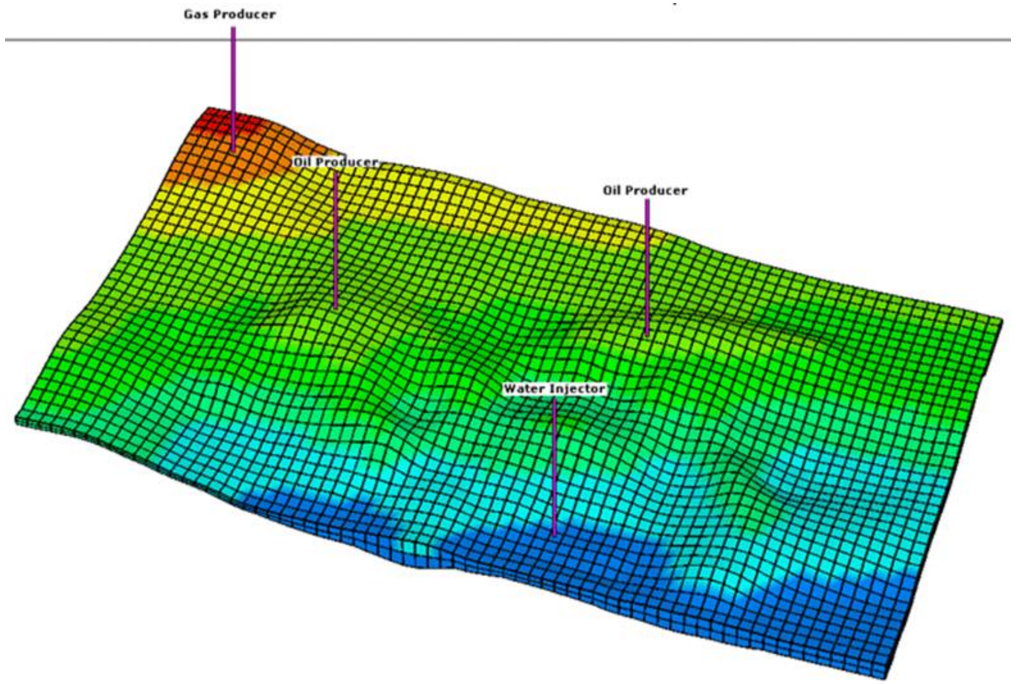
FLOW IN POROUS MEDIA



Dr. Adegboyega B EHINMOWO

Chemical and Petroleum Engineering Department
University of Lagos

Course Outline

- Fluid and rock properties (Review), Darcy's Law,
- Classification of reservoir flow System
- Steady state flow (Linear flow of incompressible and compressible fluids, Linear beds in series and parallel, Poiseuille's Law of capillary flow &Flow through fractures)
- Steady state (Radial flow of incompressible and compressible fluids, Permeability variations)
- Unsteady Flow(Flow of compressible fluids in bounded drainage areas, Average pressure in radial flow systems and re-adjustment time)
- Productivity index, Zonal damage and well stimulation, deliverability, Well spacing, and Recovery
- Displacement of oil and gas

Module 1-Introduction

Fluid Flow in Porous Media

- A complex phenomenon compared with flow in pipes
- There are no clear-cut flow paths that lend themselves to measurement whereas it is easy to measure the length and diameter of a pipe and compute its flow capacity as a function of pressure
- In this course, there will be no experimental work rather we will explore the mathematical forms of the relationships that describe the flow behavior of reservoir fluids.
- These mathematical equations are dependent on the characteristics of the reservoir
- The primary reservoir characteristics that must be considered include:

Module 1-Introduction

Fluid Flow in Porous Media

- Types of fluid in the reservoir (Incompressible, slightly compressible and Compressible fluids)
- Flow regimes (steady state, Unsteady state, Pseudo-steady state flow)
- Reservoir geometry(Radial flow, Linear flow, spherical and hemispherical flow)
- Number of flowing fluids in the reservoir(single phase, two phase, three phase)

Module 1-Introduction

Fluid Flow Equations

- Depending upon the combination of characteristics previously presented (i.e., types of flow, types of fluids, reservoir geometry, flow regime), the following fundamental equations can be combined to describe the flow of fluids in porous medium

❖ Conservation of mass equation

❖ Momentum/Transport equation-Darcy's equation

❖ Equation of state

❖ Energy Equation

Module 1-Introduction

Review : Fluid and Rock Properties

- Porosity
- Permeability
- Compressibility
- Viscosity
- Density
- Formation Volume Factors
- Solution gas: oil ratio
- ETC

Module 1-Introduction

Water viscosity

- Meehan (1980) proposed a water viscosity correlation that accounts for both the effects of pressure and salinity:

$$\mu_w = \mu_{wD}\left[1 + 3.5 * 10^{-2} P^2 (T - 40)\right]$$

$$\mu_{wD} = A + B/T$$

$$A = 4.518 * 10^{-2} + 9.313 * 10^{-7} Y - 3.93 * 10^{-12} Y^2$$

$$B = 70.634 + 9.576 * 10^{-10} Y^2$$

where μ_w = brine viscosity at p and T, cp
μ_{wD} = brine viscosity at p = 14.7, T, cp
p = pressure of interest, psia
T = temperature of interest, T °F
Y = water salinity, ppm

Module 1-Introduction

Water viscosity

- Brill and Beggs (1978) presented a simpler equation which considers only temperature effects:

$$\mu_w = exp[1.003 - 1.479 * 10^{-2} T + 1.982 * 10^{-5} T^2]$$

where Tis in °F and μw is in cp

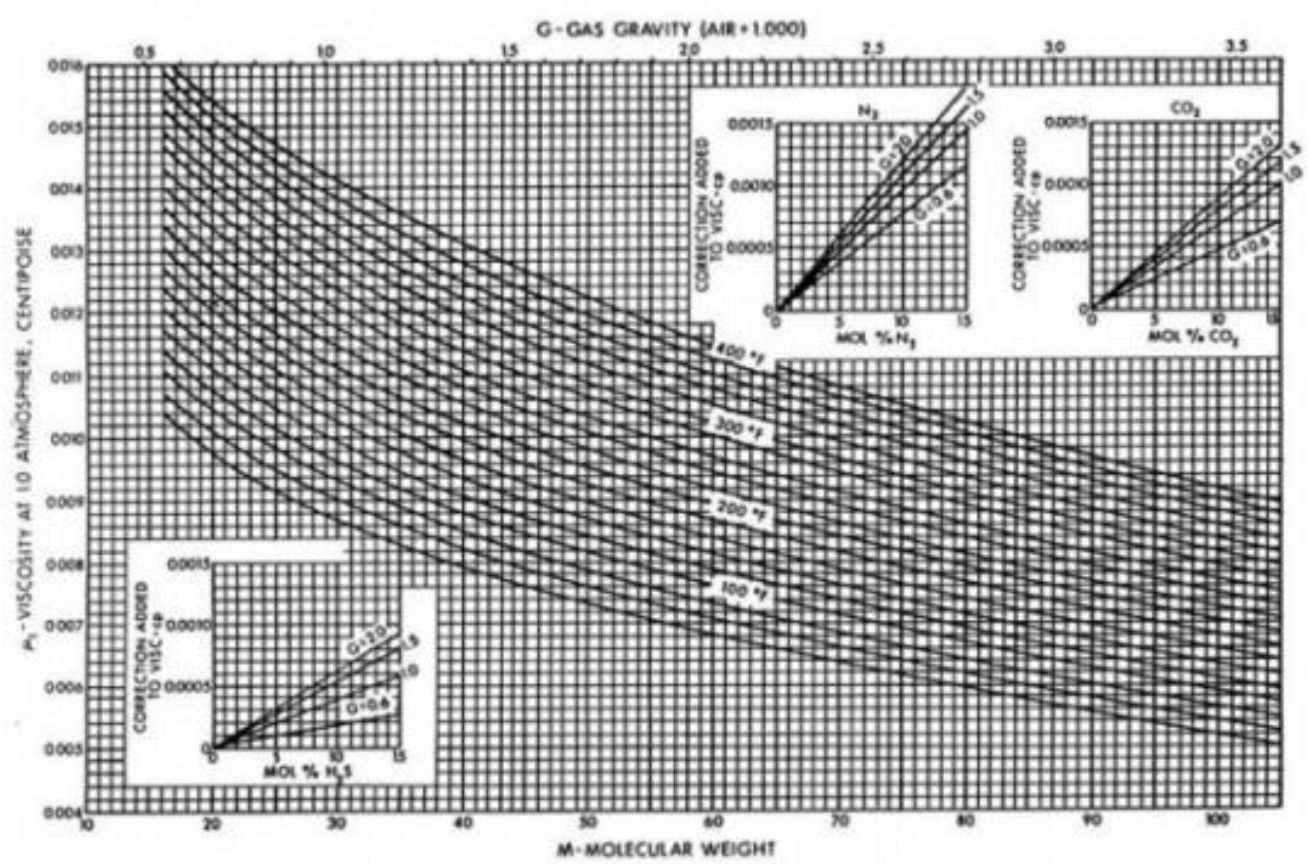
Gas viscosity

- Viscosity is a measure of a fluid’s internal resistance to flow.
- The viscosity of a natural gas, expected to increase with both pressure and temperature, is usually several orders of magnitude smaller than that of oil or water;
- Therefore, gas is much more mobile in the reservoir than either oil or water.
- If the composition of the natural gas mixture is known, then the viscosity of the mixture at given temperature and 1 atm pressure can be calculated

Gas viscosity

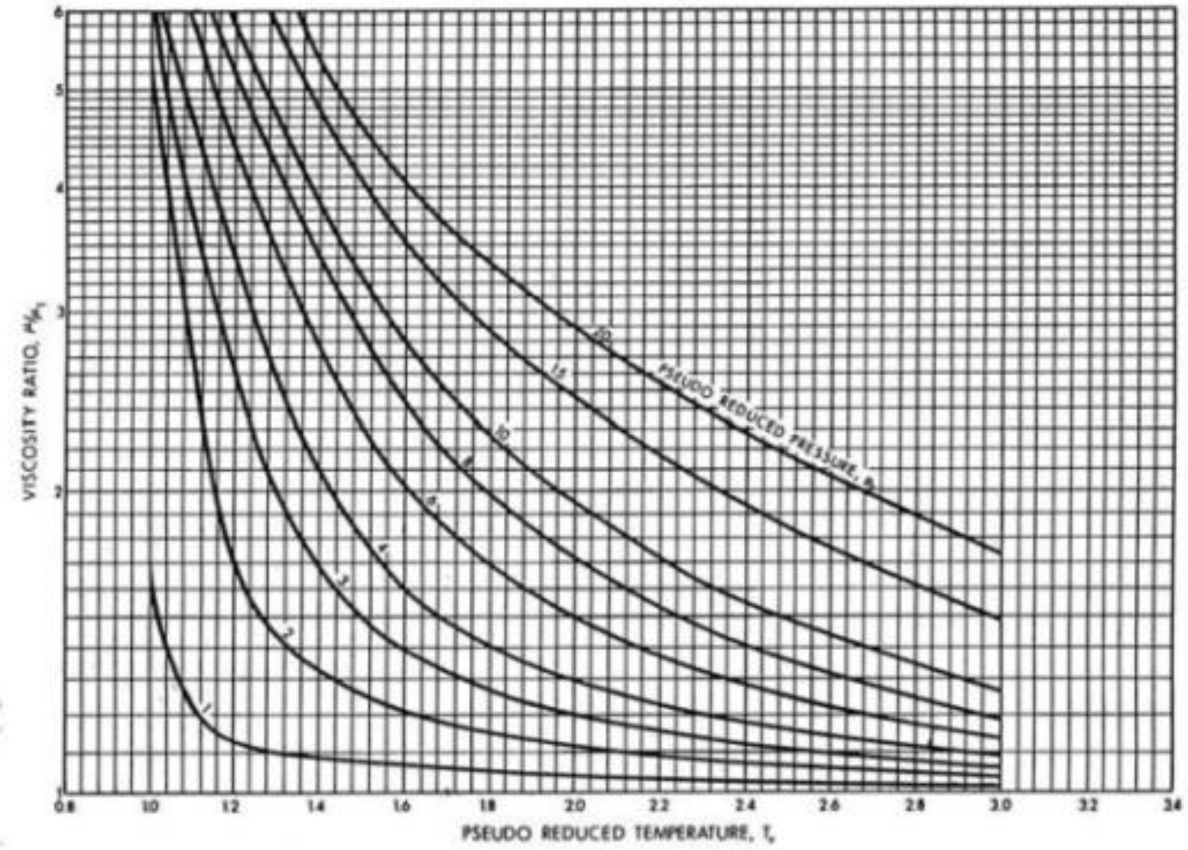
- Useful Gas viscosity correlations have been presented by a number of authors. However, the Carr, Kobayashi, and Burrows (1954) correlation presented in Figures in slides 11 and 12 has been the most popular.
- Figure in slide 11 allows the calculation of the viscosity at any temperature and at a pressure of 1 atm.
- Figure in slide 12 provides the estimation of μ / μ 1atm , which is the ratio of the viscosity at an elevated pressure to the viscosity at 1 atm

Gas viscosity



Viscosity of natural gases at 1 atm

Gas viscosity



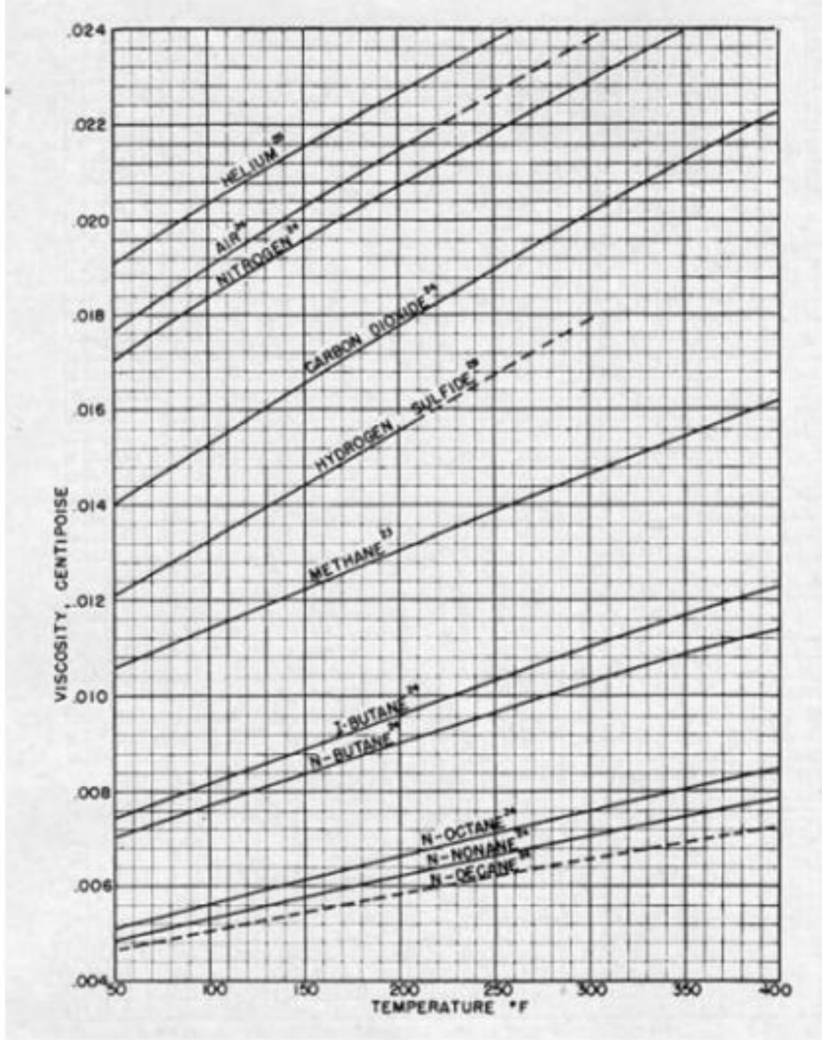
Viscosity ratio at elevated pressures and temperatures

Gas viscosity

$$\mu_g = \frac{\sum \mu_{gi} y_i MW_i^{1/2}}{\sum y_i MW_i^{1/2}}$$

- Where μ_{gi} is the viscosity of the individual component in the gas mixture at given temperature and 1 atm which can be obtained from Figure in the next slide

Gas viscosity



Gas viscosity

- With the estimation of μ / μ 1atm from Figure in slide 12, the viscosity at an elevated pressure and given temperature can be calculated.

- The most commonly used unit of viscosity is the centipoises (cp).

- 1 cp is 0.01poise (p), or 0.000672 lbm/ft-s, or 0.001 Pa-s.

Oil viscosity

- Crude oil viscosity is an important physical property that controls and influences the flow of oil through porous media and pipes.
- The oil viscosity is highly dependent on the temperature, pressure, oil,gravity, gas gravity, and gas solubility.
- Whenever possible, oil viscosity should be determined by laboratory measurements at reservoir temperature and pressure.
- The viscosity is usually reported in standard PVT analyses.

Oil viscosity

- Correlations are usually used in the absence of laboratory data
- According to the pressure, the viscosity of crude oils can be classified into three categories: Dead-Oil Viscosity, saturated-oil viscosity and under-saturated oil viscosity
- The dead-oil viscosity is defined as the viscosity of crude oil at atmospheric pressure (no gas in solution) and system temperature.
- The saturated (bubble-point)-oil viscosity is defined as the viscosity of the crude oil at the bubble-point pressure and reservoir temperature.

Oil viscosity

- The undersaturated-oil viscosity is defined as the viscosity of the crude oil at a pressure above the bubble-point and reservoir temperature

Assignment

Attempt a detailed discussion of the methods for the calculation of the viscosity of dead oil, saturated oil and undersaturated oil (Duration -1 week)

Module 1

Darcy's Law

- The fundamental law of fluid motion in porous media is Darcy's Law.
- The mathematical expression developed by Henry Darcy in 1856 states the velocity of a homogeneous fluid in a porous medium is proportional to the pressure gradient and inversely proportional to the fluid viscosity.
- For a horizontal linear system, this relationship is given as:

$$V = \frac{q}{A} = -\frac{k}{\mu}\frac{dP}{dx} \quad (1)$$

ν is the apparent velocity in centimeters per second and is equal to q/A

Module 1

Darcy's Law

- where q is the volumetric flow rate in cubic centimeters per second
- A is total cross-sectional area of the rock in square centimeters. In other words, A includes the area of the rock material as well as the area of the pore channels.
- The fluid viscosity, μ, is expressed in centipoise units
- The pressure gradient, dp/dx, is in atmospheres per centimeter, taken in the same direction as v and q.
- The proportionality constant, k, is the permeability of the rock expressed in Darcy units.

Module 1

Darcy’s Law

- Broadly speaking, Darcy’s law states that the velocity of a homogeneous fluid in porous medium is proportional to the driving force and inversely proportional to the fluid viscosity and its given as:

$$V = -0.001127\frac{k}{\mu}\left[\frac{dP}{dx} - 0.433\gamma' cos\alpha\right] \qquad (2)$$

where

- V the apparent velocity , bbls/day-ft^2
- K= permeability, millidarcies (md)
- The fluid viscosity, μ, is expressed in centipoise
- X is distance along flow path in ft

Module 1

Darcy's Law

- γ' fluid specific gravity (always relative to water)
- α = the angle measured counterclockwise from the downward vertical to the positive x direction
- The term $\left[\frac{dP}{dx} - 0.433\gamma' cos\alpha\right]$ represents the driving force. The fluid may be driven by pressure gradient $\frac{dP}{dx}$ and or hydraulic (gravitational) gradients ($0.433\gamma' cos\alpha$)
- The hydraulic gradients is usually neglected in many practical cases so equation (2) reduces to a form of eqn (1)
- But must be considered in other cases e.g production by pumping from reservoir whose pressure have been depleted and gas cap expansion reservoirs with good gravity drainage characteristics

Module 1

Darcy's Law

- The apparent velocity v is equal to qB/A, q is the volumetric flow rate in STB/day, B is the formation volume factor and A is the apparent or total cross sectional area of the rock in square feet.
- The fluid pressure gradient $\frac{dP}{dx}$ is taken in the same direction as flow, thus the negative sign in front of the constant 0.001127 indicate that if the flow is taken as positive in the positive x-direction, then the pressure decreases in that direction so that the slope $\frac{dP}{dx}$ is negative.
- Darcy's law is only applicable for laminar flow but invalid for turbulent flow since the pressure gradient increases at a greater rate than the flow rate in turbulent flow.

Module 1

Darcy's Law

- Equations (1) and (2) suggest that the velocity and pressure gradient are related by the mobility (λ).
- The mobility is the ratio of permeability to viscosity $\frac{k}{\mu}$.
- For multiphase flow, for example two-phase gas and oil flow ,it is the ratio of the mobility of the gas λ_g to that of oil λ_o which determines their individual flow rates.
- The mobility ratio M is an important factor affecting the displacement efficiency of oil by water (details in the displacement of oil & gas module)

Module 2-Classification of Reservoir flow systems

Reservoir flow systems are usually classed according to :

- The type of fluid in the reservoir (Incompressible, slightly compressible and Compressible fluids)
- The geometry of the reservoir or portion thereof (Radial flow, Linear flow, spherical and hemispherical flow)
- The relative rate at which the flow approaches a steady state condition following a perturbation. (steady state, Unsteady state, Pseudo-steady state flow)

Module 2-Classification of Reservoir flow systems

Types of fluids and compressibility

Reservoir fluid may either be incompressible, slightly compressible or compressible

- The incompressible fluid: For this type of fluid, volume does not change with pressure. This is a rather an assumption which helps simplifies the derivation and final form of many equations and are accurate for many purposes. However, in reality, there is no incompressible fluid.
- Slightly compressible fluid: This is the true description of nearly all liquids. For this type of fluid, volume change with pressure is quite small and can be expressed as:

$$V = V_R e^{c(P_R - P)} \qquad (3)$$

Module 2-Classification of Reservoir flow systems

Types of fluids and compressibility

R= the reference conditions.

Equation (3) can be derived from the integration of isothermal compressibility expression for a substance $c = -\frac{1}{v}\frac{dv}{dP}$

- But e^x can be expressed by a series expansion as

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \ldots$$

- Where x is small, the first two terms will suffice, therefore equation (3) can be written as:

$$V = V_R[1 + c(P_R - P)] \qquad (4)$$

Module 2-Classification of Reservoir flow systems

Types of fluids and compressibility

- Compressible fluid: For this type of fluid, the volume has strong dependence on pressure.
- Unlike slightly compressible fluids, the gas isothermal compressibility c_g can not be treated as a constant with varying pressure.
- The gas compressibility is given as:

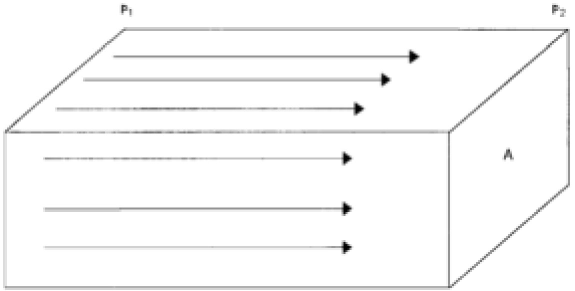
$$c_g = \frac{1}{P} - \frac{1}{Z}\frac{\partial Z}{\partial P} \qquad (4*)$$

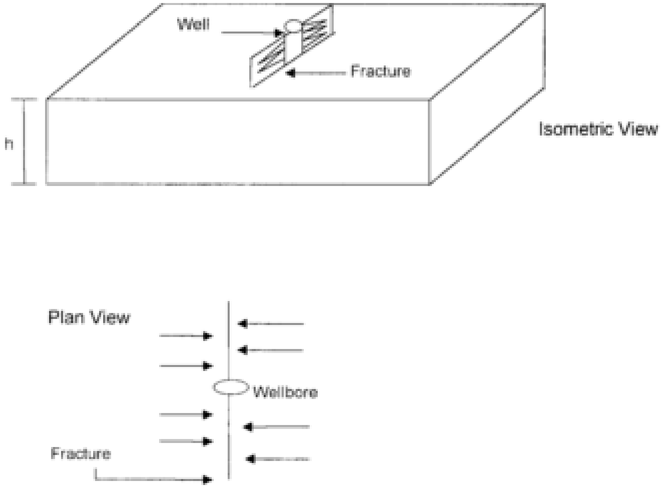
- Fluids can also be classified based on the number of phases present. It may be single or multiphase flow

Module 2-Classification of Reservoir flow systems

Classification based on geometry

- The two geometries of great practical interest are those that give rise to linear or radial flow. The spherical flow may be of interest occasionally
- Linear flow: In linear flow, the flow lines are parallel and the cross-section exposed to flow is constant



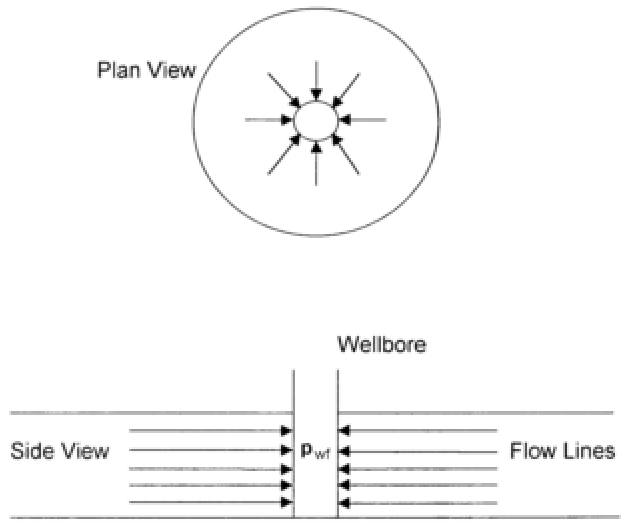


Linear Flow

Module 2-Classification of Reservoir flow systems

Classification based on geometry

- Radial flow: The flow lines are straight and converge in two dimensions toward a common center (i.e a well). The cross section exposed to flow decreases as the center is approached.



Radial Flow

Module 2-Classification of Reservoir flow systems

Classification based on geometry

- Spherical flow: The flow lines are straight and converge toward a common center in three dimensions

Spherical Flow

- The actual fluid path of fluid particles in rocks are irregular due to the shape of the pore spaces, the overall or average paths may be represented as straight lines in linear, radial or spherical flow

Module 2-Classification of Reservoir flow systems

Classification based flow regime

- Flow systems in reservoir rocks are classified according to their time dependence as steady-state, unsteady/transient, or pseudo steady state
- During the life of a well or reservoir, the system can change several times.
- It is therefore very critical to understand the flow system as much as possible in order to use the appropriate model to describe pressure and flow rate relationship

Module 2-Classification of Reservoir flow systems

Classification based flow regime- Steady state

- The flow regime is identified as a steady-state flow if the pressure at every location in the reservoir remains constant, i.e., does not change with time.
- This implies that the pressure and fluid saturations at every point through out the system adjust instantaneously to a change in pressure or flow rate in any part of the system
- Although no real system can respond instantaneously, some reservoir flow do exhibit this. This can happen for example when any production from reservoir is replaced with an equal mass of fluid from some external source

Module 2-Classification of Reservoir flow systems

Classification based flow regime- Steady state

- The external source could be a strong aquifer or pressure maintenance operations
- Mathematically, steady state flow can be describes:

$$\left(\frac{\partial P}{\partial t}\right)_i = 0 \qquad (5)$$

- The above equation states that the rate of change of pressure p with respect to time t at any location i is zero.

Module 2-Classification of Reservoir flow systems

Classification based flow regime- Unsteady state

- The unsteady-state flow (frequently called transient flow) is defined as the fluid flowing condition at which the rate of change of pressure with respect to time at any position in the reservoir is not zero or constant.
- This definition suggests that the pressure derivative with respect to time is essentially a function of both position i and time t, thus, mathematically, transient flow can be describe as:

$$\left(\frac{\partial P}{\partial t}\right)_i = f(i,t) \qquad (6)$$

Module 2-Classification of Reservoir flow systems

Classification based flow regime- Pseudo-steady state flow

- Here, the pressure at different locations in the reservoir is declining linearly as a function of time, i.e., at a constant declining rate
- Mathematically, this definition states that the rate of change of pressure with respect to time at every position is constant

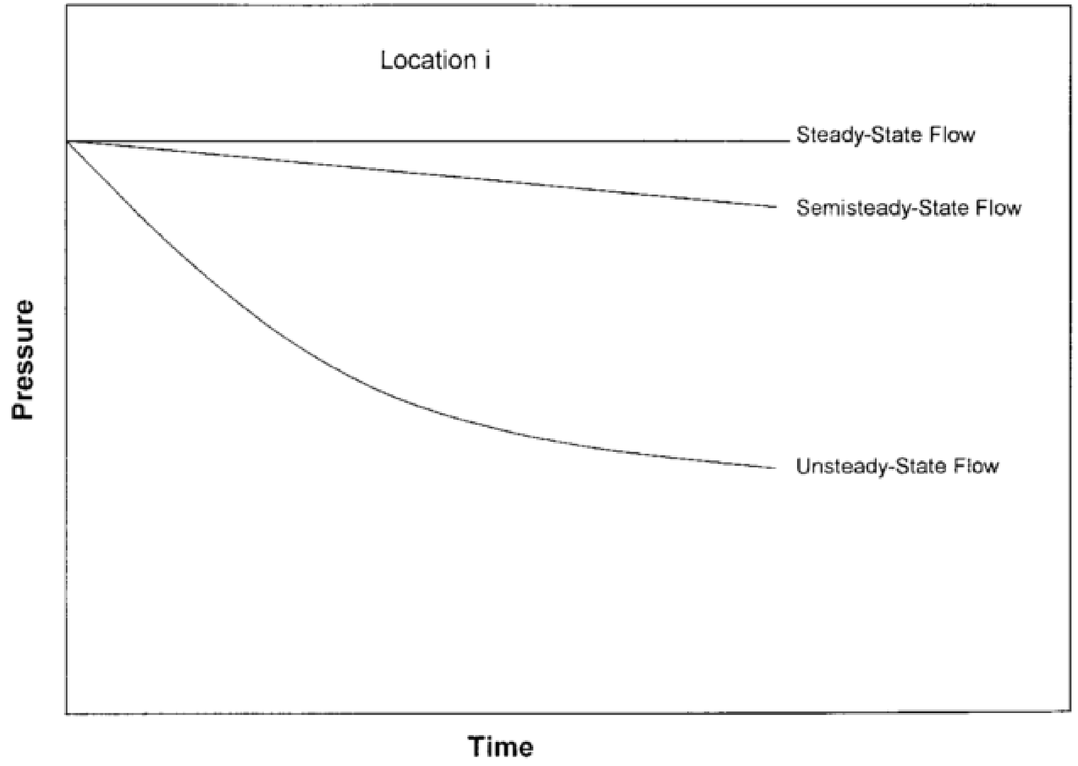
$$\left(\frac{\partial P}{\partial t}\right)_i = constant \qquad (7)$$

- It is also referred to as semisteady-state flow and quasisteady-state flow.

Module 2-Classification of Reservoir flow systems

Classification based flow regime- Pseudo-steady state flow

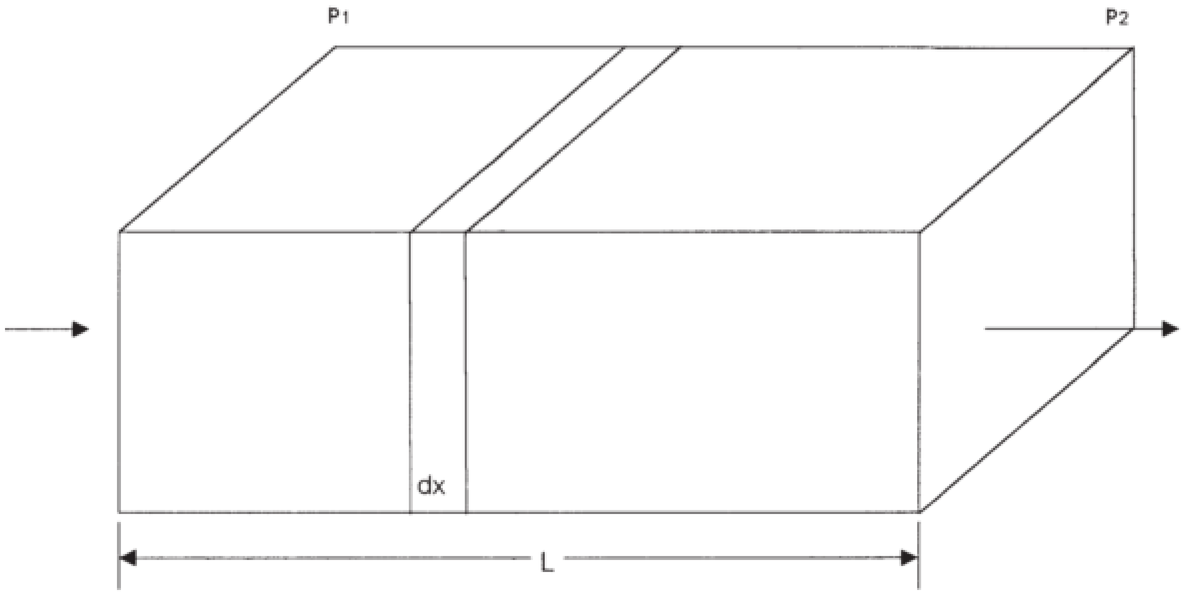
- Figure below shows a schematic comparison of the pressure declines as a function of time of the three flow regimes.



Module 3-Steady State Flow

Linear flow of Incompressible fluids

- In the linear system, it is assumed the flow occurs through a constant cross-sectional area A, where both ends are entirely open to flow. It is also assumed that no flow crosses the sides, top, or bottom as shown Figure below



Module 3-Steady State Flow

Linear flow of Incompressible fluids

- If an incompressible fluid is flowing across the element dx, then the fluid velocity v and the flow rate q are constants at all points.
- The flow behavior in this system can be expressed by the differential form of Darcy's equation, i.e., Equation(1)

$$V = \frac{q}{A} = -\frac{k}{\mu}\frac{dP}{dx}$$

- Separating the variables of Equation(1) and integrating over the length of the linear system gives

$$\frac{q}{A}\int_{0}^{L} dx = -\frac{k}{\mu}\int_{P_1}^{P_2} dP$$

Module 3-Steady State Flow

Linear flow of Incompressible fluids

This gives $q = \frac{kA(P_1 - P_2)}{\mu L}$

In customary field units this gives:

$$q = \frac{0.001127kA(P_1 - P_2)}{\mu L} \qquad (8)$$

q is the flow rate, bbl/day; k is the absolute permeability, md; p= pressure,psia; μ= viscosity,cp; L=distance,ft; A= cross sectional area,ft^2

Class exercise: Rewrite eqn (8) to get q in STB/day

Module 3-Steady State Flow

Linear flow of Incompressible fluids-example

An incompressible fluid flows in a linear porous media with the following properties:

L = 2000 ft h = 20′ width = 300′; k = 100 md Ø = 15% ;μ=2 cp;

P_1 = 2000 psi ; P_2 = 1990 psi

Calculate:

a. Flow rate in bbl/day

b. Apparent fluid velocity in ft/day

c. Actual fluid velocity in ft/day

d. Flow rate in STB/day if B is given as 1.127bbl/STB

Module 3-Steady State Flow

Linear flow of Incompressible fluids-example solution

a. Flow rate in bbl/day(calculate A and obtain answer from eqn(8)

b. Apparent fluid velocity in ft/day (q/A but conversion factor must be introduced therefore, q*5.615/A is used)

c. Actual fluid velocity in ft/day (q/ ØA; q*5.615/ Ø A

d. Flow rate in STB/day if B is given as 1.127bbl/STB (refer to class exercise

Module 3-Steady State Flow

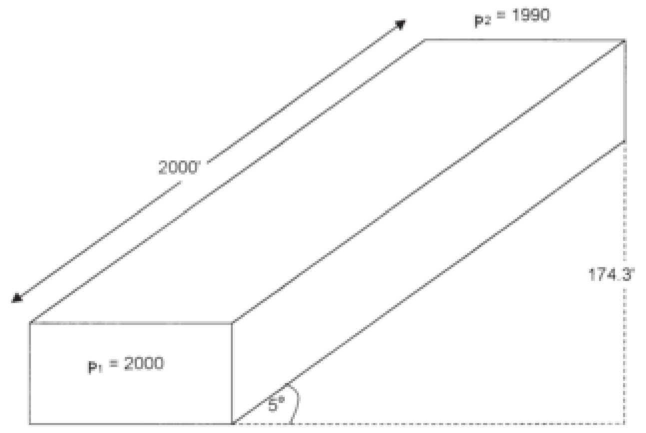
Linear flow of Incompressible fluids

- As previously discussed (see eqn (2)),the difference in the pressure ($p1-p2$) in Eqn(8) is not the only driving force in a tilted reservoir.
- The gravitational force is the other important driving force that must be accounted for to determine the direction and rate of flow.
- The fluid gradient force (gravitational force) is always directed vertically downward while the force that results from an applied pressure drop may be in any direction.

Module 3-Steady State Flow

Linear flow of Incompressible fluids-Assignment

- Assume that the porous media with the properties as given in the previous example is tilted with a dip angle of 5° as shown in Figure below. The incompressible fluid has a density of 42 lb/ft3. Resolve the previous example using this additional information.



Module 3-Steady State Flow

Linear flow of slightly compressible fluids

- Eqn (4) $V = V_R[1 + c(P_R - P)]$describes the relationship that exists between pressure and volume for slightly compressible fluid and can be written in terms of flow rate

$$q = q_R[1 + c(P_R - P)] \qquad (9)$$

- where q_R is the flow rate at some reference pressure P_R Substituting the above relationship in Darcy's equation gives:

$$\frac{q}{A} = \frac{q_R[1+c(P_R-P)]}{A} = -0.001127\frac{k}{\mu}\frac{dP}{dx} \qquad (10)$$

Module 3-Steady State Flow

Linear flow of slightly compressible fluids

- Separating variables and integrating, eqn (10) becomes

equation gives:

$$\frac{q_R}{A}\int_0^L dx = -0.001127\frac{k}{\mu}\int_{P_1}^{P_2}\frac{dP}{1+c(P_R-P)}$$

$$q_R = \frac{0.001127kA}{\mu Lc} ln\left[\frac{1+c(P_R-P_2)}{1+c(P_R-P_1)}\right] \qquad (11)$$

where q_R= flow rate at a reference pressure P_R, bbl/day

Module 3-Steady State Flow

Linear flow of slightly compressible fluids

P_1= upstream pressure, psi; P_2= downstream pressure, psi

k= permeability, md; μ=viscosity, cp; c= average liquid compressibility,psi^{-1}

- Selecting the upstream pressure P_1 as the reference pressure P_R and substituting in Eqn(11) gives the flow rate at Point 1 as:

$$q_1 = \frac{0.001127kA}{\mu Lc} ln\left[\frac{1 + c(P_1 - P_2)}{1}\right] \qquad (12)$$

Linear flow of slightly compressible fluids

- Selecting the downstream pressure P_2 as the reference pressure P_R and substituting in Eqn(11) gives the flow rate at Point 2 as:

$$q_2 = \frac{0.001127kA}{\mu Lc} ln\left[\frac{1}{1 + c(P_2 - P_1)}\right] \qquad (13)$$

where q_1 and q_2 are the flow rates at points 1 and 2, respectively.

Module 3-Steady State Flow

Linear flow of slightly compressible fluids-Example

- Consider the linear system given in previous example and, assuming a slightly compressible liquid, calculate the flow rate at both ends of the linear system. The liquid has an average compressibility of $21 \times 10^{-5}\text{psi}^{-1}$

• Choosing the upstream pressure as the reference pressure gives:

$$q_1 = \left[\frac{(0.001127)(100)(6000)}{(2)(21 \times 10^{-5})(2000)}\right] \ln[1+(21\times10^{-5})(2000-1990)]$$
$$=1.689 \text{ bbl/day}$$

• Choosing the downstream pressure, gives:

$$q_2 = \left[\frac{(0.001127)(100)(6000)}{(2)(21 \times 10^{-5})(2000)}\right] \ln\left[\frac{1}{1+(21 \times 10^{-5})(1990-2000)}\right]$$
$$=1.692 \text{ bbl/day}$$

Module 3-Steady State Flow

Linear flow of slightly compressible fluids-Example

- From the above exercise, it is shown that the values of q_1 and q_2 are not largely different
- This is due to the fact that the liquid is slightly incompressible and its volume is not a strong function of pressure.

Module 3-Steady State Flow

Linear flow of compressible fluids

- The rate of flow of gas expressed in standard cubic feet per day is the same at all cross sections in a steady state linear system.
- But the gas expands as the pressure drops, the velocity is greater at the downstream end than at the upstream end and consequently the pressure gradient increases towards the downstream end.
- The flow at any cross section X of figure shown on slide 38, where the pressure is P may be expressed in terms of the flow in standard cubic feet per day given as eqn.(13)

Module 3-Steady State Flow

Linear flow of compressible fluids

$$\left(\frac{P_{sc}}{T_{sc}}\right)\left(\frac{ZT}{P}\right)\left(\frac{Q_{sc}}{5.615}\right) = q \qquad (13)$$

- Where q= gas flow rate at pressure p, bbl/day; Q_{sc}=gas flow rate at standard conditions, scf/day; z=gas compressibility factor; T_{sc}, P_{sc}= standard temperature and pressure in ^{o}R and psia respectively
- Subsituting q into the Darcy's law, we have

$$\frac{q}{A} = \frac{Q_{sc}ZTP_{sc}}{5.615T_{sc}PA} = -0.001127\frac{k}{\mu}\frac{dP}{dx} \qquad (14)$$

Module 3-Steady State Flow

Linear flow of compressible fluids

- Separating variables and integrating; we have:

$$\frac{Q_{sc} Z T P_{sc} \mu}{5.615 T_{sc} A k * 0.001127} \int_{0}^{L} dx = -\int_{P_1}^{P_2} P dP = \frac{1}{2}\left(P_1^2 - P_2^2\right)$$

$$Q_{sc} = \frac{0.003164 T_{sc} A k \left(P_1^2 - P_2^2\right)}{Z T P_{sc} \mu L} \qquad (15)$$

We assumed constant z and μ over the specified pressures, i.e., P_1 and P_2 while integrating the eqn. to obtain eqn.(15)

Linear flow of compressible fluids

Where Q_{sc}=gas flow rate at standard conditions, scf/day; z=gas compressibility factor; T_{sc}, P_{sc}= standard temperature and pressure in °R and psia respectively; T= temperature, °R; k= permeability, md; μ= gas viscosity, cp;

A= cross-sectional area, ft²; L= total length of the linear system, ft

- Setting P_{sc}=14.7 psi and T_{sc}= 520 °R, eqn.(15) becomes:

$$Q_{sc} = \frac{0.111924Ak\left(P_1^2 - P_2^2\right)}{ZT\mu L} \qquad (16)$$

Module 3-Steady State Flow

Linear flow of compressible fluids

- But z and μ are a very strongly dependent on pressure, and we removed them from the from the integral to simplify the final form of the gas flow equation.
- Eqn.(16) is valid for applications when the pressure < 2000 psi.
- The gas properties must be evaluated at the average pressure $\bar{P}$ as defined in eqn.(17). This is reffered to as pressure-squared method

$$\bar{P} = \sqrt{\frac{P_1^2 + P_2^2}{2}} \qquad (17)$$

Module 3-Steady State Flow

Linear flow of compressible fluids-Example

A linear porous media is flowing a 0.72 specific gravity gas at 600°R. The upstream and downstream pressures are 2100 psi and 1894.73 psi, respectively. The cross-sectional area is constant at 4500 ft^2. The total length is 2500 feet with an absolute permeability of 60 md. Calculate the gas flow rate in scf/day (psc= 14.7 psia, Tsc= 520°R, z = 0.78, μ= 0.0173 cp).

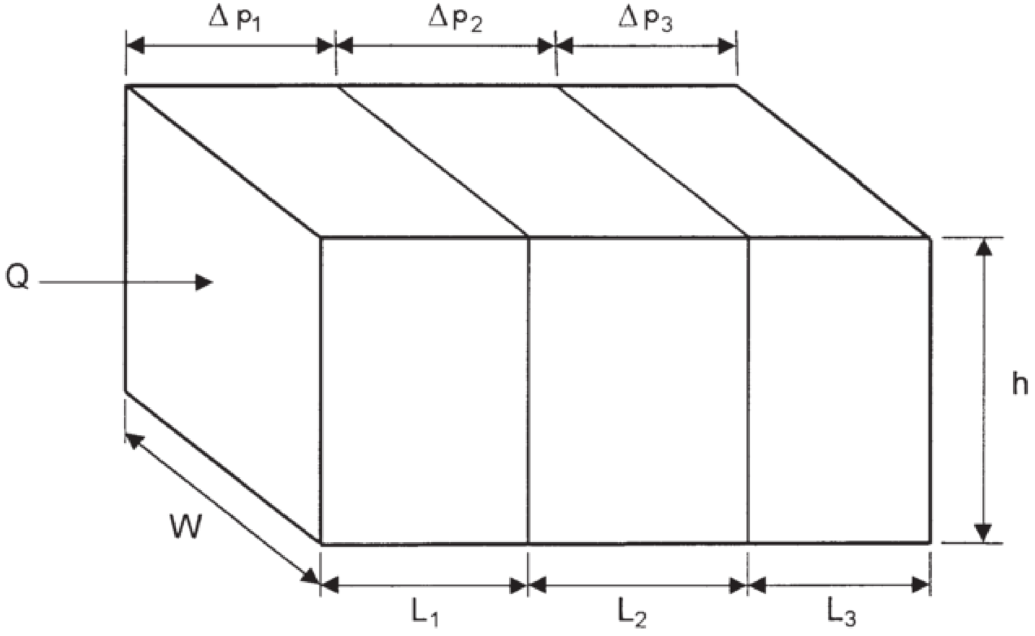
$$Q_{sc} = \frac{0.111924 Ak\left(P_1^2 - P_2^2\right)}{ZT\mu L} = \frac{0.111924 * 4500 * 60\left(2100^2 - 1894.73^2\right)}{0.78 * 600 * 0.0173 * 2500}$$

Q_{sc} =???????? Scf/day

Module 3-Steady State Flow

Fluid Flow through Linear beds in series

- Permeability variations can occur laterally in a reservoir as well as in the vicinity of a well bore.
- Consider the Figure below, which illustrates fluid flow through a series combination of beds with different permeabilities.



Module 3-Steady State Flow

Fluid Flow through Linear beds in series

- For a steady-state flow, the flow rate is constant and the total pressure drop Δp is equal to the sum of the pressure drops across each bed and mathematically described as:

$$\Delta P = \Delta P_1 + \Delta P_2 + \Delta P_3$$

- Substituting for the pressure drop in Darcy's equation

$$\frac{q_t \mu L_t}{0.001127 k_{av} A} = \frac{q_1 \mu L_1}{0.001127 k_1 A_1} + \frac{q_2 \mu L_2}{0.001127 k_2 A_2} + \frac{q_3 \mu L_3}{0.001127 k_3 A_3} \qquad (18)$$

Module 3-Steady State Flow

Fluid Flow through Linear beds in series

- But the flow rates, the cross sections, the viscosities are equal in all beds, eqn.(18) becomes eqn.(19) when rearranged:

$$\frac{L_t}{k_{av}} = \frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{L_3}{k_3}$$

$$k_{av} = \frac{L_t}{\frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{L_3}{k_3}} = \frac{\sum L_i}{\sum L_i/k_i} \qquad (19)$$

Module 3-Steady State Flow

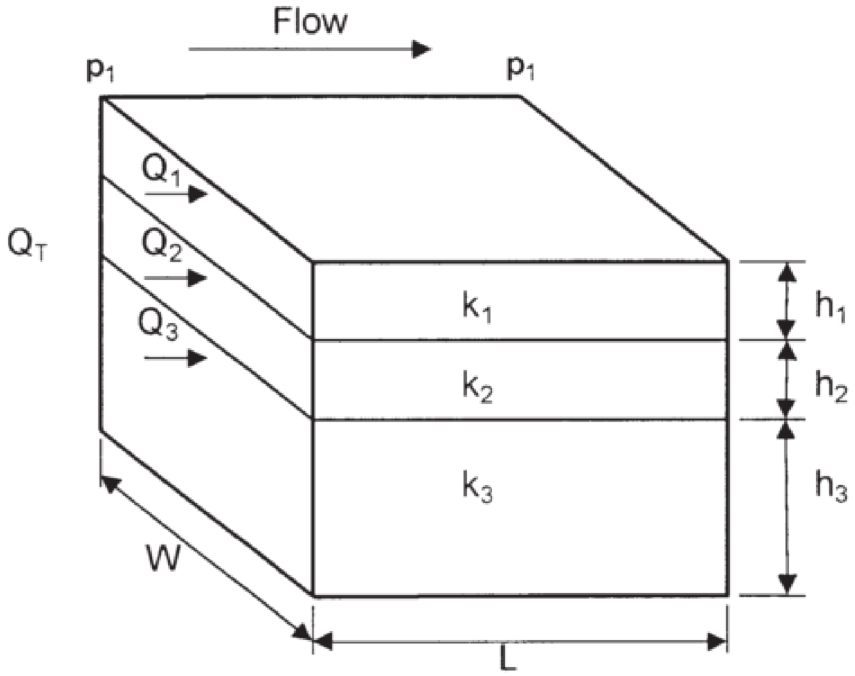
Fluid Flow through Linear beds in series

- The average permeability as defines by eqn.(19) is that permeability to which a number of beds of various geometries and permeabilities could be changed and yet the same total flow rate under the same applied pressure drop could be obtained
- Eqn.(19) was derived using incompressible fluid equation. Since the permeability is a property of the rock and not the fluid flowing through it, this eqn is also valid for compressible flow except for gases at low pressure

Module 3-Steady State Flow

Fluid Flow through Linear beds in parallel

- Consider the case where the flow system is comprised of three parallel layers that are separated from one another by thin impermeable barriers, i.e., no cross flow,as shown in the Figure below.



Module 3-Steady State Flow

Fluid Flow through Linear beds in parallel

- All the layers have the same width w with a cross-sectional area of A.
- The flow from each layer can be calculated by applying Darcy's equation in a linear form

$$q_1 = \frac{k_1 w h_1 \Delta P}{\mu L} \qquad for\ layer\ 1$$

$$q_2 = \frac{k_2 w h_2 \Delta P}{\mu L} \qquad for\ layer\ 2$$

$$q_3 = \frac{k_3 w h_3 \Delta P}{\mu L} \qquad for\ layer\ 3$$

Module 3-Steady State Flow

Fluid Flow through Linear beds in parallel

- The total flow rate from the entire system is expressed as:

$$q_t = \frac{k_{av} w h_t \Delta P}{\mu L} \qquad (20)$$

where q_t= total flow rate

k_{av}= average permeability for the entire model

w= width of the formation

Δp= P_1 - P_2

h_t= total thickness cross-sectional area of A.

Module 3-Steady State Flow

Fluid Flow through Linear beds in parallel

- The total flow rate qt is equal to the sum of the flow rates through each layer , that is:

$$q_t = q_1 + q_2 + q_3$$

Thus we have:

$$q_t = \frac{k_{av} w h_t \Delta P}{\mu L} = \frac{k_1 w h_1 \Delta P}{\mu L} + \frac{k_2 w h_2 \Delta P}{\mu L} + \frac{k_3 w h_3 \Delta P}{\mu L} \qquad (21)$$

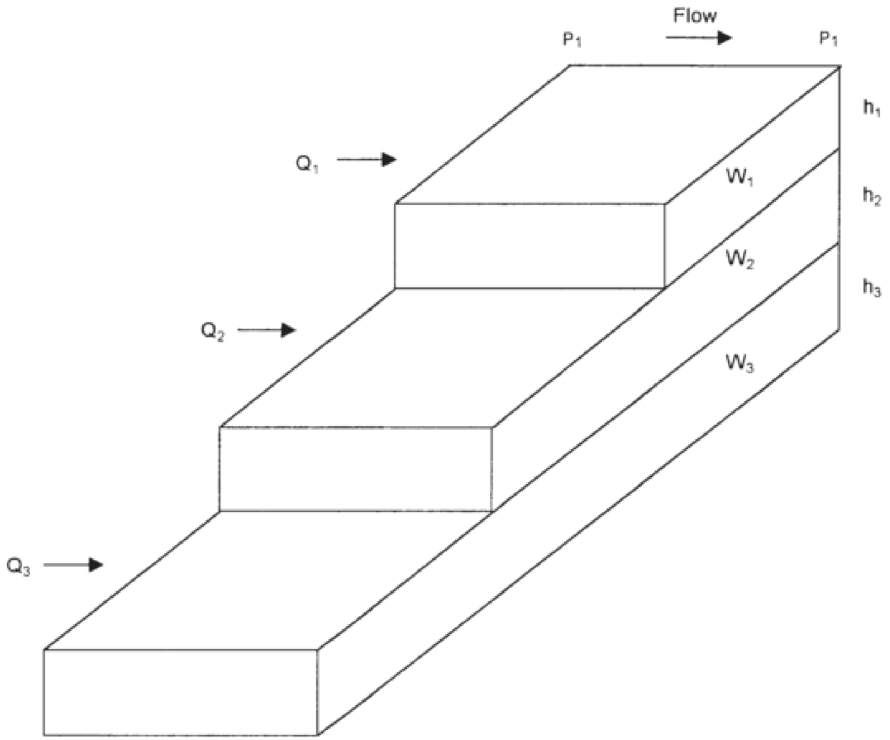
- Rearranging eqn.(21) gives the average permeability as:

$$k_{av} = \frac{k_1 h_1 + k_2 h_2 + k_3 h_3}{h_t} = \frac{\sum_{j=1}^{n} k_j h_j}{\sum_{j=1}^{n} h_j} \qquad (22)$$

Module 3-Steady State Flow

Fluid Flow through Linear beds in parallel

- Eqn.(22) is commonly used to determine the average permeability of a reservoir from core analysis data.
- The Figure below shows a similar layered system with variable layers width.



Module 3-Steady State Flow

Fluid Flow through Linear beds in parallel

- Assuming no cross-flow between the layers, the average permeability can be approximated in a manner similar to the above derivation to give:

$$k_{av} = \frac{\sum_{j=1}^{n} k_j A_j}{\sum_{j=1}^{n} A_j} \qquad (23)$$

with

A_j= $h_j w_j$; where A_j= cross-sectional area of layer j

w_j= width of layer j

Module 3-Steady State Flow

Fluid Flow through Linear beds in parallel-Example

- Given the following permeability data from a core analysis report of three parallel beds 10md,50md, and 1000md and 6 ft, 18ft and 36ft respectively in thickness but equal width, calculate the average permeability of the reservoir.
- Solution can be obtained using eqn.(22).

$$k_{av} = \frac{\sum_{j=1}^{n} k_j h_j}{\sum_{j=1}^{n} h_j} = \frac{10 * 6 + 18 * 50 + 36 * 1000}{6 + 18 + 36} = ???md$$

Module 3-Steady State Flow

Poiseuille's Law of Capillary Flow & Flow through Fractures

- Although the pore spaces within rocks seldom resemble straight, smooth-walled capillary tubes of constant diameter, the pore spaces can be treated as if they were composed of bundles of parallel capillary tubes of various diameter.
- Consider a capillary tube of length L and inside radius r_o which is flowing an incompressible fluid of μ viscosity in laminar or viscous flow under a pressure difference of $P_1 - P_2$
- From fluid dynamics, Poiseuille's law which describes the total flow rate through the capillary is given as:

Module 3-Steady State Flow

Poiseuille's Law of Capillary Flow & Flow through Fractures

$$q = 1.30(10)^{10}\frac{\pi r_o^4(P_1 - P_2)}{\mu L} \qquad (24)$$

- Darcy's law for the linear flow of incompressible fluids in permeable beds, eqn.(8) and the Poiseuille's law for incompressible fluid capillary flow, eqn.(24) are quite similar.

$$q = \frac{0.001127kA(P_1 - P_2)}{\mu L} \qquad (8)$$

- Writing $A = \pi r_o^2$ for area in eqn.(8) and equating it to eqn.(24), it can be shown that $\mathrm{k} = 1.15(10)^{13}r_o^2$

Module 3-Steady State Flow

Poiseuille's Law of Capillary Flow & Flow through Fractures

- Thus the permeability of a rock composed of closely packed capillaries, each having a radius of $4.17(10)^{-6}$ foot (0.00005 in.) is about 200md.
- And if only 25% of the rock consist of pore channels (i.e it has 25% porosity), the permeability is about one-quarter as large or about 50md.
- An equation for viscous flow of incompressible wetting fluids through smooth fractures of constant width may be obtained as:

$$q = 8.7(10)^9 \frac{W^2 A(P_1 - P_2)}{\mu L} \qquad (25)$$

Module 3-Steady State Flow

Poiseuille's Law of Capillary Flow & Flow through Fractures

- W is the width of the fracture, A is the cross sectional area of the fracture, which is equal to the product of the width W and the lateral extent of the fracture, and the pressure difference is that which exists between the ends of fracture of length L.
- Eqn.(25) may be combined with eqn.(8) to obtain an expression for the permeability of a fracture as:

$$\mathrm{k} = 7.7(10)^{12} W^2$$

Module 4-Radial Steady State Flow

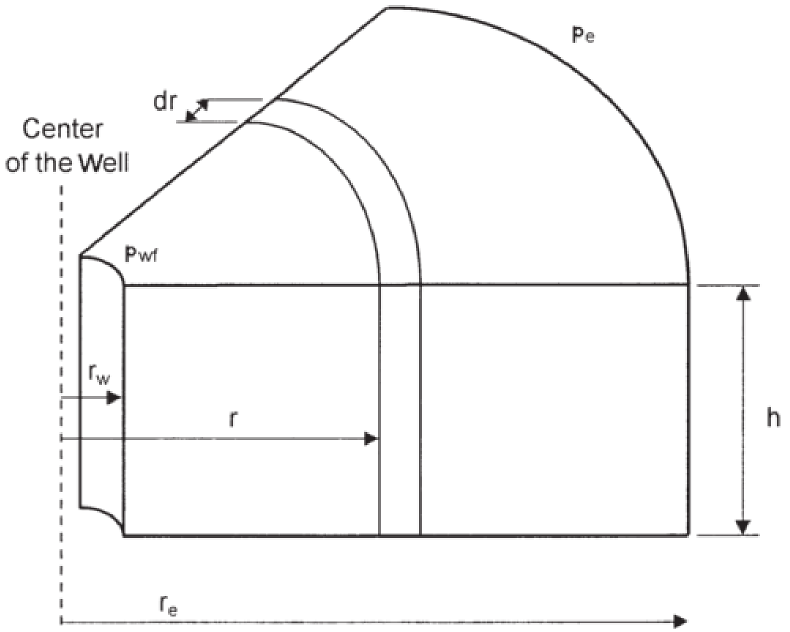
Radial flow of Incompressible fluids

- In a radial flow system, all fluids move toward the producing well from all directions.
- Pressure differential must exist for flow to occur.
- Thus, if a well is to produce oil, which implies a flow of fluids through the formation to the wellbore, the pressure in the formation at the wellbore must be less than the pressure in the formation at some distance from the well.
- The pressure in the formation at the wellbore of a producing well is know as the bottom-hole flowing pressure (flowing BHP, P_{wf}).

Module 4-Radial Steady State Flow

Radial flow of Incompressible fluids

- Consider the Figure below, which schematically illustrates the radial flow of an incompressible fluid toward a vertical well.



- The formation is considered to be of a uniform thickness h and a constant permeability k.

Module 4-Radial Steady State Flow

Radial flow of Incompressible fluids

- Let P_{wf} represent the maintained bottom-hole flowing pressure at the wellbore radius r_w and P_e denote the external pressure at the external or drainage radius.
- Darcy's equation can be used to determine the flow rate at any radius r:

$$v = \frac{qB}{A} = \frac{qB}{2\pi rh} = -0.001127\frac{kdP}{\mu dr} \qquad (26)$$

Where positive q is in the positive r direction. Separating variables and integrating between any radii, r_1 and r_2 where the pressures are P_1 and P_2 respectively. Eqn.(26) becomes:

Module 4-Radial Steady State Flow

Radial flow of Incompressible fluids

$$\int_{r_1}^{r_2} \frac{qBdr}{2\pi rh} = -0.001127 \int_{P_1}^{P_2} \frac{kdP}{\mu}$$

$$q = -\frac{0.00708kh(P_2 - P_1)}{\mu B ln\left(\frac{r_2}{r_1}\right)} \qquad (27)$$

- The minus sign is usually neglected for where P_2 is greater than P_1, the flow is known to be negative –that is, in the negative r-direction, or towards the wellbore. Eqn.(27) becomes:

$$q = \frac{0.00708kh(P_2 - P_1)}{\mu B ln\left(\frac{r_2}{r_1}\right)} \qquad (28)$$

Module 4-Radial Steady State Flow

Radial flow of Incompressible fluids

- Frequently the two radii of interest are the wellbore radius r_w and the external or drainage radius r_e. Eqn.(28) then becomes:

$$q = \frac{0.00708kh(P_e - P_{wf})}{\mu B ln\left(\frac{r_e}{r_w}\right)} \qquad (29)$$

- where q= flow rate, STB/day; P_e= external pressure, psi; P_{wf}= bottom-hole flowing pressure, psi; k= permeability, md; μ= oil viscosity, cp; B= oil formation volume factor, bbl/STB; h= thickness, ft; r_e= external or drainage radius, ft; r_w= wellbore radius, ft

Module 4-Radial Steady State Flow

Radial flow of Incompressible fluids

- The external (drainage) radius re is usually determined from the well spacing by equating the area of the well spacing with that of a circle, i.e.

$$\pi r_e^2 = 43560\text{A}.$$

$$r_e = \sqrt{\frac{43560A}{\pi}} \qquad (30)$$

where A is the well spacing in acres.

- In practice, neither the external radius nor the wellbore radius is generally known with precision. Fortunately, they enter the equation as a logarithm, so that the error in the equation will be less than the errors in the radii

Module 4-Radial Steady State Flow

Radial flow of Incompressible fluids

- Eqn.(27)can be arranged to solve for the pressure p at any radius r to give:

$$P = P_{wf} + \left[\frac{q\mu B}{0.00708kh}\right] ln\left(\frac{r}{r_w}\right) \qquad (31)$$

Exercise:

An oil well in the Niger Delta Field is producing at a stabilized rate of 600 STB/day at a stabilized bottom-hole flowing pressure of 1800 psi. Analysis of the pressure buildup test data indicates that the pay zone is characterized by a permeability of 120 md and a uniform thickness of 25ft. The well drains an area of approximately 40 acres. The following additional data is available: r_w= 0.25 ft ;A= 40 acres; B= 1.25 bbl/STB; μ= 2.5 cp. Calculate the P_e

Module 4-Radial Steady State Flow

Radial flow of slightly compressible fluids

- Eqn.(4) $V = V_R[1 + c(P_R - P)]$ is again used to express the volume dependence on pressure for slightly compressible fluids
- Substituting this equation into the radial form of Darcy's law, we have:

$$qB = \frac{q_R[1 + c(P_R - P)]}{2\pi rh} = -0.001127\frac{kdP}{\mu dr} \qquad (32)$$

- Separating variables, assuming a constant compressibility over the entire pressure drop, and integrating over the length of the porous medium we have:

$$q_R = \frac{0.00708kh}{\mu c ln\left(\frac{r_e}{r_w}\right)} ln\left[\frac{1 + c(P_R - P_2)}{1 + c(P_R - P_1)}\right] \qquad (33)$$

Module 4-Radial Steady State Flow

Radial flow of slightly compressible fluids

- where q_R is flow rate at a reference pressure P_R.
- Choosing the bottom-hole flow pressure P_{wf} as the reference pressure and expressing the flow rate in STB/day gives:

$$q = \frac{0.00708kh}{\mu Bcln\left(\frac{r_e}{r_w}\right)} ln\left[1 + c\left(P_{wf} - P_2\right)\right] \qquad (34)$$

where c = isothermal compressibility coefficient, psi^{-1}

q = oil flow rate, STB/day; k = permeability, md

Module 4-Radial Steady State Flow

Radial flow of slightly compressible fluids-Example

- The following data are available on a well in the Agbami Field:

P_e = 2506 psi; P_{wf} = 1800; r_e = 745′ ; r_w = 0.25; B = 1.25 μ = 2.5 co = 25 × 10^{-6} psi^{-1}; k = 0.12 Darcy h = 25 ft.

- Using eqns.(34) and (29) respectively the answers are obtained where the reference pressure is the P_{wf}
- Comment on the results!

Module 4-Radial Steady State Flow

Radial flow of compressible fluids

- The flow of gas at any radius r of Figure shown on slide (73), where the pressure is P, may be expressed in terms of the flow in standard cubic feet per day:

$$qB = \frac{q_{sc} Z T P_{sc}}{5.615 T_{sc} P} \qquad (35)$$

- Substituting eqn.(32) into the radial form of Darcy's law, we have:

$$\frac{Q_{sc} Z T P_{sc}}{5.615 T_{sc} P(2\pi r h)} = -0.001127 \frac{k}{\mu}\frac{dP}{dr} \qquad (36)$$

- Separating variables and integrating (36), we have:

$$Q_{sc} = \frac{0.01988 T_{sc} k h (P_1^2 - P_2^2)}{Z \mu T P_{sc} ln\left(\frac{r_2}{r_1}\right)} \qquad (37)$$

Module 4-Radial Steady State Flow

Radial flow of compressible fluids

- The product $Z\mu$ has been assumed to be constant for the derivation of eqn.(37).
- It was previously pointed out that this is usually true only for pressure less than about 1500 to 2000 psia.
- For greater pressures it was stated that a better assumption was that the product $Z\mu$/P was constant . For this case, we have:

$$Q_{sc} = \frac{0.03976T_{sc}kh(P_1 - P_2)}{(Z\mu/P)TP_{sc}ln\left(\frac{r_2}{r_1}\right)} \qquad (38)$$

- In applying eqns.(37)&(38), the products $Z\mu$ and $Z\mu/P$ should be estimated at the average pressure between $P_1 and\ P_2$. For example eqn.(17) can be used or other methods to be discussed later

Module 4-Radial Steady State Flow

Permeability Variations in Radial parallel flow

- Many producing formations are composed of strata that may vary widely in permeability and thickness
- If these strata are producing fluid into a common wellbore under the same drawdown and from the same drainage radius, the average permeability can be given as:

$$k_{av} = \frac{k_1 h_1 + k_2 h_2 + k_3 h_3 \dots}{h_t} = \frac{\sum_{j=1}^{n} k_j h_j}{\sum_{j=1}^{n} h_j} \qquad (39)$$

- Eqn.(39) is the same eqn. obtained for parallel flow in linear beds with the same bed width.
- Here again, average permeability refers to that permeability by which all beds could be replaced and still obtain the same production rate under the same drawdown.

Module 4-Radial Steady State Flow

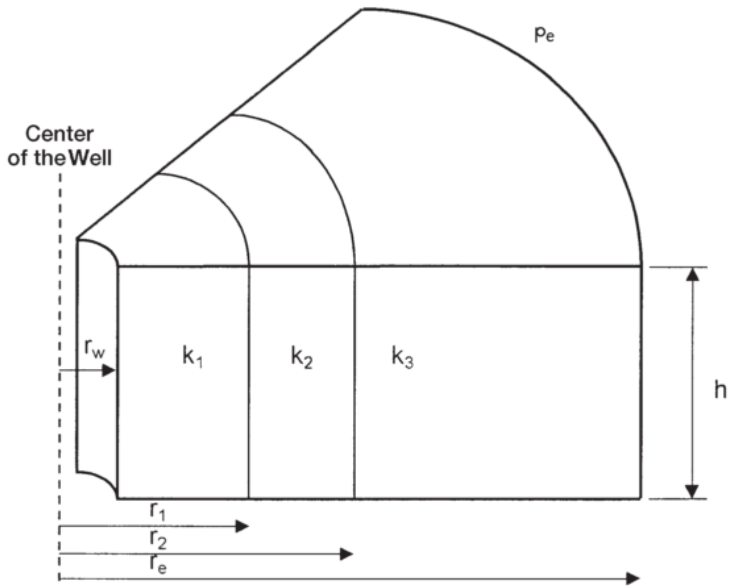
Permeability Variations in Radial flow through parallel bed

- The product kh is called the capacity of a bed or stratum and the total capacity of the formation, $\sum_{j=1}^{n} k_j h_j$, is usually expressed in millidarcy-feet.
- Because the rate of flow is directly proportional to the capacity, eqn.(29), a 10ft bed of 100md will have the same production rate as a 100ft bed of 10md permeability, other things being equal.
- There are limits of formation capacity below which production rates are not economic
- Of two formations with the same capacity and similar drawdown, the one with the lower oil viscosity will be economic but not the other

Module 4-Radial Steady State Flow

Permeability Variations in Radial flow through series beds

- In the radial system shown in Figure below, the previous averaging methodology used for the linear bed in series can be applied to produce the following generalized expression.



- Assignment: Derive the expression for the average permeability in radial flow through bed in series.

Module 5-Unsteady Radial Flow

Differential equation for radial flow system

- In this section, we will develop fundamental equations for fluid flow through porous media.
- Direct applications occur in well testing, simulation, and production decline analysis, and also form the basis for the development of secondary recovery processes.
- A mathematical description of fluid flow in a porous medium can be obtained by combining the following physical principles:

❖ The law of conservation of mass

❖ A momentum equation, e.g., Darcy's law

❖ An equation of state.

Module 5-Unsteady Radial Flow

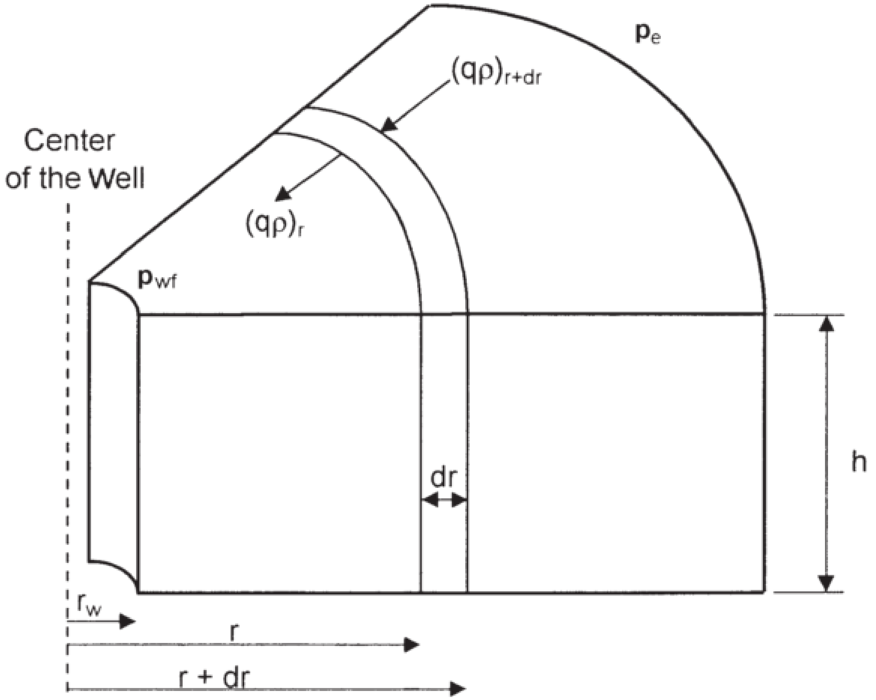
Differential equation for radial flow system-Law of conservation

- In flow of any type the conservation principle is simply a statement that some physical quantity is conserved, i.e.,neither created nor destroyed.
- In fluid flow in a porous medium, the most significant quantity conserved is mass
- Consider the volume element in the figure below. The volume element is in a reservoir of constant thickness and constant properties.
- For q=volume of flow STB/day incompressible and slightly compressible fluids and SCF/day for compressible fluids; ρ is the density of flowing fluid at reservoir condition, Lb/ft^3;

Module 5-Unsteady Radial Flow

Differential equation for radial flow system-Law of conservation

- r=distance from wellbore,ft;h is formation thickness,ft; v=velocity of flowing fluid, bbl/day-ft^2 ; μ is the fluid viscosity,cp; k is the permeability,md and Ø=porosity,fraction



Module 5-Unsteady Radial Flow

Differential equation for radial flow system-Law of conservation

- The law of conservation states that:

$$\begin{pmatrix} Mass\ entering \\ volume\ element \\ during\ interval\ \Delta t \end{pmatrix} - \begin{pmatrix} mass\ leaving \\ volume\ element \\ during\ interval\ \Delta t \end{pmatrix} = \begin{pmatrix} Rate\ at\ which \\ mass\ accumulates \\ during\ interval\ \Delta t \end{pmatrix} \quad (40)$$

- The mass entering the volume element during interval Δt is given by:

$$(qB\rho)_{r+\Delta r} = 2\pi(r+\Delta r)h\big(\rho v(5.615/24)\big)_{r+\Delta r} \qquad (41)$$

- The mass leaving the volume element during interval Δt is given by:

$$(qB\rho)_r = 2\pi rh\big(\rho v(5.615/24)\big)_r \qquad (42)$$

Module 5-Unsteady Radial Flow

Differential equation for radial flow system-Law of conservation

- The rate at which mass accumulates during the interval Δt is given by:

$$\frac{2\pi r\Delta rh[(\emptyset\rho)_{t+\Delta t} - (\emptyset\rho)_t]}{\Delta t} \qquad (43)$$

- Substituting eqns.(41,42&43) into eqn.(40) and dividing by $2\pi r\Delta rh$, and taking the limit in each term as Δr and Δt approach zero, eqn.(44) is obtained.

$$\frac{0.234}{r}\frac{\partial(r\rho v)}{\partial r} = \frac{\partial(\emptyset\rho)}{\partial t} \qquad (44)$$

- Eqn.(44) is the continuity equation and is valid for any flow system of radial geometry

Module 5-Unsteady Radial Flow

Differential equation for radial flow system-Momentum Equation

- Darcy's law expresses the fact that the volumetric rate of flow per unit cross-sectional area (v) at any point in a uniform porous medium is proportional to the pressure gradient and inversely proportional to the fluid viscosity
- This law is used to relate the fluid flow to reservoir pressure. Therefore substituting Darcy's law into eqn.(44) gives eqn.(45):

$$\frac{0.234}{r}\frac{\partial}{\partial r}\left(0.001127\frac{k}{\mu}r\rho\frac{\partial P}{\partial r}\right)=\frac{\partial(\emptyset\rho)}{\partial t} \qquad (45)$$

- The porosity from the partial derivative on the RHS of eqn.(45) is eliminated by expanding the RHS taking the indicative derivatives, we have:

Module 5-Unsteady Radial Flow

Differential equation for radial flow system-Momentum Equation

$$\frac{\partial(\emptyset\rho)}{\partial t} = \emptyset\frac{\partial\rho}{\partial t} + \rho\frac{\partial\emptyset}{\partial t} \qquad (46)$$

- But porosity is related to formation compressibility by eqn.(47):

$$c_f = \frac{1}{\emptyset}\frac{d\emptyset}{dP} \qquad (47)$$

- Applying chain rule of differentiation to $\frac{\partial\emptyset}{\partial t}$ we have:

$$\frac{\partial\emptyset}{\partial t} = \frac{d\emptyset}{dP}\frac{\partial P}{\partial t} \qquad (48)$$

- Substituting eqn.(47) into (48), we have:

$$\frac{\partial\emptyset}{\partial t} = \emptyset c_f \frac{\partial P}{\partial t} \qquad (49)$$

Module 5-Unsteady Radial Flow

General differential equation for radial flow system

- Finally substitute eqn.(49) into (46) and the result into (45), we have:

$$\frac{0.0002637}{r}\frac{\partial}{\partial r}\left(\frac{k}{\mu}r\rho\frac{\partial P}{\partial r}\right) = \rho\emptyset c_f\ \frac{\partial P}{\partial t} + \emptyset\frac{\partial \rho}{\partial t} \qquad (50)$$

- Eqn.(50) is the general partial differential equation used to describe the flow of any fluid flowing in a radial direction in porous media.
- In addition to the initial assumptions, Darcy's equation has been added, which implies that the flow is laminar. Otherwise, the equation is not restricted to any type of fluid and is equally valid for gases or liquids.
- Compressible and slightly compressible fluids, however, must be treated separately in order to develop practical equations that can be used to describe the flow behavior of these two fluids.

Module 5-Unsteady Radial Flow

Transient flow of compressible fluid in radial flow system

- To develop a solution to eqn.(50) for the compressible fluids, two additional equations are required.

❖ An equation of state, usually the real gas law, given by eqn.(51)

❖ Equation which describes the isothermal compressibility variation with pressure given by eqn. (4*) repeated here as eqn.(52)

$$PV = znRT \qquad (51)$$

$$c_g = \frac{1}{P} - \frac{1}{z}\frac{\partial z}{\partial P} \qquad (52)$$

- Combining eqns.(50,51 &52) we have:

Module 5-Unsteady Radial Flow

Transient flow of compressible fluid in radial flow system

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{P}{\mu z}\frac{\partial P}{\partial r}\right) = \frac{\emptyset c_t P}{0.0002637kz}\frac{\partial P}{\partial t} \qquad (53)$$

- c_t is the total compressibility when the reservoir contains more than one fluid, total compressibility is given as:

$$c_t = c_o S_o + c_w S_w + c_g S_g + c_f \qquad (54)$$

- Before developing solutions to the differential equations for flow through porous media, it should be noted that a differential equation describes only the physical law or laws which apply to a situation.

Module 5-Unsteady Radial Flow

Flow of compressible fluid in bounded drainage areas

- To obtain a solution to a specific flow problem, the appropriate differential equation must be solved with the boundary and initial conditions that characterize the particular situation of interest.
- Common conditions used in fluid flow are the initial and boundary conditions. The common initial condition is that the reservoir pressure P_i is same through out the reservoir at t=0 $P(r,0) = P_i$
- The cases of interest are:

❖ An infinite acting reservoir – the situation where a well is assumed to be situated in a porous medium of infinite extent

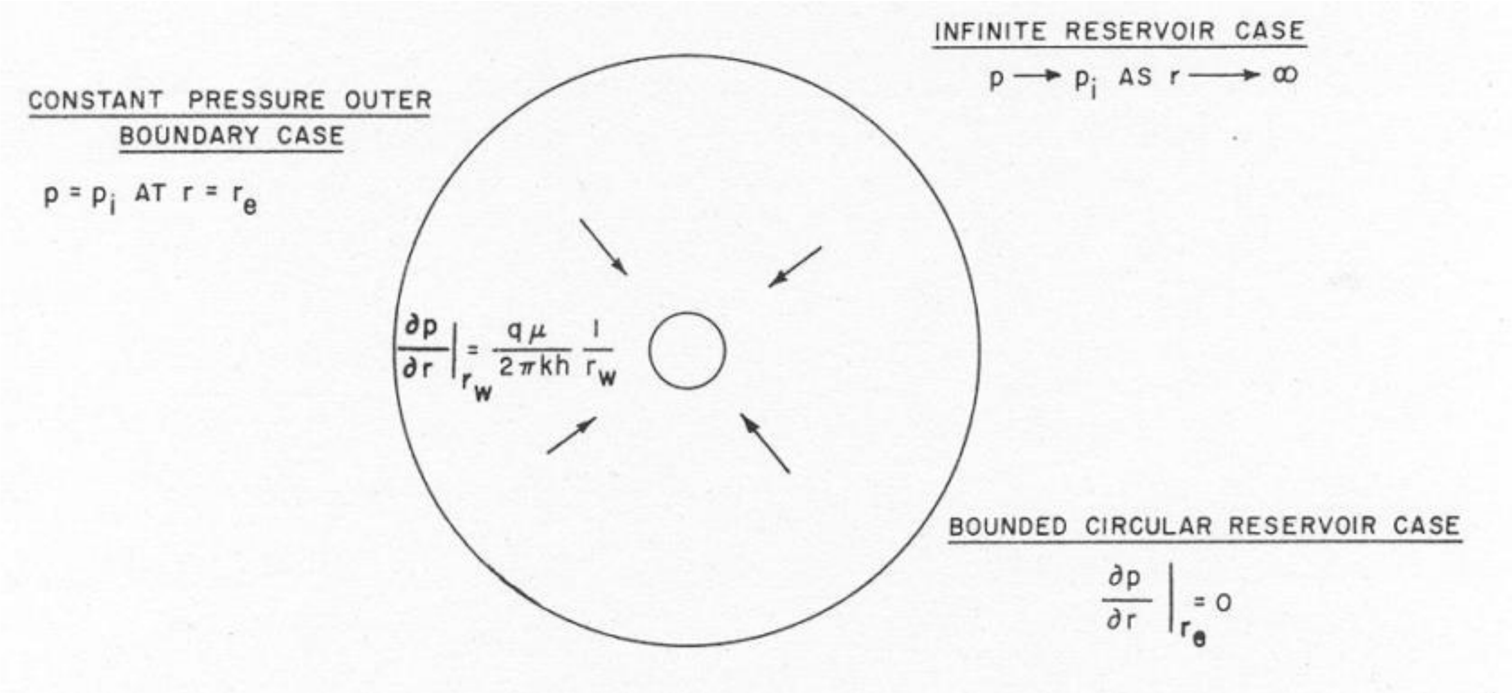
❖ Bounded reservoir – the case in which the well is assumed to be located in the center of reservoir with no flow across the external boundary which is of interest to us.

Module 5-Unsteady Radial Flow

Flow of compressible fluid in bounded drainage areas

- ❖ constant potential outer boundary - the case in which the well is assumed to be located in the center of an area with constant pressure along the external boundary.

- The Figure below is a description of these boundary conditions.



Module 5-Unsteady Radial Flow

Flow of compressible fluid in bounded drainage areas

- For Eqn.(53) has been derived on the assumption that the well was located in large reservoir. As such, the flow from or to the well would not be affected by boundaries that would inhibit the flow
- However, the time this assumption can be made is finite and often is very short in length. This is not the case in a bounded drainage areas.
- The flow feels the effect of the boundary, the flow is therefore no longer in the transient regime and the reservoir can no longer be considered infinite .
- Eqn.(53) therefore must be solved for bounded drainage areas with different assumptions which describes the pseudo steady-state flow regime

Module 5-Unsteady Radial Flow

Flow of compressible fluid in bounded drainage areas

- The initial condition remains the same as before (the reservoir pressure is P_i through out the reservoir at t=0 $\text{i.e}, P(r,0) = P_i$
- The new boundary condition used to find a solution to the radial diffusivity equation is that the outer boundary of the reservoir is a no-flow boundary i.e, $\frac{dy}{dx} = 0 \qquad at\, r = r_e$
- When these conditions are applied to eqn.(53) or its equivalent expressed in terms of Pseudo-pressure, the pseudo-steady state solution rearranged and solved for Q_{sc} is given as eqn.(54)

$$Q_{sc} = \frac{19.88(10)^{-6}T_{sc}kh}{TP_{sc}}\left[\frac{\left(m(\bar{P}) - m(P_{wf})\right)}{ln\left(\frac{r_e}{r_w}\right) - 0.75}\right] \qquad (54)$$

Module 5-Unsteady Radial Flow

Average Pressure in radial flow system

- Where m(p) is the real gas pseudo pressure with unit of $\frac{psia^2}{cp}$ in standard field units and is defined as:

$$m(p) = 2\int_{P_R}^{P} \frac{P}{\mu z} dP \qquad (55)$$

- The average pressure can be estimated using the volumetric average pressure $\bar{P}$. This is usually used to calculate fluid properties in material balance equation and is as defined in eqn.(56).

$$= \frac{\sum_{j=1}^{n} \bar{P}_j V_j}{\sum_{j=1}^{n} V_j} \qquad (56)$$

- Where $\bar{P}_j$ is the average pressure in jth drainage volume and V_j is the volume of the jth drainage volume.

Module 5-Unsteady Radial Flow

Readjustment time

- As previously discussed, when the flow rate is perturbed, there is pressure transient and the pressure wave moves from away from the well
- This movement is a diffusion phenomenon and is modelled by the diffusivity equation and the pressure moves at a rate proportional to the formation diffusivity η given as:

$$\eta = \frac{k}{\emptyset \mu c_t} \qquad (57)$$

Where k is the effective permeability of the flowing phase, Ø is the total effective porosity and μ is viscosity of the flowing phase and c_t is the total compressibility given as eqn.(54)

- When the pressure is travelling at this rate, the flow is transient

Module 5-Unsteady Radial Flow

Readjustment time

- While the flow is in transient regime, the outer boundary of the reservoir has no influence on the pressure movement and the reservoir acts as if it were infinite in size
- The late transient region is difficult to describe. It is the period after which the pressure has reached the outer boundary of the reservoir and before the pressure behavior has had time to stabilize in the reservoir. Here the pressure no longer travel at rate proportional to η
- The pseudo-steady state is the period after the pressure behavior has attained stability in the reservoir. As previously stated, during this period, the pressure at every point through out the reservoir is changing at constant rate and as linear function of time.
- The readjustment time can be defined as the time when a perturbed flow system reaches pseudo-steady state.

Module 5-Unsteady Radial Flow

Readjustment time

- The readjustment time expressed in hour can be obtained from eqn.(58).

$$t_{pss} = \frac{\emptyset \mu c_t A}{0.000264k} t_{DA} \qquad (58)$$

- Where A is the drainage area and t_{DA} has a characteristics value which depends on the drainasge shape. For a regular shape such as circle or square, it is equal to 0.1
- For a well in a 1X2 rectangle, t_{DA} is equal to 0.3 and for a 1X4 rectangle it is equal to 0.8
- Off centered wells in irregular patterns have larger values of t_{DA} ,implying that the well will feel the far-off boundaries after a significantly longer time

Module 5-Unsteady Radial Flow

Readjustment time

- If the drainage area can be approximated by a circle with equivalent drainage radius with t_{DA}=0.1, eqn.(58) becomes (59).

$$t_{pss} = \frac{1200 r_e^2}{\eta} = \frac{1200 \emptyset \mu c_t r_e^2}{k} \qquad (59)$$

Example

- Calculate the readjustment time for a well producing an oil with viscosity of 1.5cp and c_t of $15 * 10^{-6}\ psi^{-1}$, from a circular reservoir of 1000ft radius with k=100md and Ø=20%
- This means that approximately ….. Hours is required for the flow in this reservoir to reach pseudo-steady state condition after a well located in its center is opened to flow or when a change in the flow rate occurs.

Module 5-Unsteady Radial Flow

Readjustment time

- It also means that if the well is shut in, it will take approximately this time for the pressure to equalize through out the drainage area of the well, so that the measured subsurface pressure equals the average drainage area pressure of the well
- Class discussion on the readjustment time for gas reservoir:

❖ Looking at eqn.(59) and the last example, comment on the readjustment time for gas reservoir under the same condition.

❖ For gas wells drilled on wider spacings so that the value of reservoir radius is larger for gas wells than the oil well, comment on the readjustment time

Module 5-Unsteady Radial Flow

Assignment

- Using the real gas pseudo pressure m(p) , derive another form of the diffusivity equation for compressible fluid from the eqn.(53)
- From eqn.(50) derive the diffusivity equation for slightly compressible transient flow and Pseudo-steady flow

Module 6-Productivity index, zonal damage…

Productivity Index

- The ratio of the rate of production for liquid flow to the pressure drawdown is called productivity index usually represented by J.
- Mathematically productivity index is given as:

$$J = \frac{q}{\bar{P} - P_{wf}} \qquad (60)$$

- Productivity index is a measure of the well potential or the ability of the well to produce and is a commonly measured well property.
- PI can only be accurately obtained during the Pseudo-steady state regime. Why?
- Therefore to calculate productivity index from a production test, it is necessary to flow the well for a sufficiently long time to reach pseudo-steady state flow.

Module 6-Productivity index, zonal damage etc.

Productivity Index

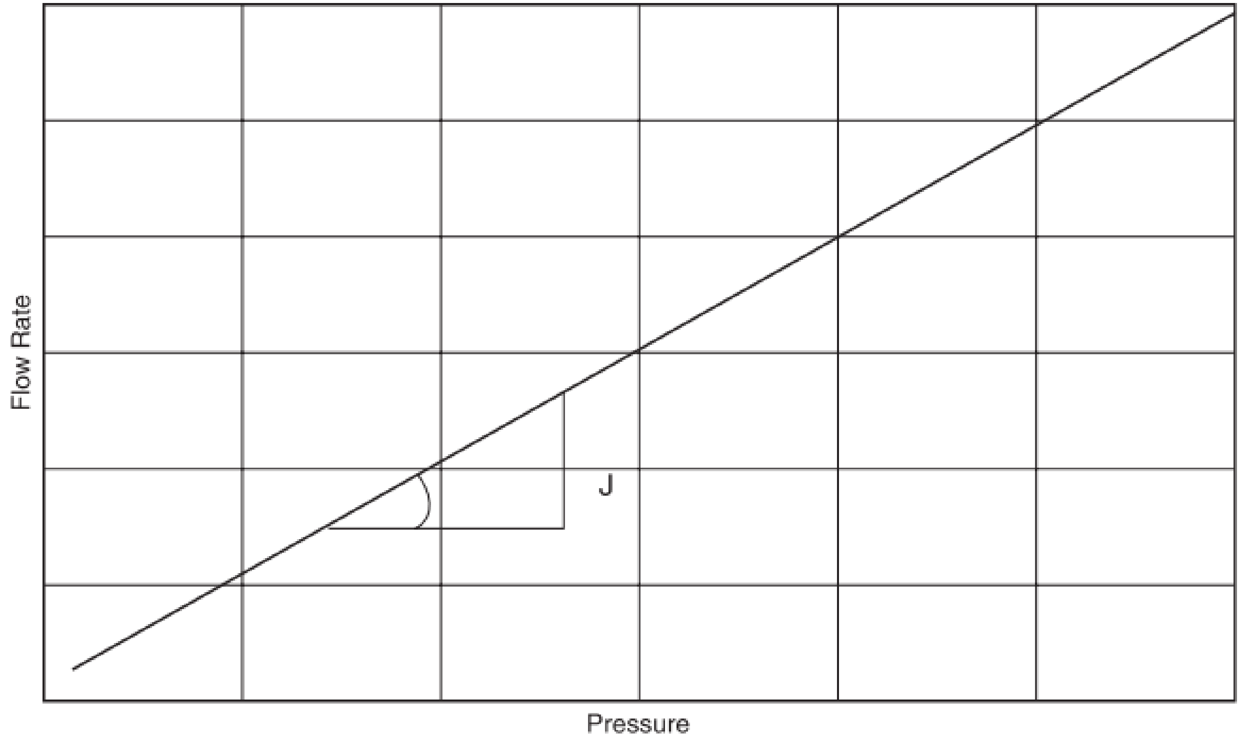
- The PI provides a valuable methodology for predicting future performance of wells since most of the well life is spent in a flow pseudo-steady state flow regime or regime that is approximating the pseudo-steady state.
- Since PI is assumed constant, eqn.(60) can be rearanged to give eqn.(61) which relates the production rate to the drawdown:

$$q = J\left(\bar{P} - P_{wf}\right) \qquad (61)$$

- Eqn.(61) indicates that the relationship between q and Δp is a straight line passing through the origin with a slope of J as shown in the Figure in the next slide

Module 6-Productivity index, zonal damage etc.

Productivity Index

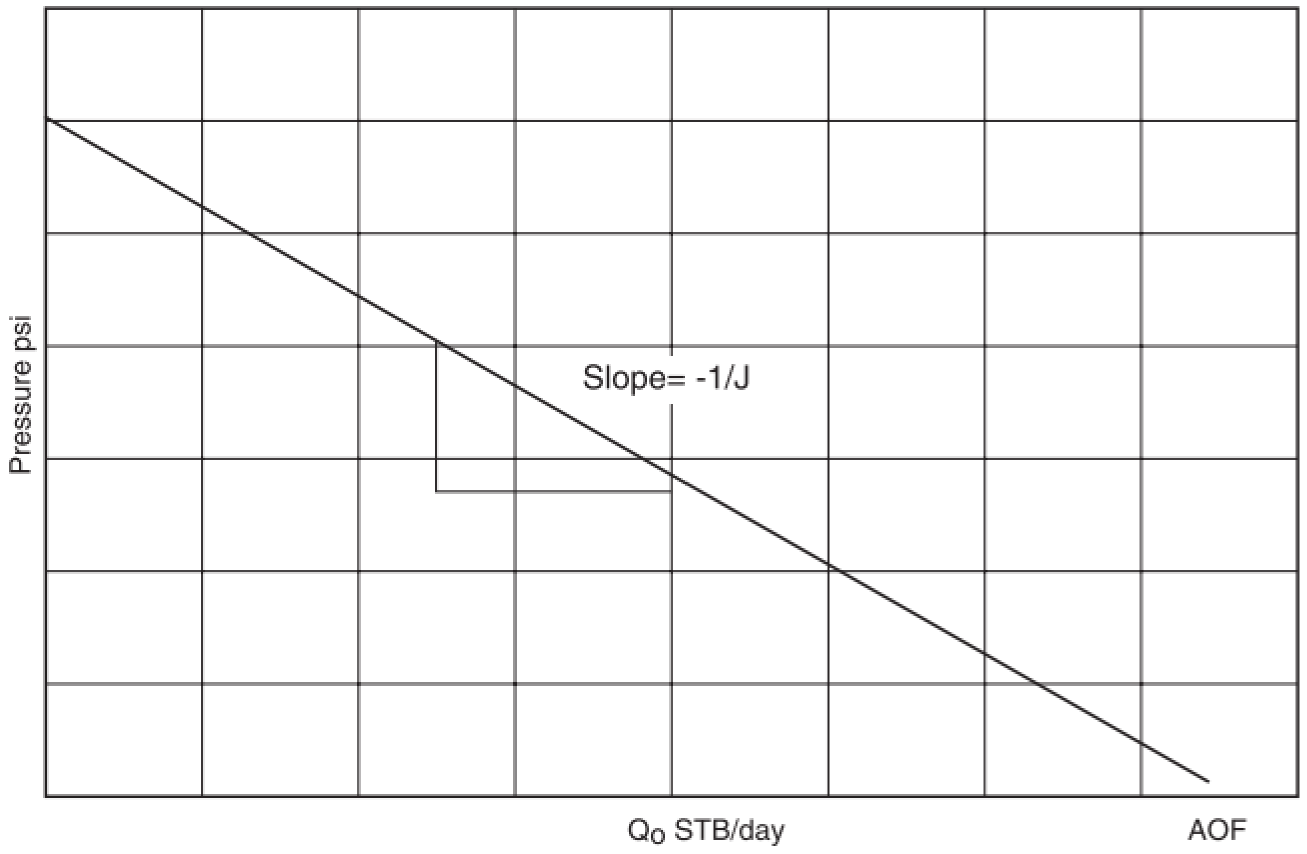


- Eqn.(60 or 61) can also be arranged to give eqn.(62) which relates the bottom hole flowing pressure to the production rate and again a straight line relationship is revealed with 1/J slope as shown below

Module 6-Productivity index, zonal damage etc.

Productivity Index

$$P_{wf} = \bar{P} - \left(\frac{1}{J}\right) q \qquad (62)$$



Module 6-Productivity index, zonal damage etc.

Productivity Index

- This graphical representation of the relationship that exists between the oil flow rate and bottom-hole flowing pressure is called the inflow performance relationship and referred to as IPR.
- Several important features of the straight-line IPR can be seen in the Figure presented in the previous slide

❖ When P_{wf} equals average reservoir pressure, the flow rate is zero due to the absence of any pressure drawdown.

❖ Maximum rate of flow occurs when P_{wf} is zero. This maximum rate is called absolute open flow and referred to as AOF.

❖ Although in practice this may not be a condition at which the well can produce, it is a useful definition that has widespread applications in the petroleum industry (e.g., comparing flow potential of different wells in the field)

Module 6-Productivity index, zonal damage etc.

Productivity Index

- The AOF is then calculated by:

$$AOF = J\bar{P} \qquad (63)$$

- The PI also provides a veritable tool to discover any well damage problem due to completion, workover, production, injection operations, or mechanical problems.
- If a measured J has an unexpected decline, one of the indicated problems should be investigated.
- A comparison of productivity indices of different wells in the same reservoir should also indicate some of the wells might have experienced unusual difficulties or damage during completion.

Module 6-Productivity index, zonal damage etc.

Productivity Index

- Since the productivity indices may vary from well to well because of the variation in thickness of the reservoir, it is helpful to normalize the indices by dividing each by the thickness of the well. This is defined as the specific productivity index J_s.

$$J_s = \frac{J}{h} = \frac{q}{h(\bar{P} - P_{wf})} \qquad (64)$$

Example: A productivity test was conducted on a well. The test results indicate that the well is capable of producing at a stabilized flow rate of 110 STB/day and a bottom-hole flowing pressure of 900 psi. After shutting the well for24 hours, the bottom-hole pressure reached a static value of 1300 psi.

Module 6-Productivity index, zonal damage etc.

Productivity Index-Example conts.

Calculate the following:

1. Productivity index
2. AOF
3. Oil flow rate at a bottom-hole flowing pressure of 600 psi
4. Wellbore flowing pressure required to produce 250 STB/day

Solution

(1) $\mathrm{J} = \frac{\mathrm{q}}{\overline{\mathrm{P}} - \mathrm{P_{wf}}} = \frac{110}{1300 - 900}$

(2) $\mathrm{AOF} = \mathrm{J}\overline{\mathrm{P}} = \mathrm{J} * 1300$

(3) $q = J(1300 - 600)$

(4) $P_{wf} = 1300 - \left(\frac{1}{J}\right) 250$

Module 6-Productivity index, zonal damage etc.

Zonal Damage and well stimulation

- Formation damage is a term that may be applied to many well productivity impairments, many of which have nothing to do with a problem in the formation.

- The goal of this module is to answer the following questions

❖ what is the damage and where is the damage located?

❖ How much damage is present?

❖ How is it affecting the productivity of the well?

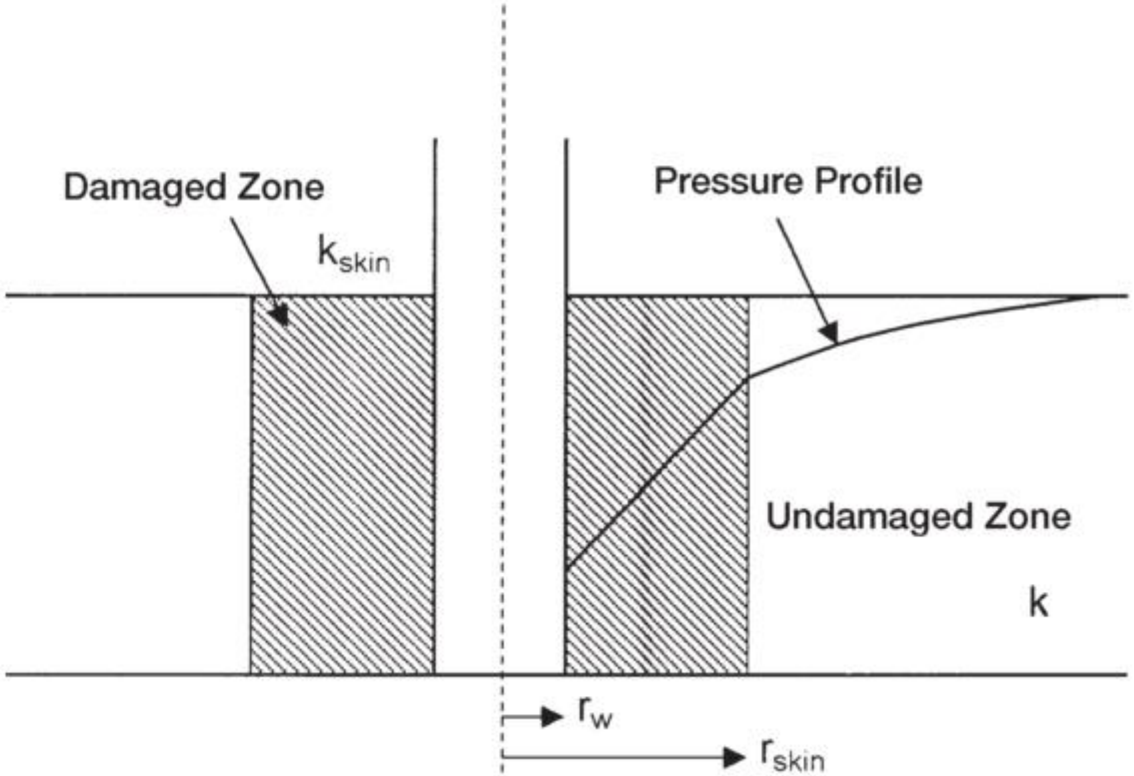
Module 6-Productivity index, zonal damage etc.

Zonal Damage and well stimulation

- Usually, materials such as mud filtrate, cement slurry, or clay particles enter the formation during drilling, completion, or workover operations and reduce the permeability around the wellbore.
- This effect is commonly referred to as a wellbore damage and the region of altered permeability is called the skin zone.
- This zone can extend from a few inches to several feet from the wellbore. Many other wells are stimulated by acidizing or fracturing, which in effect increase the permeability near the wellbore.
- Thus, the permeability near the wellbore is always different from the permeability away from the well where the formation has not been affected by drilling or stimulation.
- A schematic illustration of the skin zone is shown in the Figure below.

Module 6-Productivity index, zonal damage etc.

Zonal Damage and well stimulation



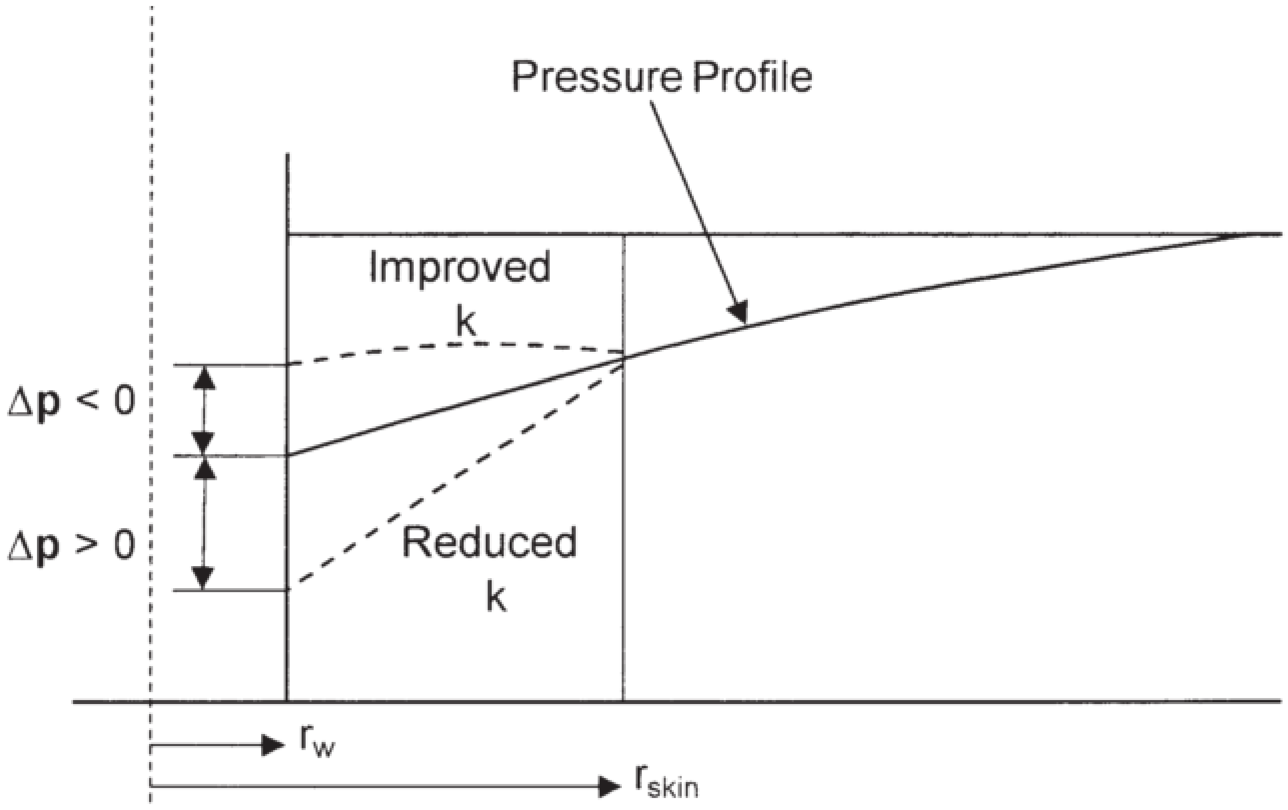
Module 6-Productivity index, zonal damage etc.

Zonal Damage and well stimulation

- Those factors that cause damage to the formation can produce additional localized pressure drop during flow.
- This additional pressure drop is commonly referred to as ΔP_{skin}.
- On the other hand, well stimulation techniques will normally enhance the properties of the formation and increase the permeability around the wellbore, so that a decrease in pressure drop is observed.
- The resulting effect of altering the permeability around the well bore is called the skin effect.
- Three possible outcomes are possible due to zonal damage and these are shown in the Figure below

Module 6-Productivity index, zonal damage etc.

Zonal Damage and well stimulation



Module 6-Productivity index, zonal damage etc.

Zonal Damage and well stimulation

- First Outcome: $\Delta P_{skin} > 0$, indicates an additional pressure drop due to wellbore damage, i.e., k_{skin} < k.
- Second Outcome: $\Delta P_{skin} < 0$, indicates less pressure drop due to wellbore improvement, i.e., k_{skin} > k.
- Third Outcome: $\Delta P_{skin} = 0$, indicates no changes in the wellbore condition, i.e., k_{skin}= k.
- Hawkins suggested that the permeability in the skin zone, i.e., k_{skin}, is uniform and the pressure drop across the zone can be approximated by Darcy's equation. Hawkins proposed the following approach:

$$\Delta P_{skin} = \begin{bmatrix} \Delta P \ in \ skin \ zone \\ due \ to \ \ k_{skin} \end{bmatrix} - \begin{bmatrix} \Delta P \ in \ skin \ zone \\ due \ to \ k \end{bmatrix} \qquad (65)$$

Module 6-Productivity index, zonal damage etc.

Zonal Damage and well stimulation

- Applying Darcy's equation, eqn.(65) becomes

$$\Delta P_{skin} = \left(\frac{q\mu B}{0.00708kh}\right)\left[\frac{k}{k_{skin}} - 1\right] ln\left(\frac{r_{skin}}{r_w}\right) \qquad (66)$$

- Where k is the permeability of the formation, md and k_{skin} is the permeability of the skin zone, md
- Eqn.(66) is the expression for determining the additional pressure drop in the skin zone and is commonly expressed in the following form:

$$\Delta P_{skin} = \left(\frac{q\mu B}{0.00708kh}\right) s = 141.2\left[\frac{q\mu B}{0.00708kh}\right] s \qquad (67)$$

Module 6-Productivity index, zonal damage etc.

Zonal Damage and well stimulation

- where s is called the skin factor and defined as

$$s = \left[\frac{k}{k_{skin}} - 1\right] ln\left(\frac{r_{skin}}{r_w}\right) \qquad (68)$$

- Eqn.(68) provides some insight into the physical significance of the sign of the skin factor. There are only three possible outcomes in evaluating the skin factor s:

❖ Positive Skin Factor, s > 0: When a damaged zone near the wellbore exists, k_{skin} is less than k and hence s is a positive number. The magnitude of the skin factor increases as k_{skin} decreases and as the depth of the damage r_{skin} increases.

Module 6-Productivity index, zonal damage etc.

Zonal Damage and well stimulation

- Negative Skin Factor, s < 0: When the permeability around the well k_{skin} is higher than that of the formation k, a negative skin factor exists. This negative factor indicates an improved wellbore condition.
- Zero Skin Factor, s = 0: Zero skin factor occurs when no alternation in the permeability around the wellbore is observed, i.e., k_{skin}= k.
- Eqn.(68) indicates that a negative skin factor will result in a negative value of ΔP_{skin}. This implies that a stimulated well will require less pressure drawdown to produce at rate q than an equivalent well with uniform permeability

Module 6-Productivity index, zonal damage etc.

Zonal Damage and well stimulation-Example

- Calculate the skin factor resulting from the invasion of the drilling fluid to a radius of 2 feet. The permeability of the skin zone is estimated at 20 md as compared with the unaffected formation permeability of 60 md. The wellbore radius is 0.25 ft.

Solution

Using eqn.(68), we have:

$$s = \left[\frac{k}{k_{skin}} - 1\right] ln\left(\frac{r_{skin}}{r_w}\right) = \left[\frac{60}{20} - 1\right] ln\left(\frac{2}{0.25}\right) = ?$$

Module 6-Productivity index, zonal damage etc.

Zonal Damage and well stimulation

- We can modify the previous flow equations based on the concept that the actual total pressure drawdown will increase or decrease by an amount of ΔP_{skin} . Assuming that ΔP_{ideal} represents the drawdown for a drainage area with a uniform permeability k, then:

$$\Delta P_{actual} = \Delta P_{ideal} + \Delta P_{skin} \qquad (69)$$

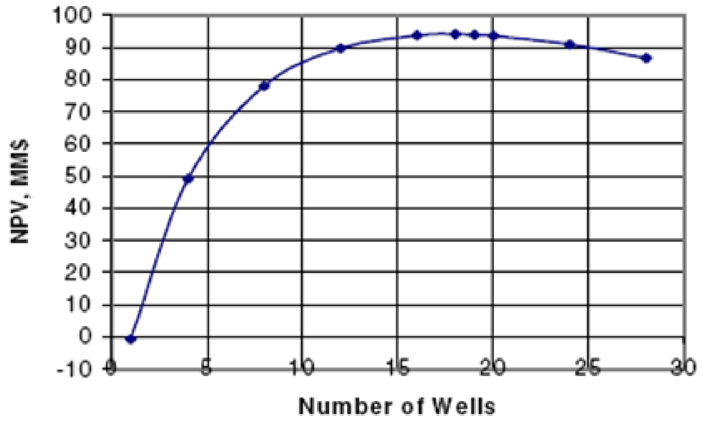
- The above concept as expressed by Eqn.(69) can be applied to all the previous flow regimes to account for the skin zone around the wellbore.

Assignment: Derive the flow equations for the following considering skin effect: steady radial incompressible flow, Unsteady radial slightly compressible and compressible flows, Pseudo-steady radial slightly compressible and compressible flows

Module 6-Productivity index, zonal damage etc

Gas well spacing

- Well spacing defines the number of subsurface drainage locations necessary to maximize hydrocarbon recovery from a pool or formation
- Appropriate well spacing is determined for maximum ultimate production for a given reservoir to avoid drilling of unnecessary wells
- This can be obtained through numerical simulation or analytical methods.e.g see the figure below



Corrie (2001) SPE 71431

Module 6-Productivity index, zonal damage etc

Gas well spacing

- Generally gas wells are of wider spacings say 160 acre spacing compared with oil wells of say 40 acre spacing.
- Well spacing is a function of geological structure, size of the reservoir.
- It therefore varies from country to country.
- In Nigeria for example based on Petroleum Act 1969 ,we have the following guidance

❖ for wells in the same pool, spacing btw wells not closer than 880yards (804metres)

❖ for wells not in the same pool, not closer than 440yards (402metres) except in cases of approved field development programme

Module 6-Productivity index, zonal damage etc.

Deliverability and recovery

- From the previous discussions and the resulting equations from the last assignment, it is clear that the bottomhole flowing pressure and indeed the pressure drawdown is an important driver of production.
- The gas well deliverability is the calculated or measured rate a gas well will produce for a given bottomhole or wellhead pressure
- This can be obtained through a deliverability test.
- Deliverability test is also referred to as productivity test
- More on deliverability will be covered under well testing

Module 7- Displacement of oil and gas

Displacement of oil and gas

- The displacement of oil and gas can happen by both external flooding processes and internal displacement processes
- This discussion is fundamental to understanding secondary and tertiary recovery techniques as well as some primary recovery mechanisms
- But to understand this module, there is the need to appreciate that in reservoir, there are usually more than one phase. Therefore a few concept must be introduced. They are:

❖ Relative permeability

❖ wettability

❖ capillary pressure

❖ Saturation

Module 7- Displacement of oil and gas

Displacement of oil and gas-Relative Permeability

- Recall that we said permeability is a property of the rock and not the fluid that flows through, but this is the case only when the fluid is single phase (100% saturation).
- This permeability at 100% saturation of a single fluid is called the absolute permeability of the rock.
- Effective permeability on the other hand, is the permeability of a given phase when more than one phase is present
- While relative permeability is the ratio of the effective permeability for a particular fluid to a reference or base permeability of the rock.

$$k_r = \frac{k_{eff}}{k_{ref}} \qquad (70)$$

Module 7- Displacement of oil and gas

Displacement of oil and gas-Relative Permeability

Example: Given a core sample $0.00215 ft^2$ in cross section and 0.1 ft long, flows a 1.0 cp brine with a formation volume factor of 1.0 bbl/STB at the rate of 0.30 STB/day under a 30 psi pressure differential. Calculate

1) absolute permeability
2) Absolute permeability if the water is relaced by an oil of 3.0 cp viscosity and 1.2 bbl/STB formation volume factor under the same pressure differential and a flow rate 0.0834 STB/day
3) If the same core sample is maintained at 70% water saturation and 30% oil saturation and at these saturations and under the same pressure drop it flows 0.18 STB/day of the brine and 0.01 STB/day of the oil, what is the effective permeability of water k_w ? Also calculate the effective permeability of oil k_o

Module 7- Displacement of oil and gas

Displacement of oil and gas-Relative Permeability

Soln:

1) $k = \dfrac{qB\mu L}{0.001127A\Delta P}$

2) $k = \dfrac{qB\mu L}{0.001127A\Delta P}$

3) (a) $k_w = \dfrac{q_w B_w \mu_w L}{0.001127A\Delta P}$

(b) $k_o = \dfrac{q_o B_o \mu_o L}{0.001127A\Delta P}$

Discussion: Sum of relative permeabilities compared with absolute permeability

Module 7- Displacement of oil and gas

Displacement of oil and gas-Relative Permeability

- In a multiphase system for example a two fluid system, their relative rates of flow are determined by their relative viscosities, their relative formation volume factors, and their relative permeability.
- From the previous example for instance, the relative permeabilities to water and to oil can be obtained as: $k_{rw} = \frac{k_w}{k}$ $and\ k_{ro} = \frac{k_o}{k}$ respectively
- The flowing water-oil ratio at reservoir conditions depends on the viscosity ratio and the effective permeability ratio (i.e on the mobility ratio already introduced in module 1). Mathematically,this can be shown as:

$$\frac{q_w B_w}{q_o B_o} = \frac{\frac{0.001127 A \Delta P k_w}{\mu_w L}}{\frac{0.001127 A \Delta P k_o}{\mu_o L}} = \frac{\frac{k_w}{\mu_w}}{\frac{k_o}{\mu_o}} = \frac{\lambda_w}{\lambda_o} = M \qquad (71)$$

Module 7- Displacement of oil and gas

Displacement of oil and gas-Relative Permeability

- From the previous example, it was observed that at 70% and 30 % water and oil saturation respectively, the water is flowing at at 14.9 times the oil rate. (use eqn. 71 to confirm this)
- Relative permeabilities may be substituted for effective permeabilities in the previous example because their ratios are equal (i.e $\frac{k_{rw}}{k_{ro}} = \frac{k_w}{k_o}$)
- However, the term relative permeability ratio is more commonly used and for the previous example, it has a value of 5
- Water flows at 14.9 times the oil rate because of viscosity ratio of 3 and a relative permeability ratio of 5

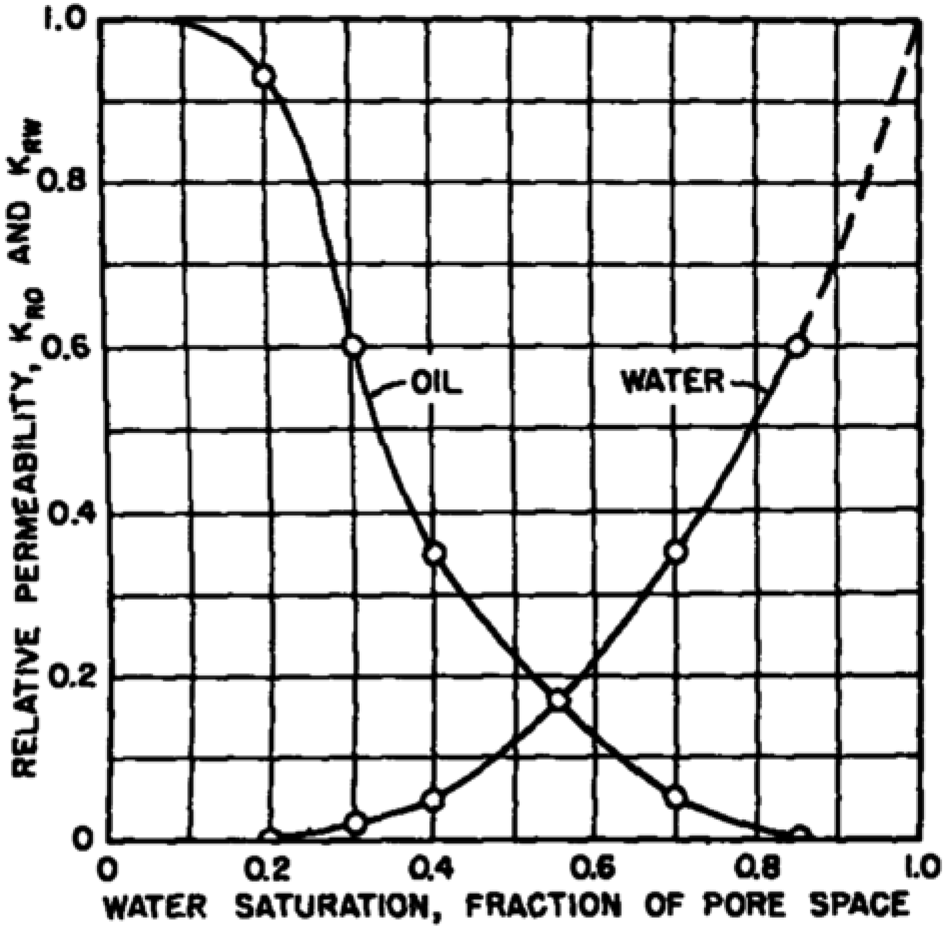
Module 7- Displacement of oil and gas

Displacement of oil and gas-Application of Relative Permeability Data

- To model a particular process, for example, fractional flow, fluid distributions, recovery and predictions
- Determination of the free water surface; i.e., the level of zero capillary pressure or the level below which fluid production is 100% water.
- Determination of residual fluid saturations
- To illustrate the application of relative permeability, consider the the Figure in the next slide which shows a plot of oil and water relative permeability curves for a particular rock as a function of water saturation
- Starting at 100% water saturation, the curves show that a decrease in water saturation to 85% (a 15% increase in oil saturation) sharply

Module 7- Displacement of oil and gas

Displacement of oil and gas-Application of Relative Permeability Data



Module 7- Displacement of oil and gas

Displacement of oil and gas-Application of Relative Permeability Data

brings the relative permeability to water from 100% down to 60%

- And at 15% oil saturation, the relative permeability to oil is zero
- This value of saturation in this case 15% is referred to as the critical saturation, the saturation at which oil first begins to flow as the oil saturation increases
- It is called residual saturation, the value below which the oil saturation can not be reduced in an oil-water system
- This explains why oil recovery by water drive is not 100% efficient
- If the initial connate water saturation is 20% for this particular rock, then the recovery from the portion of the reservoir invaded by high-pressure water influx is given as:

Module 7- Displacement of oil and gas

Displacement of oil and gas-Application of Relative Permeability Data

$$Recovery = \frac{Initial - Final}{Initial} = \frac{0.8 - 0.15}{0.8} = 81\%$$

- Also from the Figure, a further decrease in water saturation yields a a further decrease in the relative permeability to water and the relative permeability to oil increases.
- At 20% water saturation, the (connate) water is immobile and the relative permeability to oil is quite high(about 93%)
- This is due to wettability effect. That is most reservoir rocks are preferentially water wet, and at 20% water saturation, the water occupies least favourable portions of the pore spaces while oil occupies 80% of the pore spaces

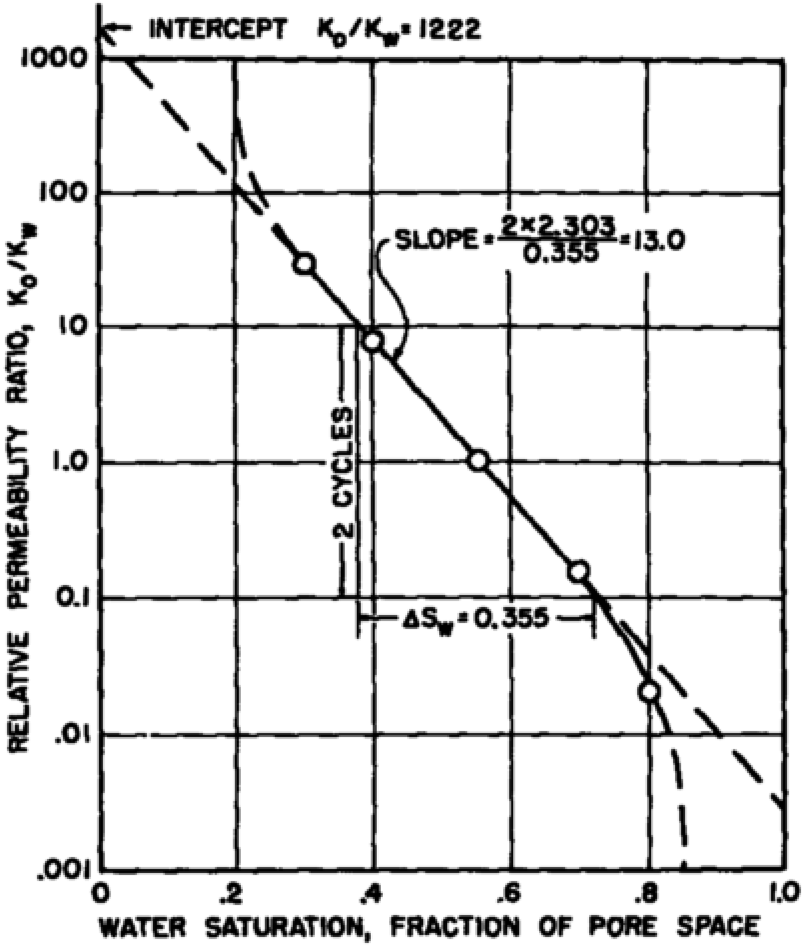
Module 7- Displacement of oil and gas

Displacement of oil and gas-Application of Relative Permeability Data

- The curves also indicate that about 10% of the pore spaces contribute nothing to the permeability, for at 10% water saturation, the relative permeability to oil is essentially 100%
- Conversely, on the other end of the curves, 15% of the pore spaces contribute 40% of the permeability
- As previously mentioned, relative permeability ratio is the preferred ratio since it is featured in the mathematical model describing two phase flow
- The Figure in the next slide therefore, shows a plot of relative permeability ratio against water saturation for the same data for the Figure previously discussed
- This plot is on a semilog scale

Module 7- Displacement of oil and gas

Displacement of oil and gas-Application of Relative Permeability Data



Module 7- Displacement of oil and gas

Displacement of oil and gas-Application of Relative Permeability Data

- The relative permeability ratio may be expressed as a function of water saturation by:

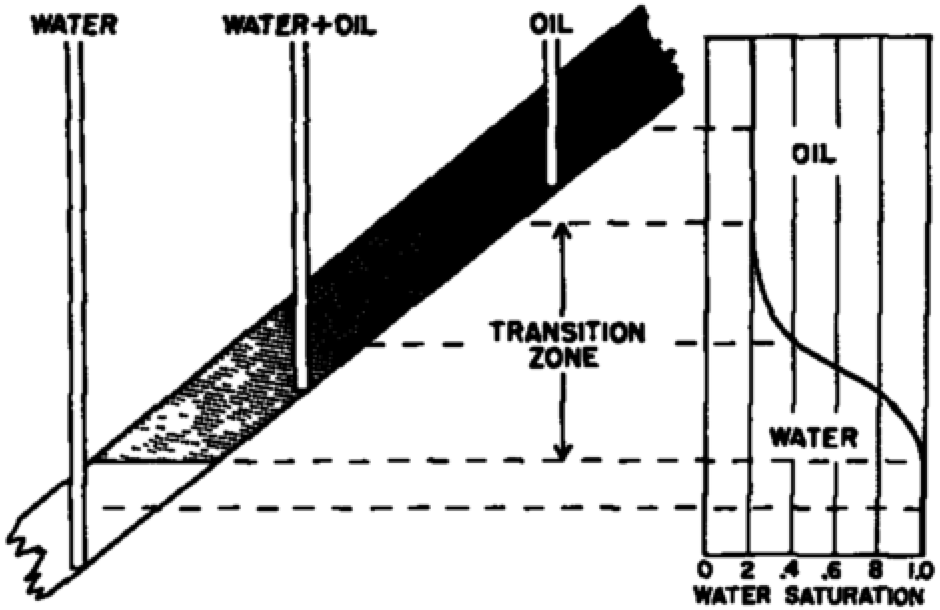
$$\frac{k_{rw}}{k_{ro}} = \frac{k_w}{k_o} = ae^{-bS_w} \qquad (72)$$

- The constants a and b may be determined from the graph or they may be determined from simultaneous equations
- Equation (72) indicates that the relative permeability ratio for a rock is a function of only the fluid saturations.
- Although it is true that the viscosities, the interfacial tensions and other factors have some effect on the relative permeability ratio, for a given rock, it is mainly a function of the fluid saturations

Module 7- Displacement of oil and gas

Displacement of oil and gas-Fractional flow curve

- In transition zone as shown in the Figure below, that is (outside true water zone or true oil zone) both oil and water will be produced and the fraction that is water will depend on the oil and water saturations at the point of completion



Module 7- Displacement of oil and gas

Displacement of oil and gas-Fractional flow curve

- If the well shown in the last slide is completed in a uniform sand at a point where oil and water saturations are 60 and 40% respectively, the fraction of water or water cut may be obtained from eqn(29) repeated here (each phase eqn. is used)

$$qB = \frac{0.00708kh(P_e - P_{wf})}{\mu ln\left(\frac{r_e}{r_w}\right)}$$

- Then water cut f_w is defined as:

$$f_w = \frac{q_w B_w}{q_w B_w + q_o B_o} \qquad (73)$$

Module 7- Displacement of oil and gas

Displacement of oil and gas-Fractional flow curve

- Combining eqn.(29) written for both phases and eqn.(73), common terms are cancelled and we have eqn.(74)

$$f_w = \frac{k_w/\mu_w}{k_w/\mu_w + k_o/\mu_o}$$

$$f_w = \frac{1}{1 + \frac{k_o \mu_w}{k_w \mu_o}} = \frac{1}{1 + \frac{k_{ro} \mu_w}{k_{rw} \mu_o}} \qquad (74)$$

- But $\frac{k_{rw}}{k_{ro}} = \frac{k_w}{k_o}$ is a function of saturation, eqn.(72) can be substituted into (74) to obtain eqn.(75)

Module 7- Displacement of oil and gas

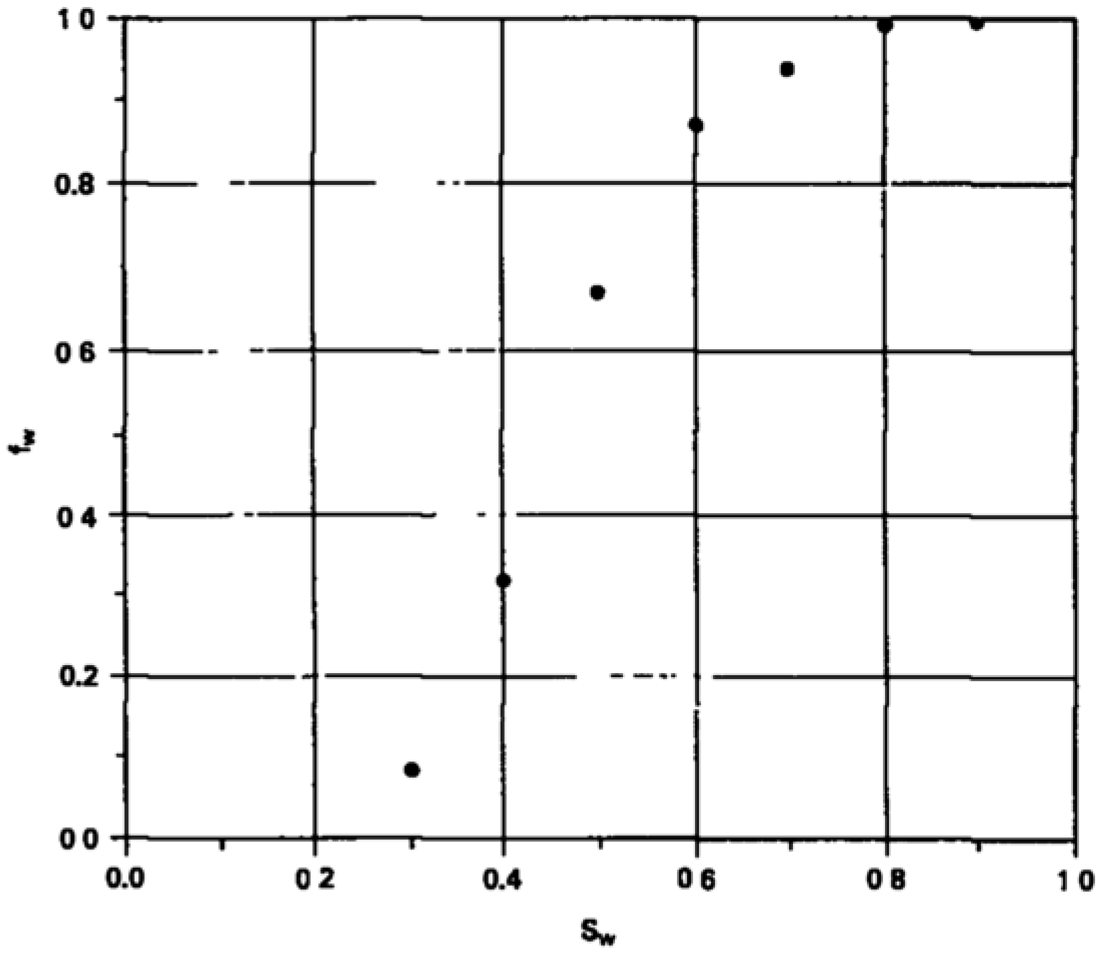
Displacement of oil and gas-Fractional flow curve

$$f_w = \frac{1}{1 + \left(\frac{\mu_w}{\mu_o}\right) a e^{-bS_w}} \qquad (75)$$

- For water cut obtained at various saturations plotted against these saturation values gives the plot in the next slide called the fractional flow curve
- This curve shows that fractional flow of water ranges from 0(for S_w less or equal to the connate water saturation) to 1 (for S_w greater or equal to one minus the residual oil saturation)

Module 7- Displacement of oil and gas

Displacement of oil and gas-Fractional flow curve



Module 7- Displacement of oil and gas

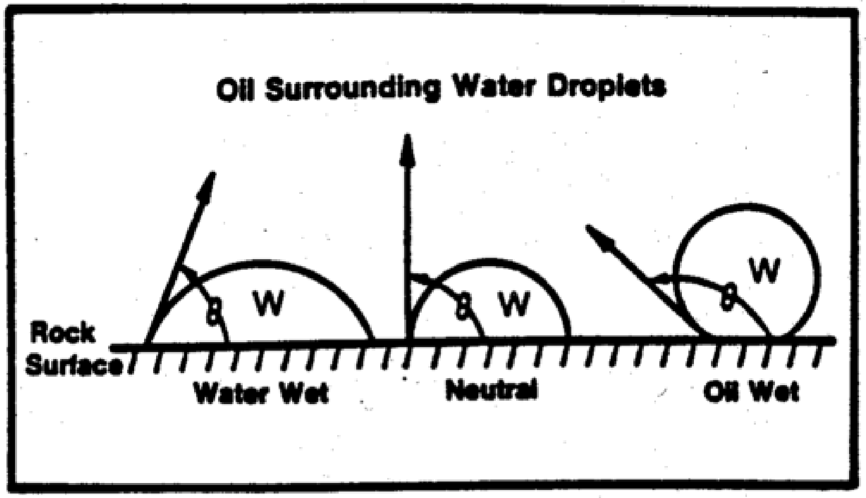
Displacement of oil and gas- Wettability

- On all interfaces between solids and fluids and between immiscible fluids, there is a surface free energy resulting from natural electrical forces.
- The forces cause molecules of the same substance to attract one another through cohesion, and molecules of unlike substances to attract one another through adhesion.
- Interfacial tension results from these molecular forces causing the surface of a liquid to form the smallest possible area and act like a membrane in tension.
- Wettability can be defined as the tendency or ability of a fluid phase to preferentially wet a solid surface in the presence of a second immiscible phase

Module 7- Displacement of oil and gas

Displacement of oil and gas- Wettability

- Three categories of wettability is generally known as shown in the Figure below:



❖ The reservoir is said to be water wet; that is, water preferentially wets the reservoir rock, when the contact angle between the rock and water is less than 90 degree.

❖ Neutral wettability : In this case, the contact angle is at 90 degree

❖ Oil wet occurs at a contact angle greater than 90 degree.

Module 7- Displacement of oil and gas

Displacement of oil and gas- Wettability

- Other lesser known types of wettability are:

❖ Intermediate wettability – no preference is shown by the rock to either fluid; i.e., equally wet.

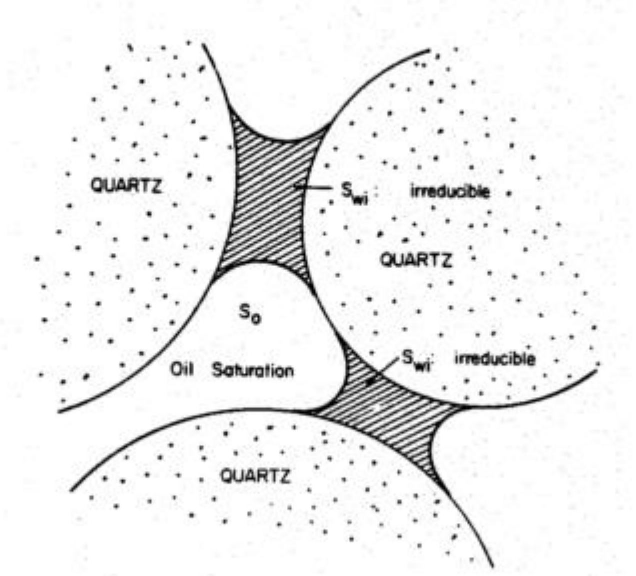
❖ Fractional wettability – heterogeneous wetting; i.e., portions of the rock are strongly oil wet, whereas other portions are strongly water wet. Occurs due to variation in minerals with different surface chemical properties.

❖ Mixed wettability – refers to small pores occupied by water and are water-wet, while larger pores are oil-wet and continuous. Subsequently, oil displacement occurs at very low oil saturations resulting in unusually low residual oil saturation

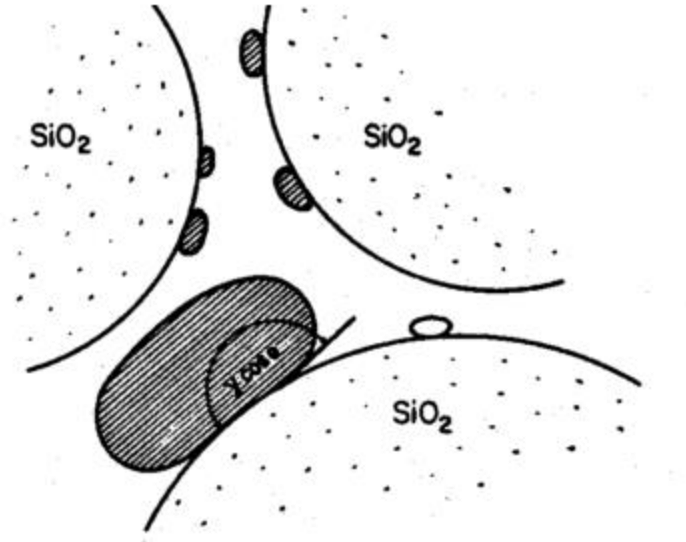
Module 7- Displacement of oil and gas

Displacement of oil and gas- Wettability

- The Figures below show the microscopic views of fluid saturation distribution in water and oil wet rock



Water wet rock



Oil wet rock

Module 7- Displacement of oil and gas

Displacement of oil and gas- Wettability

- The contact angle is a measure of the wettability of the rock-fluid system, and is related to the interfacial energies by Young's equation

$$\sigma_{os} - \sigma_{ws} = \sigma_{ow} cos\theta \qquad (76)$$

- Where:

σ_{os}=interfacial energy between oil and solid, dyne/cm

σ_{ws}=interfacial energy between water and solid, dyne/cm

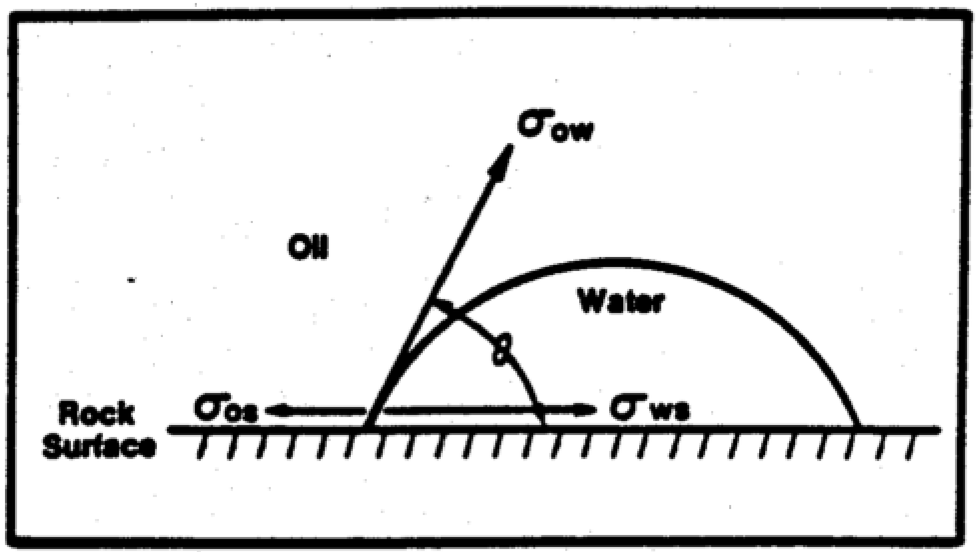
σ_{ow}=interfacial energy, or interfacial tension, between oil and water, dyne/cm

θ=contact angle at oil-water-solid interface measured through the water phase, deg

Module 7- Displacement of oil and gas

Displacement of oil and gas- Wettability

- The relationship of oil-water-solid interfacial tensions and contact angle shown mathematically in eqn.(76) can be shown graphically as given below



Module 7- Displacement of oil and gas

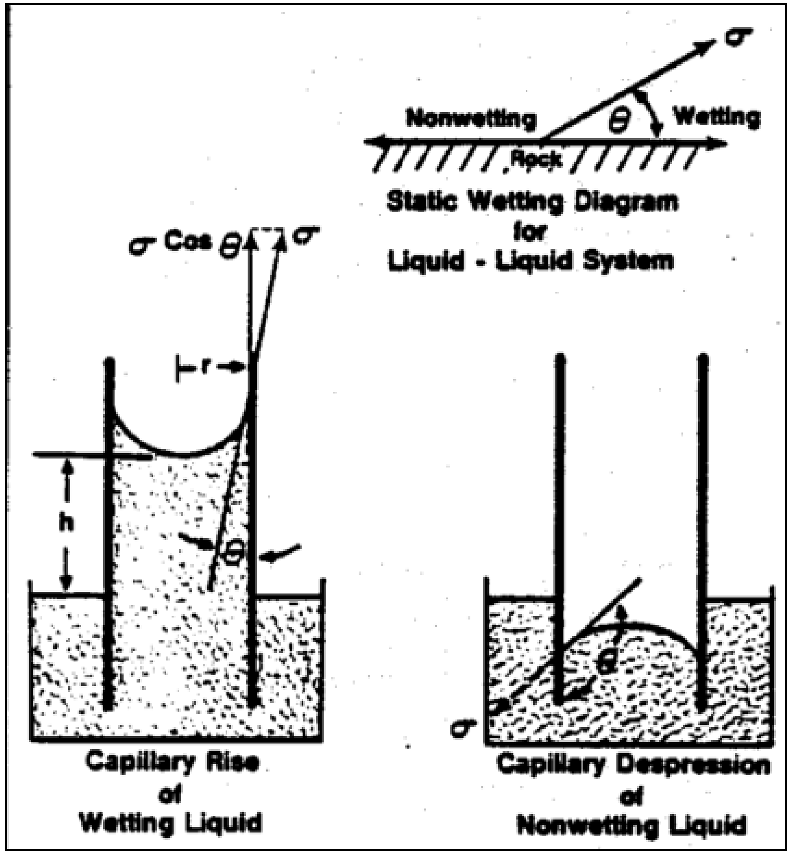
Displacement of oil and gas- Capillary pressure

- We have previously discussed that the reservoir rock can be be considered to be a complex assortment of small capillary tubes.
- The pores in typical reservoir rock are microscopic. The small pore size combined with the interfacial tension between the reservoir's immiscible fluids results in capillary pressure that is a major factor in establishing fluid distributions.
- Capillary pressure is the pressure difference across the curved interface formed by two immiscible fluids in a small capillary tube.
- To illustrate this concept, consider the Figure in the next slide where capillary tubes are immersed in wetting and nonwetting liquids.
- The capillary rise in a wetting liquid can be demonstrated with a glass capillary tube in water.

Module 7- Displacement of oil and gas

Displacement of oil and gas- Capillary pressure

- Capillary depression in a non-wetting liquid can be demonstrated by a glass tube in mercury.



Module 7- Displacement of oil and gas

Displacement of oil and gas- Capillary pressure

- Capillary pressure in a tube can be calculated from the interfacial tension between the fluids, contact angle between the rock and fluid, and the capillary tube radius See the Figure on the last slide.
- At equilibrium in the capillary tube, a force balance equation can be written as; $upward\ force = downward\ force$.
- Where the upward force for example is given as $2\sigma\pi r\ Cos\theta$
- But P_c is the force per unit area, therefore P_c is given as:

$$P_c = \frac{2\sigma\ Cos\theta}{r} \qquad (77)$$

where: r = pore radius, cm; σ = interfacial tension, dynes/cm; θ=contact angle, deg.

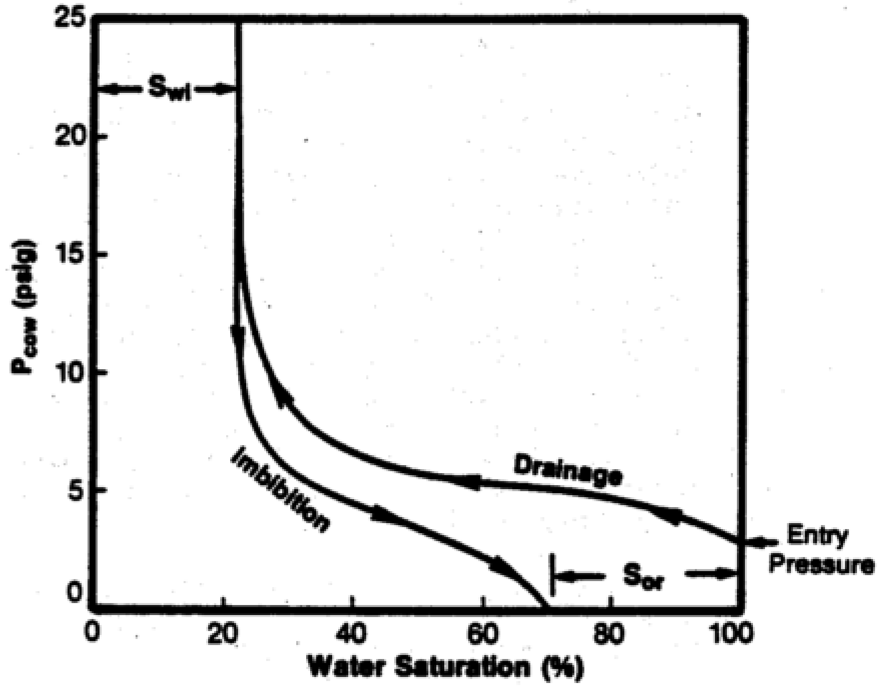
Module 7- Displacement of oil and gas

Displacement of oil and gas- Capillary pressure

- Eqn.(77) suggests that the capillary pressure in a porous medium is a function of the chemical composition of the rock and fluids, the pore size distribution of the sand grains in the rock and the saturation of the fluids in the pore
- Capillary pressures have also been found to be a function of the saturation history, although this is not reflected in eqn.(77).
- Different values of capillary pressure are obtained during the drainage process (displacing the wetting phase by the nonwetting phase) then during the imbibition process (displacing the nonwetting phase with the wetting phase)
- This historesis/hysteresis phenomenon is exhibited in all rock-fluid systems and is shown diagrammatically in the next slide

Module 7- Displacement of oil and gas

Displacement of oil and gas- Capillary pressure



- The Figure shows that the capillary pressure relationship as a function of saturation and it depends on the direction of saturation change.

Module 7- Displacement of oil and gas

Displacement of oil and gas- Capillary pressure

- The drainage curve is the relationship followed during normal migration and accumulation of oil in the reservoir. Originally the reservoir was completely filled with water (water saturation of 100%).
- The nonwetting oil phase first enters the reservoir at entry pressure- The capillary pressure required to force the first droplet of oil into the rock is called the entry pressure
- As the capillary pressure increases, so does the oil saturation. The arrows on the drainage curve show the direction of saturation change
- Because wetting-phase saturation is decreasing, it is usually called a drainage process. Water saturation continues to reduce until it reaches irreducible water saturation, Swi.

Module 7- Displacement of oil and gas

Displacement of oil and gas- Capillary pressure

- Assuming the oil reservoir was at irreducible water saturation at discovery, the imbibition curve is the path followed if water is injected into the rock starting at irreducible water saturation. This is the case during water flooding an oil reservoir.
- In Figure, the arrows show the direction of saturation change and because wetting-phase saturation is increasing (water is being imbibed into the rock), the process is called imbibition.
- The difference in paths between the drainage and imbibition curves is called historesis/hysteresis as previously defined.
- Residual oil saturation - The imbibition curve terminates at water saturation less than 100% (not all of the oil is displaced from the rock). The amount of oil trapped in an immobile state at the end of the imbibition process is called the residual oil saturation, Sor.

Module 7- Displacement of oil and gas

Displacement of oil and gas- Saturation

- A quick review of what saturation is!
- It is important to mention here that the saturation of one phase affects the permeability of the other(s). This we have previously illustrated in our discussion of relative permeability. It was clear that the sum of relative permeabilities is always less than the absolute permeability.

Module 7- Displacement of oil and gas

Displacement of oil and gas- Recovery Efficiency

- The overall recovery efficiency E of any fluid displacement process is given by the product of the macroscopic or volumetric displacement efficiency, E_v ,and the microscopic displacement efficiency, E_d.
- This is mathematically expressed as:

$$E = E_v E_d = N_p/(SoiVp/Bo) \qquad (78)$$

- The microscopic displacement efficiency is a measure of how well the displacing fluid mobilises the residual oil once it has made a contact with it. $E_d = (S_{oi} - S_{or})/S_{oi}$
- The macroscopic displacement efficiency is a measure of how well the displacing fluid has contacted the oil bearing parts of the reservoir. $E_v = E_i E_a$ (Areal & Vertical Sweep)

Module 7- Displacement of oil and gas

Displacement of oil and gas- Microscopic Displacement Efficiency

- The microscopic efficiency is affected by the following factors:
  - Interfacial and surface tension forces
  - Wettability
  - Capillary pressure
  - Relative permeability
- We have extensively discussed all these factors already

Module 7- Displacement of oil and gas

Displacement of oil and gas- Macroscopic Displacement Efficiency

- As previously mentioned, the macroscopic efficiency is a measure of how well the displacing fluid has contacted the oil bearing parts of the reservoir.
- It is made up of two other terms: the areal E_s, sweep efficiency and vertical E_i, sweep efficiency
- Macroscopic efficiency is affected by the following factors:

❖ Heterogeneities and anisotropy

❖ Mobility of the displacing fluids compared with the mobility of the displaced fluids

❖ The physical arrangement of injection and production wells

❖ The type of rock matrix in which the oil or gas exists

Module 7- Displacement of oil and gas

Macroscopic Displacement Efficiency-Heterogeneities and Anisotropy

- What is Anisotropy and Heterogeneities? How do they affect macroscopic displacement efficiency?
- Significant effect!
- Non uniform movement of fluids through heterogeneous and anisotropic hydrocarbon-bearing formation
- Large variations in properties such as porosity, permeability, and clay cement
- Presence of micro-fractures or large macro-fractures which favours fluid flow leading to substantial bypassing of HC
- Vertical and horizontal variability in permeability result in reduced vertical and areal sweep efficiencies

Module 7- Displacement of oil and gas

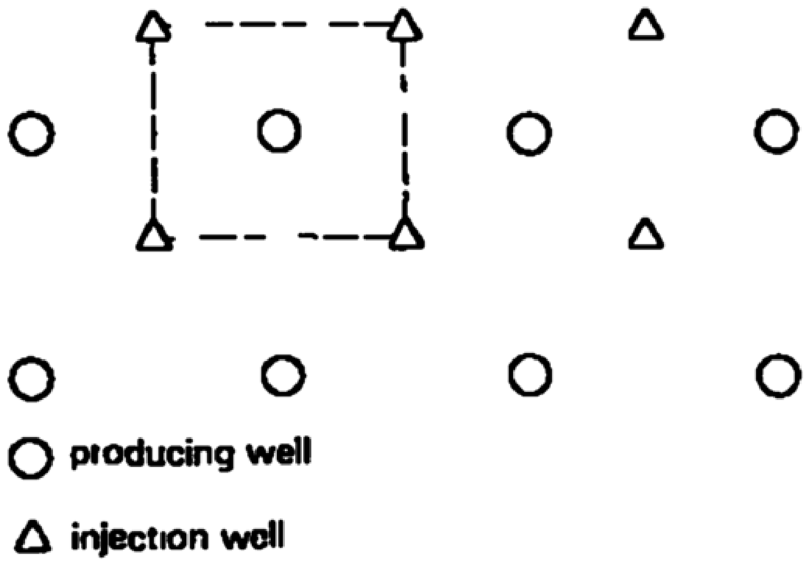
Macroscopic Displacement Efficiency-Mobility

- The relative measure of how easily a fluid moves through porous media
- A function of fluid saturations
- Areal sweep efficiencies strongly depend on the mobility ratio
- Viscous fingering can occur if the mobility of the displacing fluid is much greater than the mobility of the displaced phase
- Viscous fingering is defined as the penetration of the much more mobile displacing phase into the phase that is being displaced
- How can this be avoided?

Module 7- Displacement of oil and gas

Macroscopic Displacement Efficiency-Arrangement of Wells

- The arrangement of injection and production wells is a function of the geology of the formation and the size (areal extent) of the the reservoir

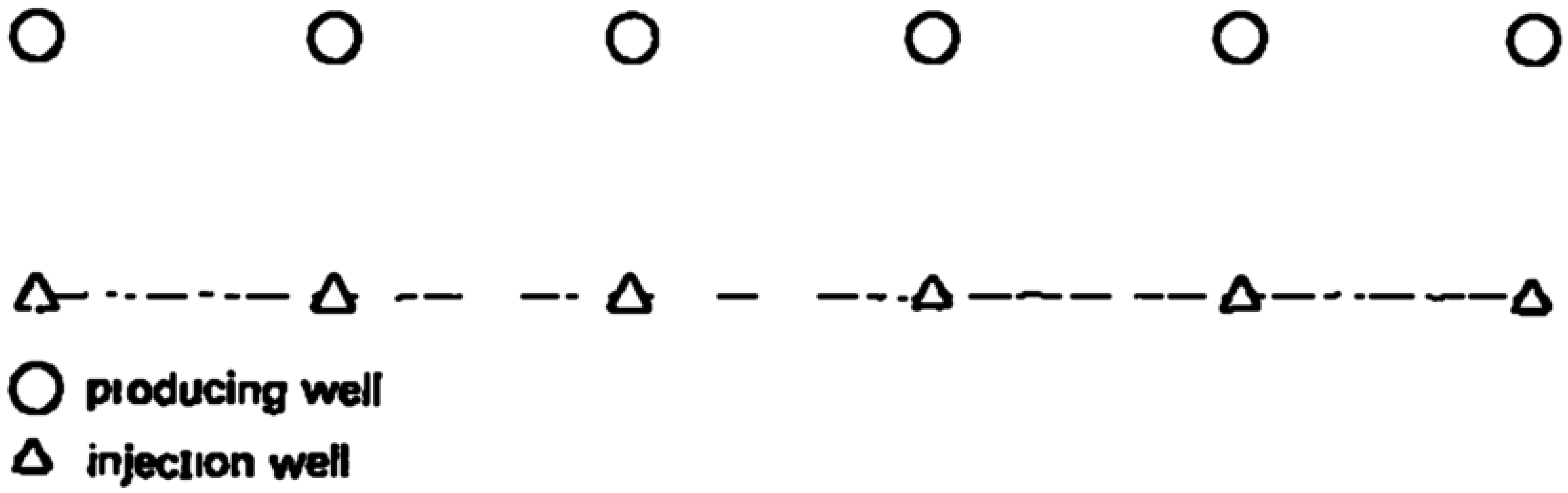


- An example of five-spot flooding network is shown above

Module 7- Displacement of oil and gas

Macroscopic Displacement Efficiency-Arrangement of Wells

- Another example of flooding network (Direct line-drive flooding network) is shown below



- Which of these networks would provide greater macroscopic efficiency?

Module 7- Displacement of oil and gas

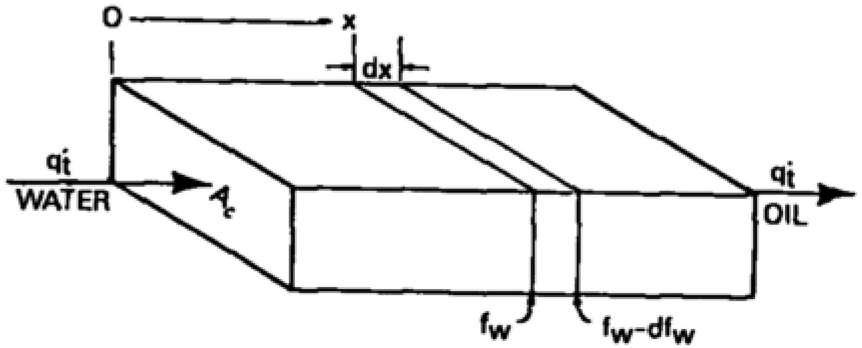
Macroscopic Displacement Efficiency-Rock Matrix type

- Sandstone or limestone?
- Sandstone formations-more uniform pore geometry than limestone but can also contain small sand grains and be so tightly packed that fluids will not readily flow through the formation
- Limestone formations- large holes (vugs), and interconnected fractures
- Their connate water have high levels of divalent ion content (Ca^{2+} and Mg^{2+}).
- Vugular porosity and high divalent ion content of limestone reservoir render it unfit for application of injection processes

Module 7- Displacement of oil and gas

Immiscible Displacement Process-The Buckley-Leverett Displacement Mechanism

- Based on the relative permeability concept, Buckley and Leverett developed a theory of oil displacement by water
- Consider a linear bed containing oil and water shown below



- Let the total throughput be the sum of water and oil flows given as:

$$q_t = q_w B_w + q_o B_o \qquad (79)$$

Module 7- Displacement of oil and gas

Immiscible Displacement Process-The Buckley-Leverett Displacement Mechanism

- Assumptions

❖ q_t in reservoir barrels is same at all cross section

❖ Gravitational and capillary forces neglected(A complete formulation can be found in the literatures e.g Engler(2010))

❖ S_w is the water saturation in any element at time t (days)

❖ Then if oil is being displaced from the element, at time (t+dt) the water saturation will be ($S_w + \mathrm{d}S_w$).

❖ If Ø is the total porosity fraction, A is the cross section in square feet and dx is the thickness of the element in feet

❖ Then the rate of increase of water in the element at time t in barrels per day is given as:

Module 7- Displacement of oil and gas

Immiscible Displacement Process-The Buckley-Leverett Displacement Mechanism

$$\frac{dW}{dt} = \frac{\emptyset A dx}{5.615}\left(\frac{\partial S_w}{\partial t}\right)_x \qquad (80)$$

- The subscript x in eqn.(80) is an element marker. This indicates that the derivative is different for each element.
- If f_w is the fraction of water in the total flow q_t barrels per day, then $f_w q_t$ is the rate of water entering the left hand face of the element dx
- The oil saturation will be slightly higher at the right hand face, so the fraction of the water flowing there will be slightly less, or $f_w - \mathrm{d}f_w$. Thus the rate of water leaving the element is $(f_w - \mathrm{d}f_w)q_t$

Module 7- Displacement of oil and gas

Immiscible Displacement Process-The Buckley-Leverett Displacement Mechanism

- Thus, the net rate of gain of water in the element at any time is:

$$\frac{dW}{dt} = (f_w - \mathrm{d}f_w)q_t - f_w q_t = -q_t df_w \qquad (81)$$

- Equating eqn.(80) and (81), we have:

$$\left(\frac{\partial S_w}{\partial t}\right)_x = -\frac{5.615 q_t}{\emptyset A}\left(\frac{\partial f_w}{\partial x}\right)_t \qquad (82)$$

- But for a given rock, assuming constant oil and water viscosities, the fraction of water f_w is a function only of the water saturation S_w as given in eqn.(75)

Module 7- Displacement of oil and gas

Immiscible Displacement Process-The Buckley-Leverett Displacement Mechanism

- However, the water saturation is a function of both time and position. That is, $f_w = \mathrm{F}(S_w) and\ S_w = G(t, x)$.Thus, we have:

$$dS_w = \left(\frac{\partial S_w}{\partial t}\right)_x dt + \left(\frac{\partial S_w}{\partial x}\right)_t dx \quad (83)$$

- We are interested in determining the rate of advance of a constant saturation plane or front, $\left(\frac{\partial x}{\partial t}\right)_{S_w}$ i.e where S_w is constant. Then from eqn.(83), we have

$$\left(\frac{\partial x}{\partial t}\right)_{S_w} = -\frac{\left(\frac{\partial S_w}{\partial t}\right)_x}{\left(\frac{\partial S_w}{\partial x}\right)_t} \qquad (84)$$

Module 7- Displacement of oil and gas

Immiscible Displacement Process-The Buckley-Leverett Displacement Mechanism

- Substituting eqn.(82) into (84), we have:

$$\left(\frac{\partial x}{\partial t}\right)_{S_w} = \frac{5.615\, q_t}{\emptyset A} \frac{\left(\frac{\partial f_w}{\partial x}\right)_t}{\left(\frac{\partial S_w}{\partial x}\right)_t} \quad (85)$$

- But

$$\frac{\left(\frac{\partial f_w}{\partial x}\right)_t}{\left(\frac{\partial S_w}{\partial x}\right)_t} = \left(\frac{\partial f_w}{\partial S_w}\right)_t \quad (86)$$

Module 7- Displacement of oil and gas

Immiscible Displacement Process-The Buckley-Leverett Displacement Mechanism

- Eqn.(85) then becomes (87). Eqn.(87) is referred to as the frontal advance equation.

$$\left(\frac{\partial x}{\partial t}\right)_{S_w} = \frac{5.615\, q_t}{\emptyset A}\left(\frac{\partial f_w}{\partial S_w}\right)_t \qquad (87)$$

- Eqn.(87) represents the velocity of the saturation front.
- Other assumptions in the derivation of eqn.(87) are:

❖ Fluids are incompressible fluid and immiscible

❖ Only oil is displaced; i.e., the initial water saturation is immobile

❖ No initial free gas saturation exists; i.e., not a depleted reservoir.

Module 7- Displacement of oil and gas

Immiscible Displacement Process-The Buckley-Leverett Displacement Mechanism

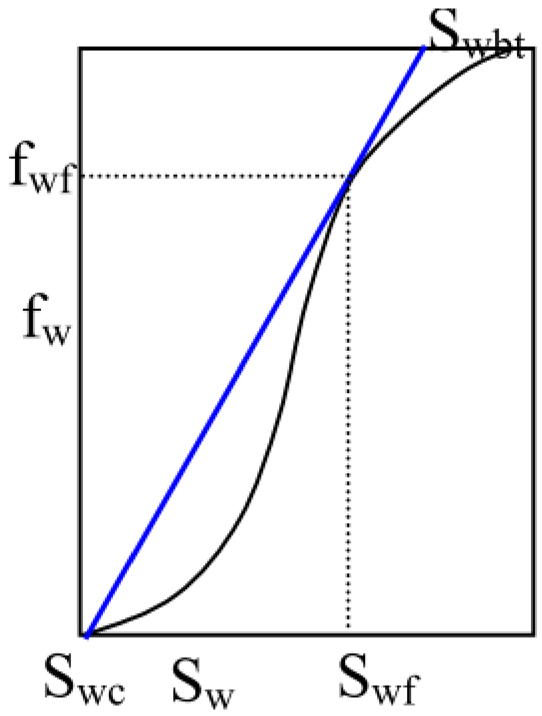
- Because the porosity, area, and throughput are constant and because for any value S_w, the derivative $\frac{\partial f_w}{\partial S_w}$ is a constant , then the rate $\frac{\partial x}{\partial t}$ is constant.
- This implies that the distance a plane of constant saturation S_w, advances is directly proportional to time and the value of the derivative $\frac{\partial f_w}{\partial S_w}$ at that saturation
- Eqn.(87) then becomes (88).

$$x = \frac{5.615\, q_t t}{\emptyset A}\left(\frac{\partial f_w}{\partial S_w}\right)_{S_w} \qquad (88)$$

Module 7- Displacement of oil and gas

Immiscible Displacement Process-The Buckley-Leverett Displacement Mechanism

- We can evaluate the derivative from the fractional flow equation (Eq. 74), either graphically or analytical.
- The figure below illustrates the graphical solution



Module 7- Displacement of oil and gas

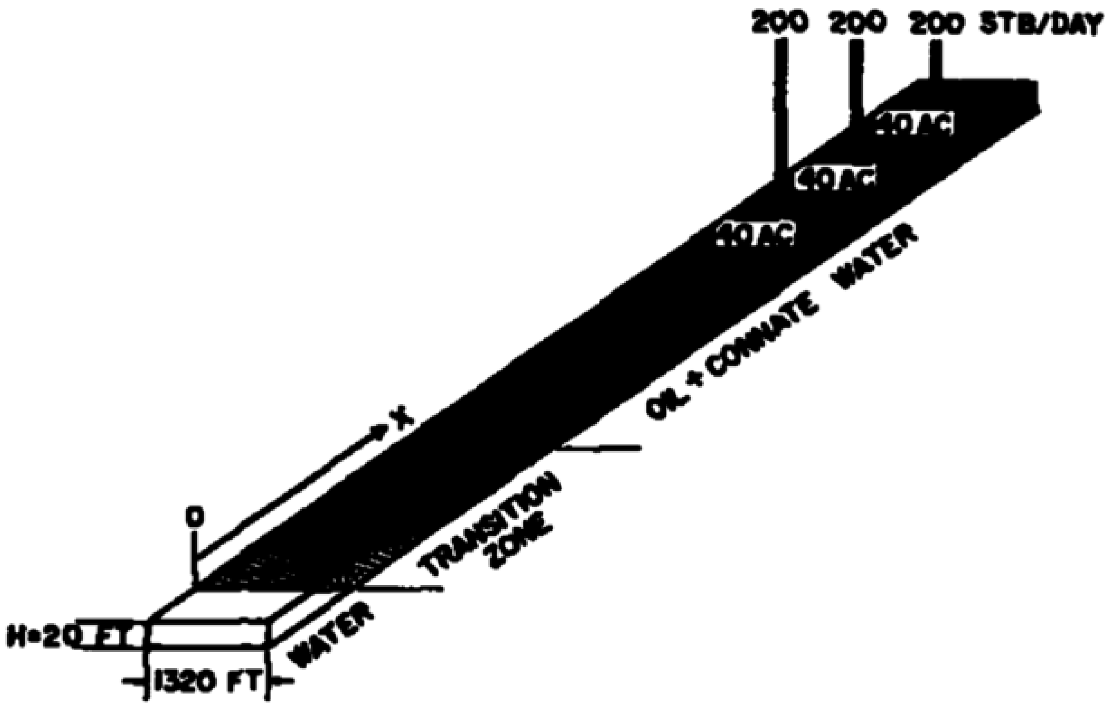
Immiscible Displacement Process-The Buckley-Leverett Displacement Mechanism

- The fractional flow of water at the front, f_{wf}, is determined from the tangent line originating at S_{wc}.
- The corresponding water saturation at the front is S_{wf}.
- The average water saturation behind the front at breakthrough, S_{wbt}, is given by the intersection at f_w =1.
- The location of the front is determined by Eqn. (88), with the slope of the tangent to the fractional curve used for the derivative function.

Module 7- Displacement of oil and gas

Immiscible Displacement Process-The Buckley-Leverett Displacement Mechanism-Exercise

- A reservoir is under active water drive and the walls are located in uniform rows along the strike on 40 ac spacing as shown in the Figure below



Module 7- Displacement of oil and gas

Immiscible Displacement Process-The Buckley-Leverett Displacement Mechanism-Exercise

- This gives rise to approximate linear flow and if the daily production of each of the three wells located along the deep is 200STB of oil per day, then for an active water drive and oil volume factor of 1.5bbl/STB. Given the porosity of 25%

❖ Calculate the total reservoir throughput

❖ Estimate the distances the various constant water saturation planes will travel in 60, 120 and 240 days.

❖ Given that the ratio of viscosity of water to the oil $\frac{\mu_w}{\mu_o}$ is constant at 0.5, using the table below, plot the fractional flow curve and estimate the various derivative $\frac{\partial f_w}{\partial S_w}$ at that various saturations

Module 7- Displacement of oil and gas

Immiscible Displacement Process-The Buckley-Leverett Displacement Mechanism-Exercise

S_w	$\frac{k_o}{k_w}$
0.20	Inf.
0.30	17.0
0.40	5.50
0.50	1.70
0.60	0.55
0.70	0.17
0.80	0.0055
0.90	0.00

Module 7- Displacement of oil and gas

Immiscible Displacement Process-The Buckley-Leverett Displacement Mechanism-Solution guide

- Total throughput is q_t and it is the total production from the three wells (200* 3* 1.5)
- The various distances are obtained using eqn.(88)
- To plot the fractional flow curve and estimate the various derivative, we need eqn.(74)

Assignment

1.Submit a complete solution of this exercise (1 week)

2. Attempt a review on gas displacement

Module 8- Waterflooding

Introduction

- Waterflooding is the most used secondary recovery method worldwide to improve oil recovery from reservoirs.

- It is the use of injection water to produce reservoir oil.

- The performance of a waterflood project in terms of oil recovery depends on many factors, such as
  - Rock wettability,
  - Rock and fluid properties,
  - Formation heterogeneities,
  - Flood patterns,
  - Composition of injection water,
  - Water injection rates,
  - Fluid production rates, etc.

- These factors should be considered in the design and installation of a waterflood project.
  - Waterflood Patterns
  - Waterflood Design

Module 8- Waterflooding

Waterflood Patterns

- Waterflood patterns are based on the arrangement of injectors and producers in a waterflooded reservoir.
- This is aimed at improving areal sweep efficiency.
- The criteria for converting these oil fields to waterflooding were to minimize drilling of new wells, maximize water injection, and improve oil recovery.
- The waterflood pattern installed in these oil fields depended on the density of existing wells, reservoir properties, and projected rate of oil recovery.
- Pattern flooding is not efficient for development of Deepwater reservoirs or reservoirs found in similar environments with high development costs.
- The ratio of water injectors to producers, and the location of these wells are usually optimized using reservoir simulation models.

Module 8- Waterflooding

Common Waterflood Patterns

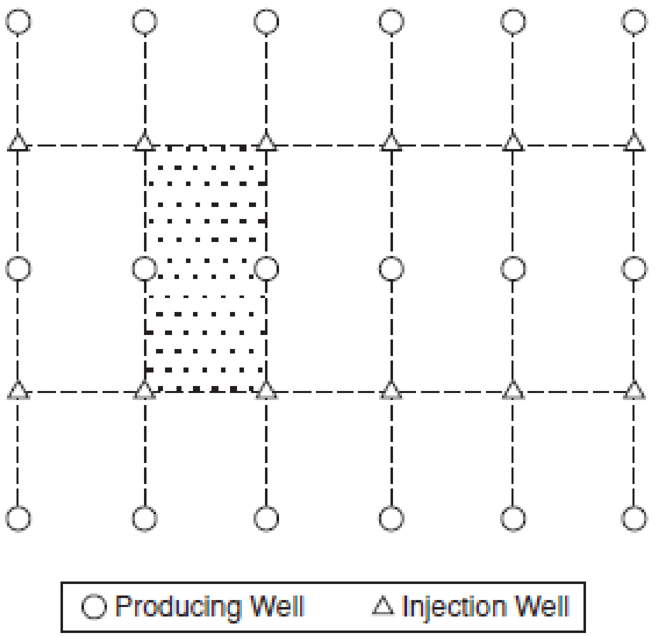
- The direct line drive,
- The staggered line drive,
- The five-spot,
- The nine-spot patterns.
- Peripheral flood pattern

Module 8- Waterflooding

Direct Line Drive

- In Direct Line Drive, the injectors and producers offset each other directly.

- Could lead to an early water breakthrough if the distances between the injectors and producers are small.



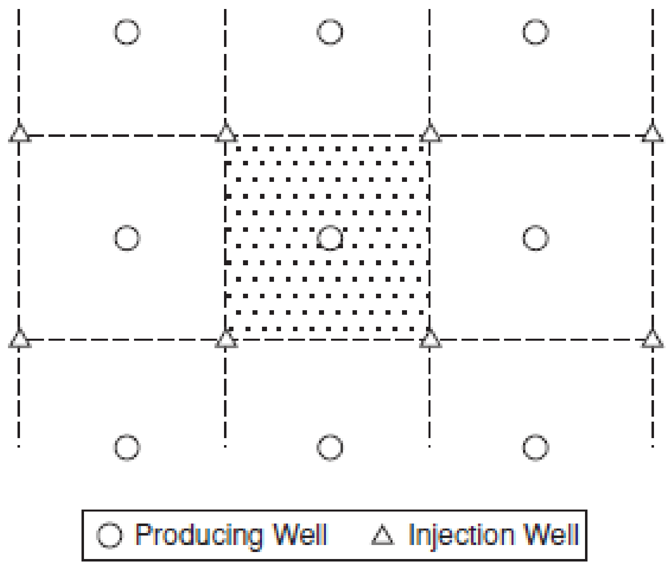
Direct line drive pattern.

Module 8- Waterflooding

Staggered line drive pattern

- In Staggered Line Drive, the injectors and producers are placed in an alternate rows not to offset each other directly.

- The aim is to reduce the early water breakthrough on sweep efficiency as encounter in the Direct Drive.

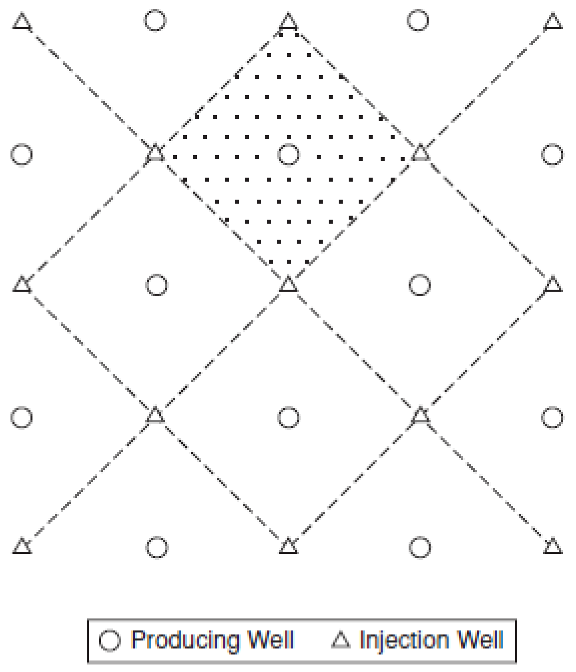


Staggered line drive pattern.

Module 8- Waterflooding

Normal five-spot pattern

- The five-spot pattern has been used widely in many field applications and model studies.
- This pattern generally results in good areal sweep efficiency.
- The ratio of injectors to producers in a five-spot pattern is also one.



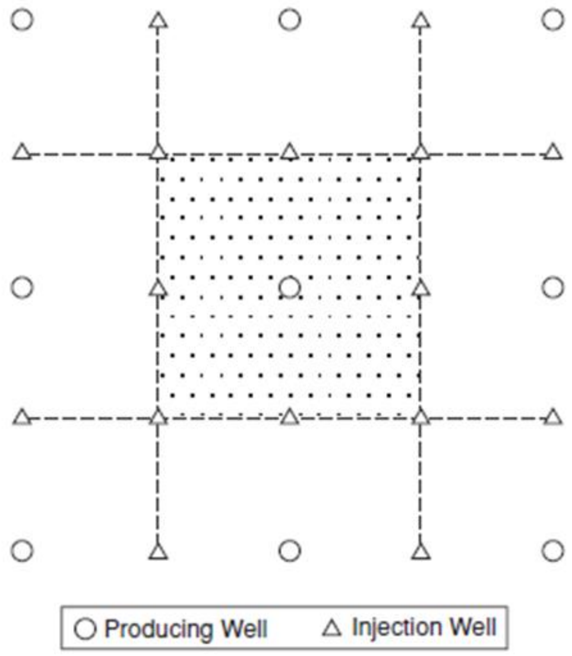
Normal five-spot pattern.

Module 8- Waterflooding

Normal nine-spot pattern

- This pattern is used in low permeability reservoirs where high water injection volumes are desired.

- The ratio of injectors to producers in a nine-spot pattern is three.



Normal nine-spot pattern.

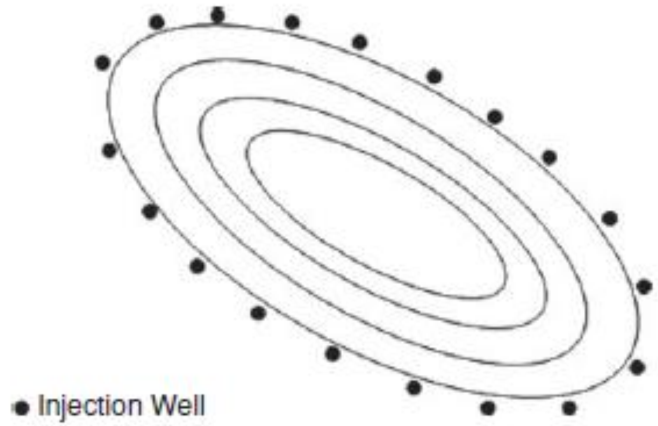
Module 8- Waterflooding

Peripheral flood pattern

- This is when injectors are located around part or the entire boundary of the reservoir.

- If the injectors are located along one side of the reservoir or the middle of the reservoir, it is called a line flood.

- Peripheral flooding is used with these objectives
  - To maintain reservoir pressure,
  - Minimize water production, and
  - Improve oil recovery.



Peripheral flood pattern.

Module 8- Waterflooding

Recommended Steps in Waterflood Design

- Construct a geologic model of the reservoir or project area
- Analyze rock/fluid properties data.
- Construct reservoir flow model with data from Steps 1 and 2.
- Run prediction cases.
- Optimize waterflood design.
- Perform sensitivity analyses.
- Conduct a pilot waterflood project.

Module 8- Waterflooding

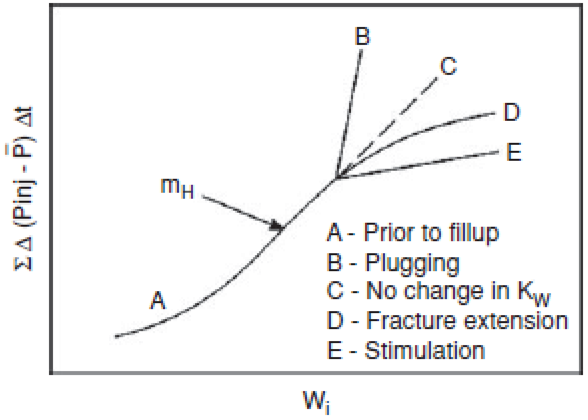
Waterflood Management

- Injection wells,
- Injection water,
- Production wells,
- Production water,
- Facilities, and
- The reservoir.

Module 8- Waterflooding

Injection Wells

- Maintaining the desired levels of water injection rates is key to a successful waterflood.
- Scale build-up and plugging of the formation from solid deposits may be exhibited and will lead to reduced injectivity over time.
- Fall-off test is used to gauge well injectivity (Build-up Test for well Productivity).
- The conditions of an injection well can be monitored periodically by **Hall Plot** on the performance of the injector.



Characteristic Hall plot.

Module 8- Waterflooding

Injection Wells

Darcy's Law for injection well is expressed below

$$q_w = \frac{0.00708 k_w h (p_{inj} - \overline{p})}{\mu_w \ln\left(\frac{r_e}{r_w}\right)} \qquad 89$$

Where

q_w = water injection rate, BWPD;

k_w = effective permeability to water, md;

h = formation thickness, ft;

p_{inj} = bottom-hole injection pressure, psa;

$\dot{P}$ = average reservoir pressure, psia;

U_w = water viscosity, cp;

r_e = external radius, ft; and

r_w = wellbore radius, ft.

The cumulative water injection, W_i, can be expressed as:

$$W_i = \int_0^t q_w dt \qquad 90$$

Module 8- Waterflooding

Injection Wells

Substituting Eq. 89 into Eq. 90

$$\int_0^t (p_{inj} - \overline{p})\, dt = \left(\frac{141.2\mu_w \ln\left(\frac{r_e}{r_w}\right)}{k_w h} \right) W_i \qquad 91$$

By integration, eq. 91 becomes Eq. 92:

$$\sum \Delta(p_{inj} - \overline{p}) \times \Delta t = \left(\frac{141.2\mu_w \ln\left(\frac{r_e}{r_w}\right)}{k_w h} \right) W_i \qquad 92$$

If it is assumed that all terms on the RHS of Eq.92 are constant except W_i, then Eq 92 can be written in an equivalent form.

$$\sum \Delta(p_{inj} - \overline{p}) \times \Delta t = m_H W_i, \quad \text{where} \quad m_H = \left(\frac{141.2\mu_w \ln(r_e/r_w)}{k_w h} \right) \qquad 93$$

Eq. 93 can be used for Hall Plot as shown previous.

Module 8- Waterflooding

Injection Well Class Exercises

- Prepare a Hall plot from the data for an injection well as shown in the Table below. Determine the condition of the well from Hall plot. Estimate permeability of the well. Other well and reservoir data are as follows:

Formation thickness, h	50ft,
Reservoir Radius, r_e	1000ft,
Well radius, r_w	0.5ft
Water viscosity, U_w	1.15cp

N	Δt (days)	Injection Pressure, p_{inj} (psia)	Average Pressure, $\bar{p}$ (psia)	Injection volume, (bbls)
1	25	4260	3205	25,000
2	20	4295	3215	19,800
3	30	4285	3210	29,250
4	22	4279	3211	22,220
5	28	4295	3225	27,500
6	30	4305	3230	25,500
7	31	4322	3232	24,025
8	29	4328	3233	22,765
9	24	4343	3227	19,200
10	27	4338	3229	21,465
11	30	4353	3234	23,400
12	21	4360	3236	16,695

Module 8- Waterflooding

Injection Water

- The source and quality of injection water with its sufficient is a component in the management of a waterflood project.

- The water analysis on the compatibility of source water with reservoir formation water and swelling clays or other minerals in the reservoir rock, the type of injection facility (open or closed), and the extent of water treatment required before the water can be used safely for injection.

- Extensive water treatment should be avoided because it could be expensive, and adversely impact the economics of the project.

Module 8- Waterflooding

Injection Water - Effects of untreated Injection Water

- Corrosion
  - Dissolved gases in the form of oxygen, carbon dioxide, and hydrogen sulfide
  - Precipitate ferrous hydroxide if injection water contains iron.

- Solid precipitation
  - Presence of iron, manganese, and barium in the injection water are likely to precipitate barium sulfate.
  - Barium sulfate can cause severe plugging of production wells and production facilities

- Promote growth of bacteria:
  - Iron bacteria and sulfate reducing bacteria.
  - Controlled by treating injection water with chlorine or other bactericides
  - The iron bacteria cause precipitation of iron and manganese hydroxides which can plug the formation.
  - The sulfate reducing bacteria react with sulfate ion to create iron sulfide which can also plug the formation.
  - Hydrogen sulfide is a by-product, very corrosive and toxic substance that can pose serious problems.

Module 8- Waterflooding

Production Wells

- Each production well should be tested frequently to measure fluid production rates.
- Typical graphs commonly used are plots of fluid production rates versus time, percent oil cut versus cumulative oil produced, or water-oil ratio versus cumulative oil produced.
- If the production well appears to be damaged, a pressure build-up test should be conducted on the well to determine the extent of damage.
- Reduction in well productivity can be commonly caused by the deposition of scale on the sandface and in the production string.
- Some types of scale can be removed by simulation of the wells with acid.
- Profile (fluid entry) logs should be run periodically on production wells.

Module 8- Waterflooding

Production Water

- Produced water analysis is a means of monitoring the progress of the waterflood, and tracking potential problems with solids deposition caused by mixing of source water and formation water.

- At early stages of the waterflood project, produced water consists mostly of formation water.

- Later, as production wells are affected by water breakthrough, produced water will contain increasing amounts of injected water.

- The proportion of injected water in produced water is sometimes used as an indicator in monitoring the location of the floodfront (Tracer can be used for the purpose).

Module 8- Waterflooding

Facilities

- Should be able to withstand potential threat from massive corrosion caused by the water.

- Protection of injection and production facilities from corrosion begins with careful water analysis, and selection of equipment with proper metallurgy.

- The corrosion engineers should routinely monitor all facilities used in the project for corrosion, and implement preventative measures to forestall onset of corrosion in these facilities.

Module 8- Waterflooding

Management of Waterflooded Reservoirs

- The reservoir should be monitored regularly to
  - Determine the location of the floodfront,
  - Evaluate the distributions of reservoir pressure and fluid saturations, and
  - exploit opportunities for re-alignment of injectors and producers to improve sweep efficiency and hydrocarbon recovery.

Module 9 – Enhanced Oil Recovery

Enhanced Oil Recovery - Introduction

- Enhanced oil recovery (EOR) is sometimes called tertiary oil recovery or improved oil recovery.

- It defines all processes that increase oil recovery beyond primary or secondary recovery processes.

- A typical EOR process involves injecting a fluid other than water into the reservoir and it is very sensitive to oil prices.

- The price of oil on a sustainable basis must exceed the cost of the injectant plus operating costs by a sizeable margin for an EOR process to be considered economical.

- An EOR process must be efficient in terms of cost per barrel of oil recovered and also effective in substantially increasing the volume of oil recovered beyond the current recovery process.

- Economic evaluation is the key important step in the selection of an EOR process and is emphasized throughout the selection process.

- Roughly 65% of the original oil in place (OOIP) is estimated to remain in the reservoir after primary and secondary recoveries.

Module 9 – Enhanced Oil Recovery

EOR Processes

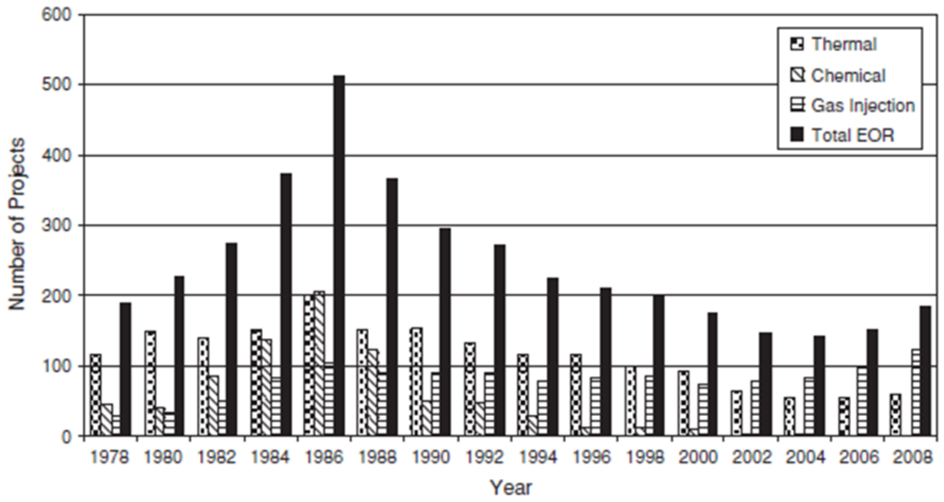
EOR processes can be classified under three main groups.

- Miscible Gas Injection Processes
  - Nitrogen injection
  - Hydrocarbon (HC) gas injection
  - Carbon dioxide (CO2) injection
  - Sour gas, flue gas, etc. injection
- Chemical Processes
  - Polymer flooding
  - Polymer/surfactant flooding
  - Alkali-Surfactant-Polymer (ASP) flooding
  - Microbial
- Thermal Processes
  - In-Situ Combustion (ISC) or High Pressure Air Injection (HPAI)
  - Steam/Hot water injection
  - Steam Assisted Gravity Drainage (SAGD)

Module 9 – Enhanced Oil Recovery

USA STATISTICS ON EOR PROJECTS

- This chart shows the total number of EOR projects in the United States of America from 1978 to 2008.
- From 1978 to 1986, the number of thermal and chemical projects rose each year, but has been on decline since 1988.
- No active commercial chemical project in the United States since 2006.
- Gas injection projects have been on a relatively steady growth in the United States since 1978.
- Gas injection projects have surpassed both thermal and chemical projects as EOR processes in the United States since 2002.
- It is projected that gas injection as an EOR process will continue to grow in the United States in future years because it can be applied to a wide range of reservoirs with favourable economic outcomes.

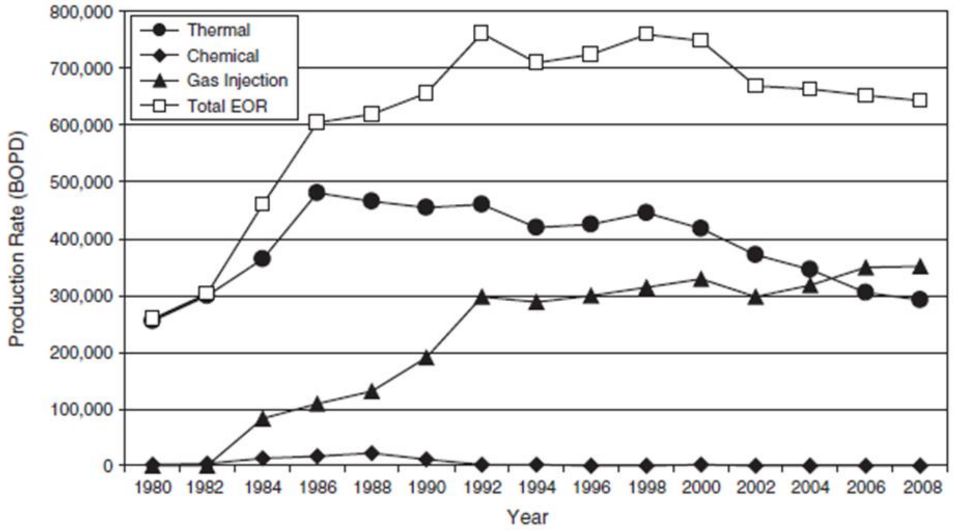


Number of EOR projects in the United States from 1978 to 2008.

Module 9 – Enhanced Oil Recovery

USA STATISTICS ON OIL PRODUCTION FROM EOR PROJECTS

- The oil production associated with the three EOR processes, including the total EOR production in the United States from 1980 to 2008 are shown in this chart.
- Thermal processes produced more oil in the United States than any other EOR process from 1980 to 2004.
- However from 2006, gas injection projects produced more oil than thermal projects.
- In the United States, the contribution to oil production from chemical projects is very limited.



Oil production from EOR projects in United States from 1978 to 2008.

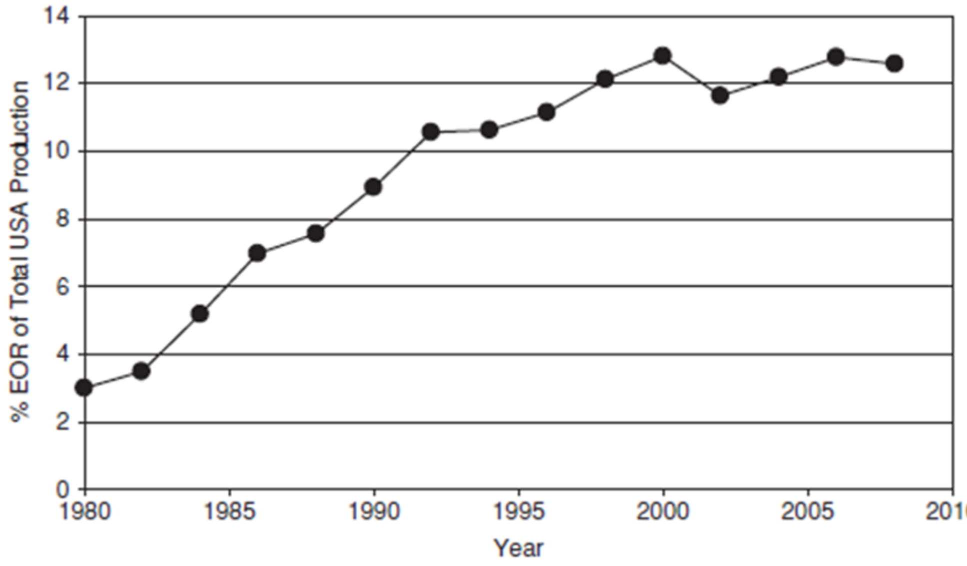
Module 9 – Enhanced Oil Recovery

USA STATISTICS ON THE PERCENTAGE EOR OIL PRODUCTION

- This chart shows the percentage contribution of EOR projects to total oil production in the United States since 1980.

- The percentage of total United States oil production due to EOR projects has increased since 1980, and appears to have stabilized at around 12.4% since 1998.

- Oil prices have substantial influence on the initiation and execution of EOR projects.

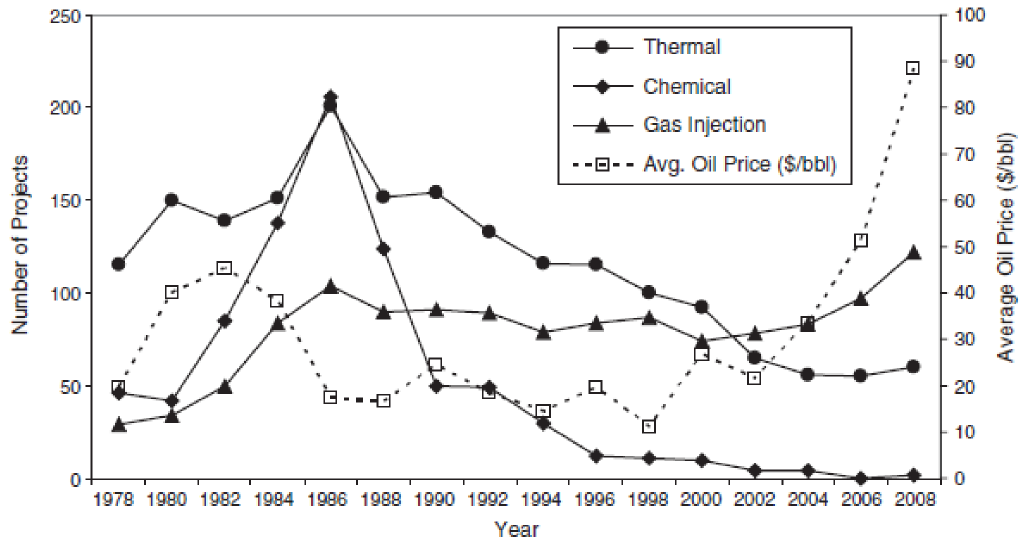


Percentage EOR oil Production in United States from 1980 to 2010.

Module 9 – Enhanced Oil Recovery

EOR PROJECTS VERSUS OIL PRICES

- This chart shows the variation of average oil prices with the number of active EOR projects in the United States from 1978 to 2008.
- All EOR projects increased between 1978 and 1986 when there was an increase in average oil prices in the United States.
- The numbers of EOR projects were flat or decreasing between 1988 and 2000 when average oil prices were relatively stable.
- Since 2000, gas injection projects have increased in the United States with increase in average oil price.
- During this period, the number of thermal projects has stabilized while the number of chemical projects has declined almost to zero.



EOR projects versus oil prices in the United States from 1978 to 2008

Module 9 – Enhanced Oil Recovery

Other Countries EOR Prospects

- Canada has substantial oil production from gas injection and thermal projects.
- Indonesia has the single largest EOR project in the world, with production at 190,000 BOPD in 2008 from steam injection in the Duri field.
- China has numerous EOR projects consisting mainly of chemical and thermal processes.
- Other countries that have many EOR projects include Brazil, Oman, Mexico, Trinidad, and Venezuela.

Module 9 – Enhanced Oil Recovery

EOR Screening Criteria

- The application of EOR processes are both reservoir-specific and reservoir fluid-specific.
- This literally means that each EOR process must be specifically evaluated before it can be applied to a reservoir.
- The evaluation process is typically extensive and may include laboratory work, geologic and reservoir modelling, economic analyses, and in many cases field trial in the form of a pilot test.
- The selection criteria presented here are meant to serve as the first-pass screening procedures that compare the candidate reservoir with other reservoirs that have been produced with an EOR process.
- They should not replace the rigorous evaluation procedure that each EOR process must undergo before it is actually implemented in the field.

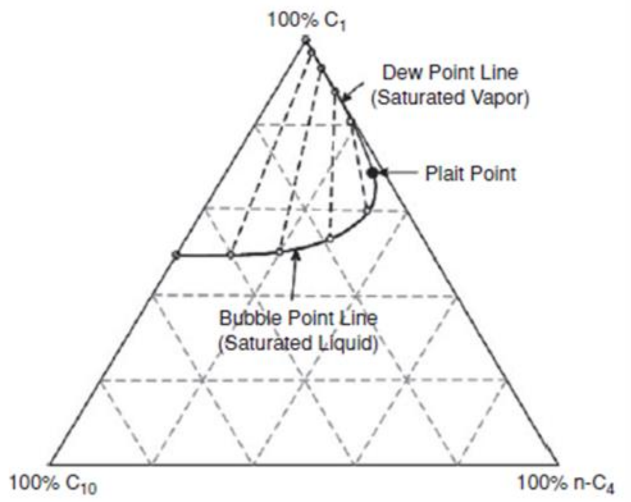
Module 9 – Enhanced Oil Recovery

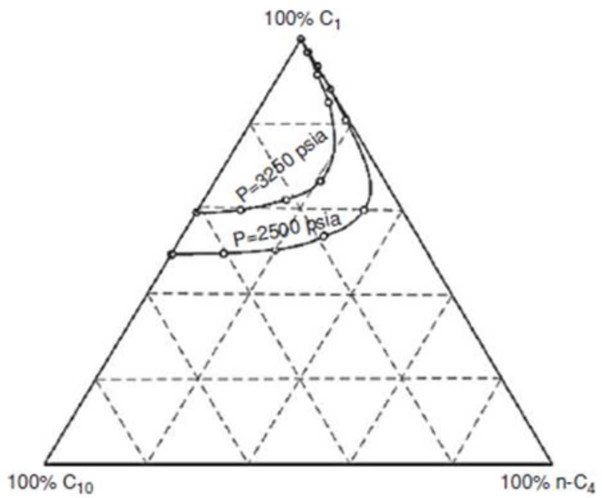
EOR Screening Criteria

EOR Type	Fluid Properties			Reservoir Properties				
	Gravity (API)	Viscosity (cp)	Temp. (F)	Porosity (%)	Perm (md)	Oil Sat. at start (%PV)	Lithology	Depth (ft)
Nitrogen/flue gas	>30	<0.5	>250	>10%	>30	>50	Carbonate or Sandstone	>7000
Hydrocarbon Gas (HC)	21-57	0.1-1.3	136-290	4-26	10-5000	30-98	Carbonate or Sandstone	4000 – 14500
Carbon Dioxide (CO_2)	28-44	0.4-3.0	100-250	4-26	2-500	25-90	Carbonate or Sandstone	2000-12000
Polymer/ Surfactant Flooding	14-34	5-80	80 – 160	20 -30	170 – 900	60-75	Sandstone	1300 – 4600
Alkali/Surfactant	14-34	5-80	80 – 160	20 -30	170 – 900	60-75	Sandstone	1300 – 4600
Polymer Flooding	14-34	5-80	80 – 160	20 -30	170 – 900	60-75	Sandstone	1300 – 4600
Microbial (MEOR)*	-	5 – 50	<176	>=20	>50	-	Carbonate or Sandstone	<7700
ISC or HPAI	19-33	2-660	110 -230	17-32	10-1265	50-94	Carbonate or Sandstone	400 -8300
Steam	8 -30	50 – 500,000	45-290	15-65	100-10000	44-90	Carbonate or Tripolite	200 – 3600

Module 9 – Enhanced Oil Recovery

Miscible Gas Injection Processes

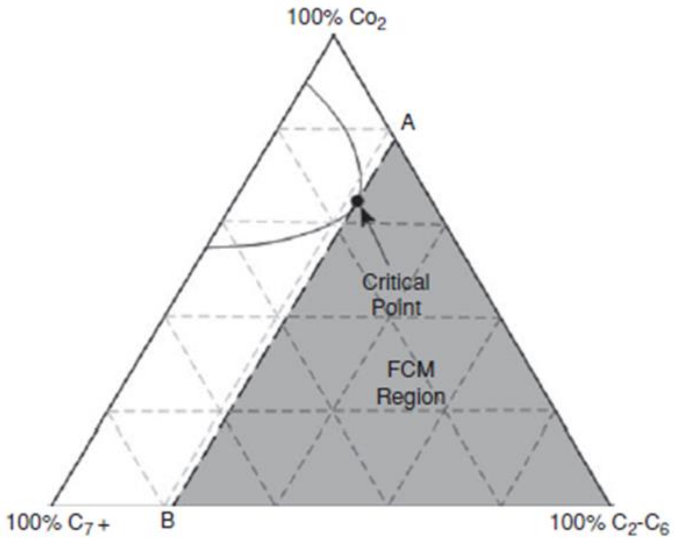




- The concept of miscibility in a gas displacement process can be demonstrated with a ternary-phase diagram.
- A ternary-phase diagram for the methane/n-butane/decane system at 2500 psia and 160^{o}F.
- This system has a two-phase region which is enclosed by a bubble point line and a dew point line that meet at the plait (or critical) point.
- At 2500 psia and 160^{o}F, mixtures of this system with compositions within the two-phase envelope are immiscible, while mixtures of the system with composition outside the two-phase envelope are miscible.
- The relative size of the two-phase envelope depends on temperature and pressure of the system.
- For instance, at constant temperature of 160^{o}F, increasing the pressure of the system from 2500 to 3250 psia will shrink the two-phase envelope as shown in the second diagram.
- This also means that at higher pressures more compositions of the methane/n-butane/decane system are miscible at 160 ^{o}F.

Module 9 – Enhanced Oil Recovery

First-Contact Miscibility (FCM)



Pseudoternary diagram for CO_2-hydrocarbon system.

- The phenomenon of first-contact miscibility (FCM) occurs when the injected fluid is completely miscible with the reservoir fluid under the existing reservoir pressure and temperature.
- Under FCM, the injected fluid completely mixes with the reservoir fluid to create a transition fluid that miscibly displaces the reservoir fluid.
- FCM can be considered as an instantaneous process of complete mixing of the injected fluid with the reservoir fluid.
- The factors that determine FCM are reservoir temperature and pressure, composition of the injected fluid, and composition of the reservoir fluid.
- Line AB is the critical tie line drawn through the critical point of the phase envelope.
- As a simple illustration of FCM, pure CO2 will achieve FCM with reservoir fluid in the shaded region

Module 9 – Enhanced Oil Recovery

Multiple-Contact Miscibility (MCM)

- Multiple-contact miscibility (MCM) is achieved as the result of repeated contacts in the reservoir between the reservoir oil and the injected fluid, and also fluids generated in-situ by interactions between the injected fluid and the reservoir oil.

- Two Main Processes
  - Vaporizing gas Drive
    - In the vaporizing gas drive, the composition of the injected fluid is progressively modified by vaporization of the intermediate hydrocarbon components (C2 to C6) in the reservoir oil until the modified fluid becomes ultimately miscible with the original reservoir oil.

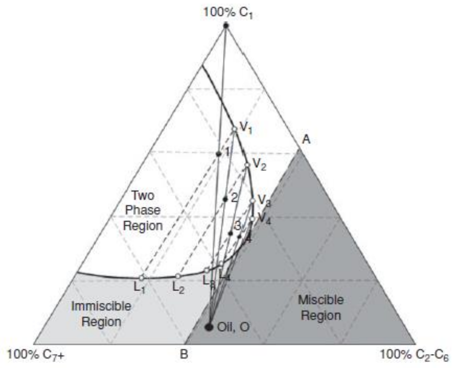
  - Condensing Gas drive
    - In the condensing gas drive, intermediate hydrocarbon components in the injection fluid condense in the reservoir oil to generate a modified fluid that becomes miscible with the injection fluid.

  - Combined condensing and vaporizing (CV) gas drives.

- The type of MCM process that is dominant in a displacement drive depends on the composition of the reservoir oil, the composition of the injection fluid, and the temperature and pressure of the reservoir.

Module 9 – Enhanced Oil Recovery

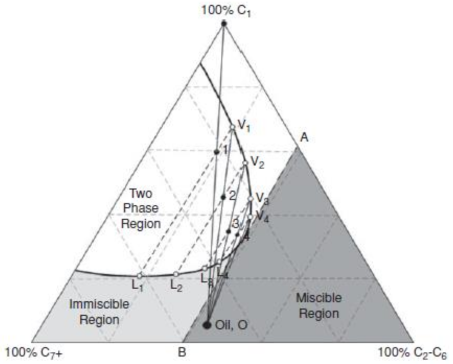
Vaporizing Gas Drive MCM Process



- The mechanism for achieving MCM by vaporizing gas drive is best illustrated with a pseudoternary diagram.
- The pseudoternary diagram is simplified to represent a system composed of methane (C1)/ethane-hexane (C2-C6)/Heptane-plus (C7+).
- Suppose a lean gas composed entirely of methane is injected into a reservoir containing oil with composition represented at point O
- The methane gas will vaporize the intermediate components of the oil to form a fluid with composition represented at point 1 in the two-phase region.
- This fluid will consists of a vapor phase shown as V1, and a liquid phase shown as L1, on the phase envelope.
- As the displacement drive progresses, the vapor phase, V1, advances to contact additional original reservoir oil, and again vaporize more intermediate components to form another fluid with composition represented at point 2 within the two-phase envelope.
- This fluid separates into a vapor phase, V2, and a liquid phase, L2. The vapor phase, V2, advances to contact more original oil to form a third fluid with composition represented at point 3 within the two-phase envelope.
- The vapor phase, V3, from the fluid at point 3 contacts more original oil to form a fluid represented at point 4, whose vapor phase, V4, achieves miscibility with original oil.

Module 9 – Enhanced Oil Recovery

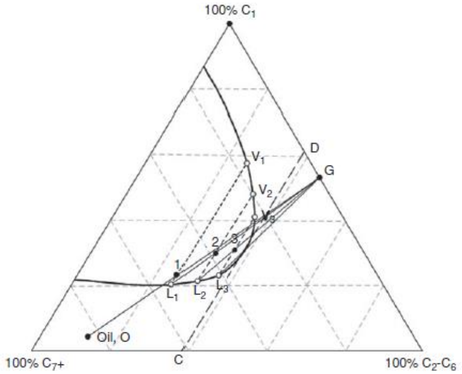
Vaporizing Gas Drive MCM Process



- The vaporizing gas drive MCM process is a mechanism whereby the vapor phase of the displacing fluid is progressively enriched through vaporization of the intermediate components in the original reservoir oil by repeated contacts until a fluid miscible with the original reservoir oil is achieved.
- Note that reservoir temperature and pressure are assumed to be constant, and local thermodynamic equilibrium is assumed to have been achieved between the intermediate phases during this process.
- The line AB drawn through the critical point on the two-phase envelope is called the critical tie-line.
- For the fluid system at fixed conditions represented in Figure, reservoir oil compositions on the critical tie-line or to the right of the critical tie-line are miscible with injection gas composition to the left of point A on the critical tie-line.
- The figure shows that reservoir oil composition to the left of the critical tie-line will be immiscible with injection gas composition to the left of point A on the critical tie-line.
- Note that these descriptive representations on the pseudoternary diagram are applicable only to vaporizing gas drive processes.
- The critical tie-line can be used also to define the minimum miscibility pressure (MMP) for a fluid system at fixed conditions. The MMP is defined as the minimum pressure at which the reservoir oil composition lies on a critical tie-line extension for a specified fluid system.
- The MMP represents the minimum pressure at which MCM can be achieved in the reservoir for the fluid system. At pressures less than MMP, the displacement process will be immiscible.

Module 9 – Enhanced Oil Recovery

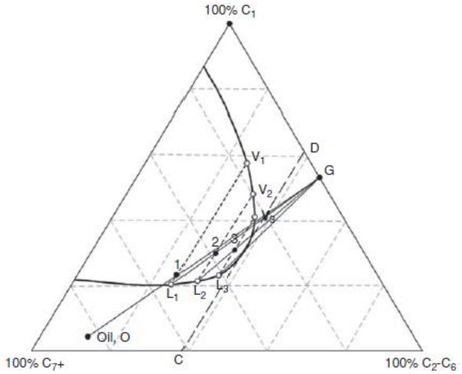
Condensing Gas Drive MCM Process



- The mechanism of attaining MCM by condensing gas drive is illustrated with a pseudoternary diagram.
- The injected fluid is represented by point G, and the reservoir oil is shown as point O. To initiate the multiple contact process, the intermediate components in the enriched injection gas will condense in the reservoir oil to form a fluid within the two-phase envelope with composition represented at point 1.
- This fluid is composed of a vapor phase, V1, and a liquid phase, L1. As the displacement advances, the liquid phase, L1, is contacted by more injection gas to form a fluid with composition shown at point 2.
- This fluid is composed of a vapor phase, V2, and a liquid phase, L2. The liquid phase, L2, is contacted by more injection gas to form a fluid shown at point 3 composed of a vapor phase, V3, and liquid phase, L3.
- Further contact between the liquid, L3, and the injection gas produces a fluid which is ultimately miscible with the injection gas.
- The condensing gas drive MCM process can be described as the mechanism of progressive enrichment of the condensed liquid phase with intermediate hydrocarbons in the injection gas resulting from multiple contacts that ultimately produces a fluid that is miscible with the injection gas.
- Note that this description of the condensing gas drive MCM process assumes that reservoir conditions are fixed, and local thermodynamic equilibrium is attained between the intermediate liquid and vapor phases.

Module 9 – Enhanced Oil Recovery

Condensing Gas Drive MCM Process



- In Figure, the critical tie-line is shown as line CD. Any injection fluid composition to the left of point D will not achieve miscibility with the oil composition to the left of point C.
- The oil composition to the left of point C can achieve MCM with injection fluid composition at point D or to the right of point D. For the condensing gas drive, the MMP is defined as the minimum pressure at which the injection fluid composition lies on the extension of a critical tie-line.
- For the condensing gas drive, the injection fluid composition must lie on the critical tie-line or to the right of the critical tie-line and the oil composition must lie on the opposite (left) side of the critical tie-line as shown in Figure for MCM to be achieved.
- Another important parameter that can be used to define the condensing gas MCM process is the minimum miscibility enrichment (MME).
- This describes the process of enriching the composition of the injection gas with intermediate hydrocarbons (C2 to C6) to achieve miscibility with the reservoir oil.
- The MME at a given pressure is defined as the minimum enrichment of the injection gas composition, such that the gas composition lies on an extension of a critical tie-line.
- Thus for a condensing gas drive process, miscibility can be achieved either by increasing pressure or by enriching the composition of the injection fluid with intermediate hydrocarbons.

Module 9 – Enhanced Oil Recovery

Combined Condensing/Vaporizing (CV) Gas Drive MCM Process

- Zick reported that a combined condensing/vaporizing gas drive mechanism rather than the traditional condensing gas drive alone is the key mechanism in the displacement of real reservoir oil by enriched fluids. This observation was supported by the work done by Stalkup.

- The importance of the observation is that MMPs and MMEs estimated by pseudoternary methods are different from MMPs and MMEs observed for combined CV drives.

- This development should be expected since pseudoternary diagrams are inadequate for representation of a complex fluid system, such as reservoir hydrocarbon systems.

Module 9 – Enhanced Oil Recovery

Methods for Determination of MMP or MME for Gasfloods

- The two main techniques used in the determination of MMP or MME can be broadly classified as analytical techniques and experimental methods.
- Analytical techniques, which are based on numerical computations, can be subdivided further into empirical correlations and compositional simulations.
  - Empirical correlations are based on fitting equations to MMP data obtained from experimental methods, and/or computed from compositional simulations.
  - Compositional simulations methods generally use equations of state to predict MMP or MME from the compositions of the injection gas and the reservoir oil.
- The three common analytical techniques
  - the tie-line method,
  - the 1-D slimtube simulation, and
  - the mixing cell method.
- The two most common experimental methods
  - the slimtube method and the rising
  - bubble apparatus method.

Module 9 – Enhanced Oil Recovery

Empirical Correlations for MMP (Hydrocarbon Injection Gases)

Glaso presented set of equations for the prediction of MMPs.

$$MMP_{x=34} = 6329.0 - 25.410y - z(46.745 - 0.185y) + T\left(1.127 \times 10^{-12} y^{5.258} e^{319.8zy^{-1.703}}\right) \quad 94$$

$$MMP_{x=44} = 5503.0 - 19.238y - z(80.913 - 0.273y) + T\left(1.700 \times 10^{-9} y^{3.730} e^{13.567zy^{-1.058}}\right) \quad 95$$

$$MMP_{x=54} = 7437.0 - 25.703y - z(73.515 - 0.214y) + T\left(4.920 \times 10^{-14} y^{5.520} e^{21.706zy^{-1.109}}\right) \quad 96$$

Where MMP = Minimum miscibility pressure, psig

X = Molecular weight of C_2 through C_6 in the injection gas, lbm/lb-mole

Y = Corrected molecular weight of C_{7+}= $(2.622/\gamma_{o,C7+}{}^{-0.846})^{6.588}$

$\gamma_{o,C7+}$ = the specific gravity of stock tank oil

Z = Methane in injection gas , mole percent

Module 9 – Enhanced Oil Recovery

Empirical Correlations for MMP Exercises 1

- Calculate the MMP for a reservoir oil and hydrocarbon gas mixture with the following properties:

Molecular weight of C2 through C6 in injection gas	32lbm/lb-mole
Mole percent of methane in injection gas	65.2
Corrected molecular weight of C7+ in stock-tank Oil	224 lbm/lb-mole
Reservoir Temperature	160^{o}F

- Compare your answer with the experimental MMP reported in the literature for this system by Glasø is 3700 psig.

Module 9 – Enhanced Oil Recovery

Empirical Correlations for MMP (Nitrogen Injection Gases)

Glaso presented set of equations for the prediction of MMPs for Nitrogen gas.

For molecular weight of C7+ > 160 and mole percent of intermediates > 28, use eq. 97 below

$$MMP = 6364.0 - 12.090M_{C_{7+}} + T\left(1.127 \times 10^{-12} M_{C_{7+}}^{5.258} e^{23,025.0 M_{C_{7+}}^{-1.703}} - 20.80\right) \quad 97$$

For molecular weight of C7+ < 160 and mole percent of intermediates > 28, use eq. 98 below

$$MMP = 7695.1 - 12.090M_{C_{7+}} + T\left(1.127 \times 10^{-12} M_{C_{7+}}^{5.258} e^{23,025.0 M_{C_{7+}}^{-1.703}} - 39.77\right) \quad 98$$

If the mole percent of intermediates < 28, use eq. 99 below

$$MMP = 9364.0 - 12.090M_{C_{7+}} + T\left(1.127 \times 10^{-12} M_{C_{7+}}^{5.258} e^{23,025.0 M_{C_{7+}}^{-1.703}} - 20.80\right) - 99.3C_{2-6} \quad 99$$

Module 9 – Enhanced Oil Recovery

Empirical Correlations for MMP (Nitrogen Injection Gases)

The Firoozabadi and Aziz correlation for prediction of MMP for nitrogen or lean gas injection is:

$$MMP = 9433 - 188 \times 10^3\left(\frac{C_{2-5}}{M_{C_{7+}}T^{0.25}}\right) + 1430 \times 10^3\left(\frac{C_{2-5}}{M_{C_{7+}}T^{0.25}}\right)^2 \qquad 100$$

Where $C_{2\text{-}6}$ = mole percent of the intermediates (C2 through C6) in the reservoir oil.

$C_{2\text{-}6}$ = mole percent of C_2 through C_5 including CO2 and H2S in the reservoir fluid.

Module 9 – Enhanced Oil Recovery

Empirical Correlations for MMP Exercises 2

- Calculate the MMP for a reservoir oil and Nitrogen gas injection mixture with the following properties:

Molecular weight of C7+ in Reservoir oil	193.3lbm/lb-mole
Mole percent of C2 through C5 in reservoir	25.17
Reservoir Temperature	160^{o}F
Bubble point pressure of reservoir oil	4000psi

- Use the Glasø and Firoozabadi and Aziz Correlations for Nitrogen Gas Injection.
- Compare your answers with the reported experimental MMP for this system is 4280 psi.

Module 9 – Enhanced Oil Recovery

Empirical Correlations for MMP (Pure Carbon Dioxide Injection Gases)

Glaso presented set of equations for the prediction of MMPs for pure CO2-Oil Systems;

For mole composition of C_2 through C_6> 18%, use eq. 101 below

$$MMP_{pure} = 810.0 - 3.404M_{C_{7+}} + T\left(1.700 \times 10^{-9}M_{C_{7+}}^{3.730}e^{786.8M_{C_{7+}}^{-1.058}}\right) \quad (101)$$

For mole composition of C_2 through C_6< 18%, use eq. 102 below

$$MMP_{pure} = 2947.9 - 3.404M_{C_{7+}} + T\left(1.700 \times 10^{-9}M_{C_{7+}}^{3.730}e^{786.8M_{C_{7+}}^{-1.058}}\right) - 121.2C_{2-6} \quad (102)$$

The Emera presented correlation for pure CO2 injection

$$MMP_{pure} = 5.0093 \times 10^{-5} \times (1.8T + 32)^{1.164} \times M_{C5+}^{1.2785} \times \left(\frac{C_{C_1+N_2}}{C_{C_{2-4}+H_2S+CO_2}}\right)^{0.1073} \quad (103)$$

Module 9 – Enhanced Oil Recovery

Empirical Correlations for MMP (Pure Carbon Dioxide Injection Gases)

If the bubble point pressure (P_b) of the oil is less than 50psi, **Emera correlation** becomes:

$$MMP_{pure} = 5.0093 \times 10^{-5} \times (1.8T + 32)^{1.164} \times M_{C_{5+}}^{1.2785} \qquad 104$$

The Yuan et al.17 correlation is based on analytical MMPs for pure CO_2

$$MMP_{pure} = a_1 + a_2 M_{C_{7+}} + a_3 P_{C_{2-6}} + \left(a_4 + a_5 M_{C_{7+}} + a_6 \frac{P_{C_{2-6}}}{M^2_{C_{7+}}}\right)T + \left(a_7 + a_8 M_{C_{7+}} + a_9 M^2_{C_{7+}} + a_{10} P_{C_{2-6}}\right)T^2 \qquad 105$$

Where

The coefficients in Eq. 105 are: $a_1 = -1.4364\text{E} + 03$; $a_2 = 0.6612\text{ E} + 01$; $a_3 = -4.4979\text{ E} + 01$; $a_4 = 0.2139\text{ E} + 01$; $a_5 = 1.1667\text{ E} - 01$; $a_6 = 8.1661\text{ E} + 03$; $a_7 = -1.2258\text{ E} - 01$; $a_8 = 1.2883\text{ E} - 03$; $a_9 = -4.0152\text{ E} - 06$; $a_{10} = -9.2577\text{ E} - 04$.

Module 9 – Enhanced Oil Recovery

Empirical Correlations for MMP Exercises 3

- Calculate the MMP for a reservoir oil and Nitrogen gas injection mixture with the following properties:

Molecular weight of C7+ in Reservoir oil	234lbm/lb-mole
Molecular weight of C5+ in Reservoir oil	185lbm/lb-mole
Mole percent of C1 in reservoir oil	4.4
Mole percent of C2 through C6 in reservoir	24.0
Reservoir Temperature	180^oF

- Use the Glasø, Emera and Yuan Correlations to calculate MMP for pure CO_2 Gas Injection.
- Compare your answers with the reported experimental MMP for this system is 3250 psi.

Module 9 – Enhanced Oil Recovery

Empirical Correlations for MMP (Impure Carbon Dioxide Injection Gases)

Glaso presented set of equations for the prediction of MMPs for pure CO2-Oil Systems;

For mole composition of C_2 through $C_6 > 18\%$, use eq. 101 below

$$MMP_{pure} = 810.0 - 3.404 M_{C_{7+}} + T\left(1.700 \times 10^{-9} M_{C_{7+}}^{3.730} e^{786.8 M_{C_{7+}}^{-1.058}}\right) \qquad 101$$

For mole composition of C_2 through $C_6 < 18\%$, use eq. 102 below

$$MMP_{pure} = 2947.9 - 3.404 M_{C_{7+}} + T\left(1.700 \times 10^{-9} M_{C_{7+}}^{3.730} e^{786.8 M_{C_{7+}}^{-1.058}}\right) - 121.2 C_{2-6} \qquad 102$$

The Emera presented correlation for pure CO2 injection

$$MMP_{pure} = 5.0093 \times 10^{-5} \times (1.8T + 32)^{1.164} \times M_{C_{5+}}^{1.2785} \times \left(\frac{C_{C_1 + N_2}}{C_{C_{2-4}} + H_2S + CO_2}\right)^{0.1073} \qquad 103$$

If the bubble point pressure (P_b) of the oil is less than 50psi, Emera correlation becomes:

$$MMP_{pure} = 5.0093 \times 10^{-5} \times (1.8T + 32)^{1.164} \times M_{C_{5+}}^{1.2785} \qquad 104$$

The Yuan et al.17 correlation is based on analytical MMPs for pure CO_2

$$MMP_{pure} = a_1 + a_2 M_{C_{7+}} + a_3 P_{C_{2-6}} + \left(a_4 + a_5 M_{C_{7+}} + a_6 \frac{P_{C_{2-6}}}{M_{C_{7+}}^2}\right)T + \left(a_7 + a_8 M_{C_{7+}} + a_9 M_{C_{7+}}^2 + a_{10} P_{C_{2-6}}\right)T^2 \qquad 105$$

Where The coefficients in Eq. 105 are: $a_1 = -1.4364\text{E}+03$; $a_2 = 0.6612\text{E}+01$; $a_3 = -4.4979\text{E}+01$; $a_4 = 0.2139\text{E}+01$; $a_5 = 1.1667\text{E}-01$; $a_6 = 8.1661\text{E}+03$; $a_7 = -1.2258\text{E}-01$; $a_8 = 1.2883\text{E}-03$; $a_9 = -4.0152\text{E}-06$; $a_{10} = -9.2577\text{E}-04$.

Module 9 – Enhanced Oil Recovery

Chemical Flooding Processes

- They are displacement processes that improve oil recovery from reservoirs with the aid of chemicals that are generally called surfactants, polymers, and alkalis eg polymer/surfactant (micellar) process, and alkali/surfactant/polymer (ASP) process.

- Only polymers are injected for mobility control and improved oil recovery in polymer flooding.

- Chemical flooding processes improve oil recovery
  - by lowering the interfacial tension (IFT) between oil and water,
  - solubilization of the oil in the micelles,
  - emulsification of the oil and water,
  - alteration of the wettability of the rock, and
  - enhancement of the mobility of the displacing fluids.

Module 9 – Enhanced Oil Recovery

Chemical Flooding Processes

- For each application of chemical flooding, the chemicals must be formulated and tailored to the properties of the reservoir rock and its fluid system.
- This usually requires extensive testing of various combinations of the chemicals in laboratory coreflood tests followed with a pilot test in the field.
- There are few field-wide chemical flooding projects in the world in comparison with either CO2 or HC gas injections projects. The decline of chemical flooding can be attributed to the high cost of the chemicals and higher risk associated with achieving predicted oil recoveries with field-wide projects.
- The chemical flooding processes that have been applied in the field are surfactant/polymer floods, ASP floods, and polymer floods. In some fields, microbial flooding has been tested.
- These chemical flooding processes are discussed in more detail in the following sections.

Module 9 – Enhanced Oil Recovery

Types of Chemical Flooding Processes

- Polymer/Surfactant Flooding
- Alkali/Surfactant/Polymer (ASP) Flooding
- Polymer Flooding
- Microbial Enhanced Oil Recovery (MEOR)

Module 9 – Enhanced Oil Recovery

Polymer/Surfactant Flooding

- Also called micellar/polymer flood which is a slug of micellar solution (consisting of water, surfactant, electrolytes, etc.) is injected into the reservoir.
- The size of the micellar slug can vary from 3% to 30% of the flood pattern pore volume (PV) depending on the flood design.
- The micellar slug is followed with a mobility buffer slug made up of polymer and water for mobility control. The entire micellar/polymer slug is then chased with drive water.
- The polymer solution sometimes contains a biocide to control microorganisms that can degrade the polymer and reduce the viscosity of the solution.
- The micellar slug is formulated such that a favorable mobility ratio exists between the slug and the oil bank that is formed by the flooding process.

Module 9 – Enhanced Oil Recovery

Polymer/Surfactant Flooding

- This sometimes requires the addition of small amounts of polymers in the composition of the micellar solution.

- The micellar slug is usually the most expensive solution used in micellar/polymer flooding.

- The surfactants in the micellar slug are degraded with the advancement of the flood due to adsorption on rock surfaces and trapping of some of the micellar solution as a residual phase.

- Adsorption of surfactants on rock surfaces can be reduced by preflushing the flood zone with fresh water. Preflushing with fresh water also reduces incompatibilities between the surfactant and formation brine.

- The application of micellar/polymer floods have declined because of the high cost of micellar solution and limited economic success of the process in field tests.

Module 9 – Enhanced Oil Recovery

Alkali/Surfactant/Polymer (ASP) Flooding

- Alkali/surfactant/polymer (ASP) flooding is more widely used than micellar/polymer flooding in field applications.
- ASP formulation utilizes the favorable mechanisms of its key components (alkali, surfactant and polymer) to improve oil recovery.
- The main functions of the alkaline component are to promote emulsification of the crude oil, reduce IFT, reduce adsorption of the surfactants, and regulate phase behavior of the mixtures.
- The main function of the surfactants is to reduce IFT between the oil and the injected slugs.
- The polymers are used for mobility control and improvement of sweep efficiencies.
- The favorable attributes of ASP formulations have been tested and proven in many laboratory and field tests.
- ASP flooding has been applied in most cases to sandstone reservoirs but laboratory and pilot tests suggest that it can also be applied to carbonate reservoirs.

Module 9 – Enhanced Oil Recovery

Polymer Flooding

- Polymer flooding is the most widely used chemical flooding process for both sandstone and carbonate reservoirs.
- Polymer flooding can be considered as a waterflooding process in which polymers have been added to the solution to improve mobility control.
- Thus, polymer flooding is able to achieve higher oil recovery than ordinary waterflooding due to improved volumetric sweep efficiency resulting from better mobility control.
- Factors affecting the application of polymer flooding are the cost of the polymers, the permeability of the reservoir, and the temperature of the reservoir.
- The cost of polymers limits the concentration and slug size of the polymers that can be used to achieve higher oil recovery.
- Low rock permeability can limit the use of more viscous polymers to improve mobility control, and can cause low injectivity.
- Polymers are not stable at reservoir temperatures higher than 175 ^{o}F.
- Many field applications of polymer flooding have been reported as technical and economic success.

Module 9 – Enhanced Oil Recovery

Microbial Enhanced Oil Recovery (MEOR)

- Microbial activities in the reservoir are believed to increase oil recovery by altering the wettability of the rock, reducing interfacial tension, reducing oil viscosity, generation of gases such as carbon dioxide, production of surfactants, etc.
- These mechanisms, which combine to improve oil recovery by the MEOR process, are complex and not yet fully understood.
- Consequently, after many years of laboratory and pilot tests, MEOR processes have not been applied on a large scale in the field in comparison with other EOR processes.
- There are no active MEOR projects in the United States in 2008,5 and small-scale MEOR projects are in operation in China,5 and the North Sea.
- The screening criteria for microbial processes are presented in Table above.
- Generally, MEOR processes are applicable to reservoirs with injected and connate water salinities 150,000 ppm; oil viscosity in the 5–50 cp range; reservoir temperature 176 F; rock porosity 20%; rock permeability 50 md.; and reservoir depth 7700 ft.

Module 9 – Enhanced Oil Recovery

Microbial Enhanced Oil Recovery (MEOR)

- Most MEOR tests have been on sandstone reservoirs, although there have been tests in carbonate reservors. The bacteria source in MEOR processes can be indigenous or exogenous.
- Indigenous bacteria are already present in the reservoir while exogenous bacteria are cultivated and injected into the reservoir.
- Microbial processes in the reservoir can be aerobic or anaerobic.
- For aerobic processes, oxygen must be injected to maintain the required concentration of bacteria. Injection of oxygen into the wellbore may cause severe corrosion problems.
- Anaerobic processes require injection of large amounts of nutrients in the form of sugars to sustain the growth of bacteria.
- This will increase the cost of the MEOR process due to the additional costs of purchasing, processing, and injecting the nutrients.
- Large-scale field utilization of MEOR processes have not occurred in the industry because of limited knowledge of the process, and lack of models to predict potential oil recovery from application of the technology.

Module 9 – Enhanced Oil Recovery

Thermal Processes

- Thermal EOR processes are defined to include all processes that supply heat energy to the rocks and fluids contained in a reservoir thereby enhancing the ability of oil (including other fluids) to flow by primarily reducing its viscosity.
- The oil caused to flow by the supply of thermal energy is produced through nearby wells.
- The two main thermal processes that are presented in this section are steamflooding and in-situ combustion
- Steamflooding has achieved substantial commercial successes around the world and is ranked as one of the top EOR processes.
- In-situ combustion process, by comparison, has attained limited commercial success, and is actively applied in few reservoirs around the world.
- The main obstacle to the application of in-situ combustion technology is ability to manage and control the advancement and progression of the combustion process.
- Steamflooding has been used on highly viscous, shallow reservoirs at depths of 200 to 3600 feet while in-situ combustion can be applied to less viscous, deeper reservoirs at depths reaching 8300 feet.

Module 9 – Enhanced Oil Recovery

Steamflooding Methods

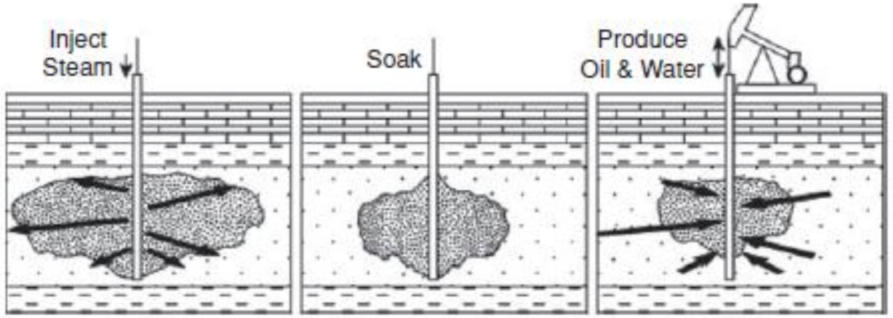
- Steamflooding is an established EOR technique that has been applied successfully on many heavy oil reservoirs around the world.
- The process started in early 1960 with cyclic steam injection in the Tia Juana Field in Venezuela, the Schoonebeek Field in the Netherlands, and the Yorba Linda Field in California.

- **Steamflooding Methods**
  - **Cyclic Steam Stimulation (CSS)**
  - **Steam Drive (SD)**
  - **Steam-Assisted Gravity Drainage (SAGD)**
  - **Water-Alternating-Steam Process (WASP)**

Module 9 – Enhanced Oil Recovery

Cyclic Steam Stimulation (CSS)

- Cyclic steam stimulation (CSS) is a simple and cheap method of applying thermal recovery process on a reservoir.
- It involves injecting steam into a well for several weeks, shutting the well in as long as necessary to allow the steam to heat the oil in the areas around the well, and putting the well back on production to recover the heated oil.

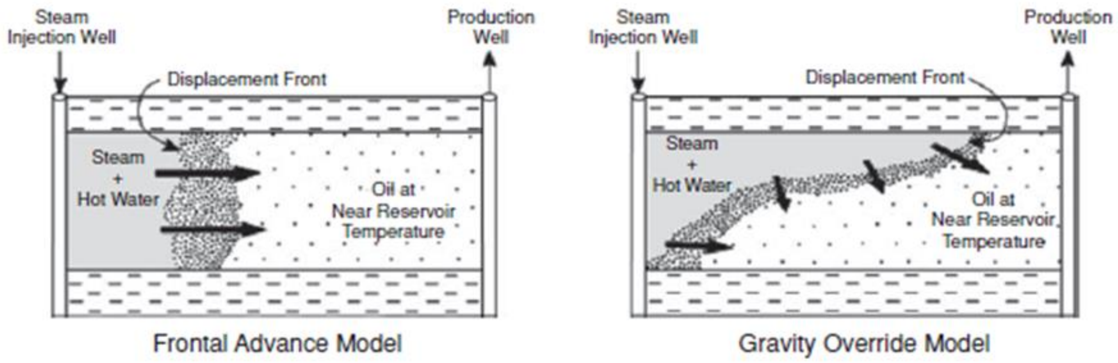


- This process is repeated when the production from the well declines to a low level.
- The cycle is repeated many times until the ratio of oil produced to steam injected termed oil-steam ratio (OSR) drops to a level that is considered uneconomic.
  -

Module 9 – Enhanced Oil Recovery

Steam Drive (SD)

- In steam drives, steam is injected continuously at injectors with the aim of driving oil towards producers.
- Typically in most projects, steam injection is organized in patterns. For instance, in a normal five-spot pattern, the producer is located in the middle of the pattern and surrounded by four steam injectors.

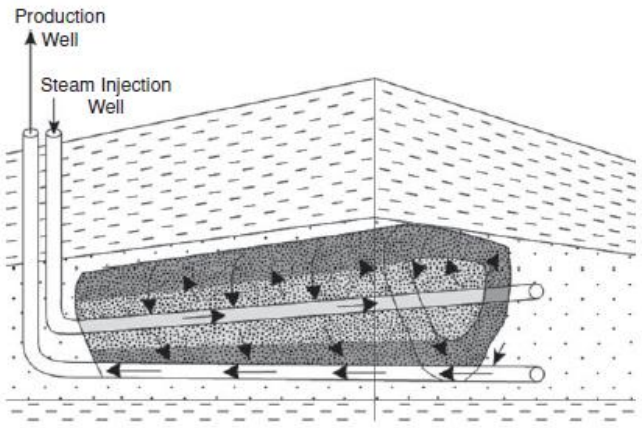


- Various pattern configurations of injectors and producers have been used in many fields.
- The selection of the pattern that is best suited for a particular reservoir is an essential part of the design and pilot testing process, before field implementation of the project.
- Steam drive was initially described as essentially a frontal displacement process similar to waterflooding, whereby steam displaces oil towards a producer.
- Later, experimental and field evidence showed that the mechanism of oil recovery by steam is dominated by gravity drainage caused by steam rising towards the top of the formation due to gravity override
- The accumulation of steam at the top of the formation creates a steam zone which continues to expand with more steam injection.
- This steam zone can propagate and breakthrough at the producers to cause inefficient utilization of the energy from the steam due to cycling.
- The main challenges of managing a steam drive project are monitoring the size and growth of the steam zone, monitoring producers for early steam breakthrough, adjusting steam inject rates, and measuring the trend of oil-steam ratio for each pattern.

Module 9 – Enhanced Oil Recovery

Steam-Assisted Gravity Drainage (SAGD)

- The concept of steam-assisted gravity drainage (SAGD) was tested and shown to be successful by the Alberta Oil Sands Technology and Research Authority (AOSTRA) at the Underground Test Facility in Fort McMurray, Alberta, Canada between 1987 and 2004.
- In 2008, more than ten commercial SAGD projects were in operation in the Athabasca area of Canada, and SAGD has quickly become the preferred thermal recovery process for oil sands in the area.



- The SAGD process is applicable to other heavy oil and oil sands deposits in other countries such as China, Venezuela, and Russia.
- The SAGD process consists of two horizontal wells about 15 feet apart located close to the bottom of the formation.
- Steam is injected into the top horizontal well, while the horizontal well below it functions as the producer. The steam creates an expanding steam chamber around the injector as more steam is injected.
- Within the steam chamber and at its boundaries, as the viscosity of the oil is reduced, its mobility increases causing it to drain under gravity towards the production well.
- The SAGD process should be applied to reservoirs with formation thickness greater than 50 feet, good vertical permeability, and absence of thief zones.
- In some SAGD projects, a hydrocarbon solvent (such as methane) is added to the steam to improve oil recovery.
- This variation of the SAGD process has been described as the expanding solvent-SAGD (ES-SAGD).

Module 9 – Enhanced Oil Recovery

Water-Alternating-Steam Process (WASP)

- Water-alternating-gas (WAG) is an injection strategy used in gasfloods to reduce viscous fingering, improve vertical sweep, and thereby increase oil recovery.

- This concept is also used in steamflooding by injecting water alternating with steam to reduce fingering of steam, and improve its vertical sweep efficiency.
-
- This strategy is called water-alternating-steam process or WASP.

- The main benefit of WASP over continuous steam injection is reduction or elimination of premature steam breakthrough at producing wells.

- Early steam breakthrough at producers wastes heat energy, and reduces productivity of the wells.

Module 9 – Enhanced Oil Recovery

Steamflood Models

- Steamflood models are used to predict the performance of the process in terms of oil production rates, oil-steam ratios, cumulative oil recovery, steam zone size, etc.
- Consequently, steamflood models are essential tools used in the screening, pilot testing, field implementation, and management of steamflood projects.
- The two main mathematical steamflood models widely used in the industry are analytical models and numerical simulation models.
- Analytical steamflood models are based on heat flow and energy balance equations, assuming uniform reservoir properties and shape of the steam zone.
- Numerical steamflood simulation models are based on simultaneous solutions of material balance and energy balance equations, including possibly an equation of state for component phase equilibrium calculations.
- Numerical simulation models can incorporate most of the known geologic features (such as structures, faults, permeability barriers, etc.) and spatial variation of rock and fluid properties for the reservoir.
- Numerical models are more rigorous than analytical models in representing the physical processes of steamflooding, and achieving results that include the impact of geological heterogeneities on the performance of steamfloods.
- Consequently, numerical simulation models are recommended as the preferred method for prediction of steamflood performance.

Module 9 – Enhanced Oil Recovery

Types of Steamflood Models

- Analytical Steamflood Models

- Numerical Steamflood Simulation Models

Module 9 – Enhanced Oil Recovery

Implementation of EOR Projects

- Process screening and selection.
- Quick economic evaluation of selected processes.
- Geologic and reservoir modelling of selected processes.
- Expanded economic evaluation of processes.
- Pilot testing, if necessary.
- Update/upgrade simulation models with pilot test data/results.
- Detailed economic evaluation of selected process.
- Field implementation of process.
- Project management.

References

1. Craft et al. 1991. Applied Petroleum Reservoir Engineering 2nd Ed
2. Ahmed T 2006 Reservoir Engineering Handbook, 3rd Ed
3. Ezekwe N, 2011 Petroleum Reservoir Engineering Practice
4. Economides et al1994, Petroleum Production systems
5. Ertekin,T., Abou-Kassem J.H., King,G.R. Basic Applied Reservoir Simulation, SPE, 2001
6. Dake, L.P 1978. Fundamental of reservoir engineering
7. Engler,T.W 2010. Fluid Flow in porous media